

PHY4P01
PROJECT REPORT

STUDY OF NUCLEAR FISSION OF $^{235}_{92}\text{U}$ WITH THERMAL NEUTRONS INTO Xe and Sr ISOTOPE FRAGMENTS

A Dissertation Submitted
in Partial Fulfillment of the Requirements for the Degree of

MASTER OF SCIENCE
in
PHYSICS

Submitted by
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KEEZHAL MUKKU, VADAKARA

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Signature of the examiners

- 1.
- 2.

CALICUT UNIVERSITY

BONAFIDE CERTIFICATE

This is to certify that the project entitled “**STUDY OF NUCLEAR FISSION OF $^{235}_{92}\text{U}$ WITH THERMAL NEUTRONS INTO Xe and Sr ISOTOPE FRAGMENTS**” in partial fulfillment for the award of the Degree of Master of Science in Physics, to the Calicut University is a bonafide record of project work done by **ANURIYA RAVEENDRAN (SBAWMPH001)** during the academic year 2022-2024. The result embodied in this project report have not been submitted to any other university or institute for the award of degree or diploma.

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BONAFIDE CERTIFICATE

This is to certify that the project entitled “**STUDY OF NUCLEAR FISSION OF $^{235}_{92}\text{U}$ WITH THERMAL NEUTRONS INTO Xe and Sr ISOTOPE FRAGMENTS**” Submitted in fulfillment of the requirement for the fourth semester project of Master of Science in Physics is result of the bonafide work carried out by **ANURIYA RAVEENDRAN** during the academic year 2023-24.

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DECLARATION

I **ANURIYA RAVEENDRAN**, hereby declare that the project report entitled “**Study Of Nuclear Fission Of $^{235}_{92}\text{U}$ With Thermal Neutrons into Xe and Sr Isotope Fragments**” is submitted to Calicut University under the guidance of **M.G.V SANKARACHARYULU** in the fulfillment of the requirements for the award of Master of Science. This is a bonafide work carried out by me and the results embodied in this project report have not been reproduced/copied from any source. The results embodied in this project report have not been submitted to any other university or institution for the award of any other degree or diploma.

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I would like to express gratitude to my family and friends for their support in completion of this work.

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ABSTRACT

Induced fission of $^{235}_{92}\text{U}$ with thermal neutrons into $^{140}_{54}\text{Xe}$ and $^{94}_{38}\text{Sr}$ is investigated using liquid drop model and quantum mechanical model with shell energy corrections. The critical energy is measured using liquid drop model. Liquid drop model does not consider the spin of fragments and their coupling.

Quantum mechanical model is used to study nuclear potential. The yield distribution of mass number of fragments, neutron number of fragments and proton number of fragments is measured and plotted.

The neutron distribution curves are plotted at different temperatures of fissile nucleus. The distribution of neutrons and the yield of mass fragments is measured and plotted. The effect of deformation of fragments is also studied.

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Chapter 1

INTRODUCTION

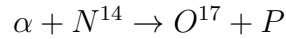
In the last two decades, through technological, experimental and theoretical advances, the situation in experimental fission studies has changed dramatically. Current studies in the area of nuclear fission are focused on addressing various challenges and opportunities related to nuclear energy production, safety, waste management, non-proliferation, advanced reactor designs such as small modular reactors, molten salt reactors and high temperature gas-cooled reactors, exploring alternative fuel forms such as accident-tolerant fuels and thorium based fuels. Research also focuses on severe accident management strategies and development of next generation reactor safety standards. Efforts are ongoing to develop a sustainable solution for the management and disposal of radioactive waste generated by nuclear fission reactors that includes research on advanced waste treatment technologies, geological disposal methods and long - term monitoring strategies. These ongoing research efforts seek to advance the state-of-the-art in nuclear fission technology while addressing key challenges and ensuring the safe, secure and sustainable use of nuclear energy resources.

1.1 BEGINNING OF NUCLEAR PHYSICS

The beginning of nuclear physics as a distinct scientific discipline can be traced back to the late 19th and early 20th centuries with the discovery of radioactivity. In 1896, Henri Becquerel accidentally discovered radioactivity while studying phosphorescent materials. He found that uranium salts emitted invisible, penetrating rays that could expose photographic plates. This discovery led them to further investigations into the nature of radioactive emissions. In 1897, J.J Thomson discovered the electrons, a fundamental particle with a negative electron charge. This discovery laid the groundwork for understanding atomic structure and provided for the foundation for later developments in nuclear physics. In 1909 Ernest Rutherford conducted the famous gold foil experiment, which led to discovery of the atomic nucleus. Rutherford's model of the atom proposed that the atom consist of a small, dense nucleus surrounded by orbiting electrons. This revolutionary concept marked the beginning of modern atomic theory and nuclear physics. In the early 20th century, further experiments identified the proton as a positively charged particles in the atomic nucleus. The neutron, a neutral particle, was discovered by James Chadwick in 1932. These discoveries provided crucial insights in to the composition and structure of atomic nuclei. In 1938 Otto Hahn, Fritz Strassmann, Lise Meitner, and Otto Frisch discovered nuclear fission, the process by which heavy atomic nuclei split in to lighter fragments, releasing a large amount of energy. This discovery marked a significant break through in nuclear physics and paid the way for the development of nuclear weapons and nuclear energy. These early discoveries and developments laid the foundations for the emergence of nuclear physics as the distant scientific discipline, which continues to explore the fundamental properties and applications of atomic nuclei and nuclear reactions.

1.2 NUCLEAR REACTIONS

The process of production of new nuclei and elementary particles in collision of particles and nuclei is called **nuclear reaction**. If upon collision original nuclei and particles remain and no new nuclei or particles are produced, then the process is called **scattering**. **Rutherford** was the first to observe nuclear reaction in 1919, bombarding nitrogen atomic nuclei by alpha particles. Nuclear reaction was detected by the appearance of secondary ionising particles having the path in gas longer than alpha particle path and identified as protons. Subsequently, the photographs of the process were taken with the help of the cloud chamber. This reaction is represented by the expression,



1.2.1 Excited states of nuclei

The nuclear reactions to be considered may be represented by set of equation,

$$x+X \rightarrow \begin{cases} X + x \\ X^* + x \\ Y + y \\ Z + z \end{cases}$$

If the outgoing particle is of the same kind as the incident particle, the process is called **scattering**. The **elastic scattering** in which the total kinetic energy of the system, projectile + target, is same before the collision as after. Some kinetic energy is usually transferred from the projectile to the target nucleus, but the latter is left in the same internal, or nuclear state as before the collision. A collision between two billiard balls, in which neither ball is damaged or otherwise changed, is an example of elastic collision. The second reaction represents **inelastic scattering**, in which the target nucleus X is raised to the excited state X^* (STAR), and the total kinetic energy of the system is decreased by the amount of the excited energy given to the target nucleus. The other reaction of the set represents different possible nuclear transmutations in which the product nuclei may be formed in their ground states or more often in excited states. The excited product nucleus usually decays very quickly to the ground state with the emission of gamma rays.

1.2.2 Classification of nuclear reactions

. Nuclear reactions are classified as: scattering process, knockout reaction, transfer reaction, direct reaction, compound reaction, resonance reaction.

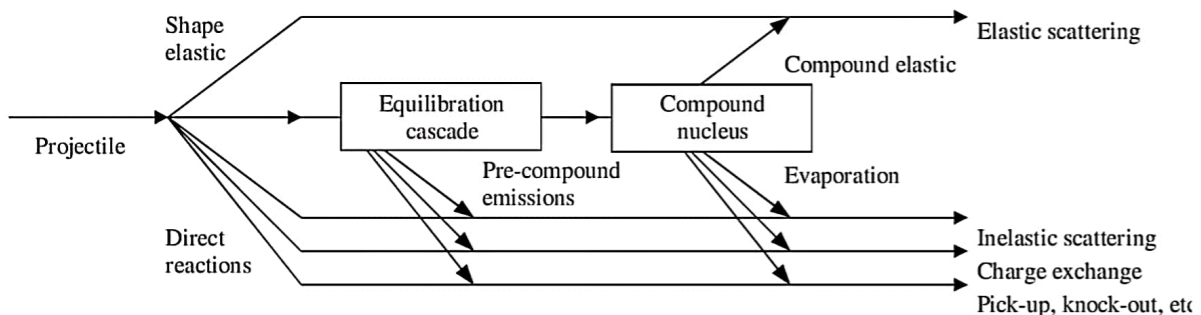


Figure 1.1:

Scattering process

A nuclear reaction is called a scattering process, if the incident and out going particles are same ($a = b$ & $X = Y$). The scattering is elastic if Y and b are in their ground state, inelastic, if Y and b are in excited state.

Knockout process

A nuclear reaction is called out knock out reaction if a and b are same but the reaction causes yet another nucleon to be ejected separately.

Transfer process

A nuclear reaction is called transfer reaction if one or two nucleons are transferred between projectile and target.

Direct reaction

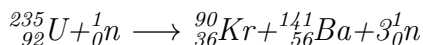
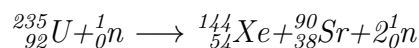
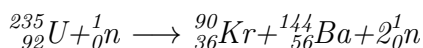
A nuclear reaction is called a direct reaction if only a few nucleons take part in the reaction. Such reactions might insert or remove a single nucleon from a shell model state and might therefore serve as ways to explore the shell structure of nuclei. Many excited states of Y can be reached in these reactions.

Compound reaction

A nuclear reaction is called a compound reaction if the incoming and target nuclei merge briefly for a complete sharing of energy before the out going nucleon is ejected.

Resonance reaction

A nuclear reaction is called a resonance reaction if the incoming particles forms a quasi bound state before the outgoing particle is ejected. subsectionNuclear Fission In the fission process, certain heavy elements, such as some forms of Uranium, are split when a neutron strikes them. When they split, they release energy in the form of kinetic energy (heat) and radiation. The process not only produces energy but also additional neutrons that can be used to fission other Uranium nuclei and start a chain reaction. Fission is a nuclear reaction in which an atomic nucleus splits, or fission, into fragments, usually two fragments of comparable mass, with the release of large amount of energy in the form of heat and radiation.



The neutron was discovered by Chadwick in 1932. Enrico Fermi and his co-workers exposed many elements to neutron capture. In 1939, Hans and Strassmann showed that barium is produced when a uranium nucleus is irradiated with a neutron. In 1939, Meitner and Frisch proposed that the uranium nuclei after a neutron capture is highly unstable and may split into two, which is called as fission. Fission process is important only for heavy nuclei (thorium and beyond). A fission process produces two heavy fragments and several neutrons. These neutrons can produce further fission process. Thus chain reaction can occur very rapidly. Using a nuclear reactors, controlled nuclear reactions can be produced to obtain energy for producing electricity.

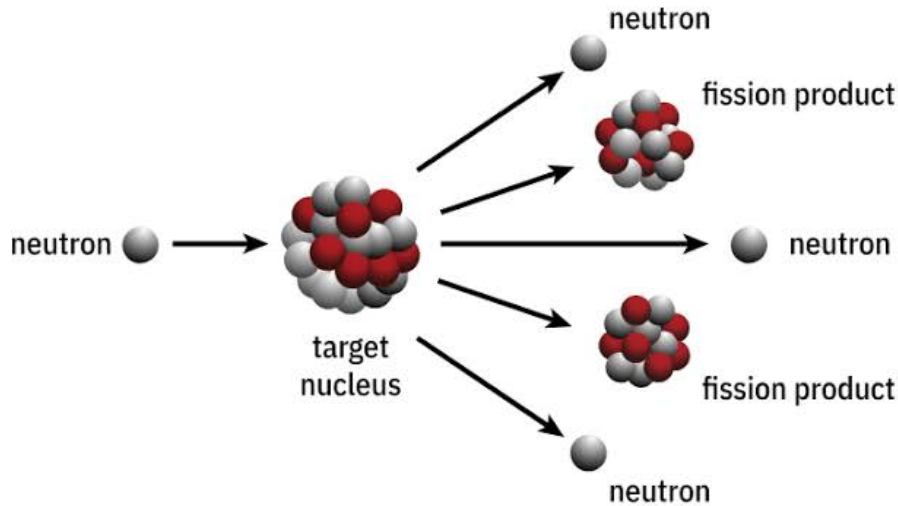


Figure 1.2:

1.2.3 Nuclear Fusion

Nuclear Fusion reactions power the Sun and other stars. In a fusion reaction, two light nuclei merge to form a single heavier nucleus. The process releases energy because the total mass of the resulting single nucleus is less than the mass of the two original nuclei. The leftover mass becomes energy. Einstein's equation ($E=mc^2$), which says in part that mass and energy can be converted into each other, explains why this process occurs. If scientists develop a way to harness energy from fusion in machines on Earth, it could be an important method of energy production.

Fusion reactions are inhibited by the electrical repulsive force, called the Coulomb force, that acts between two positively charged nuclei. For fusion to occur, the two nuclei must approach each other at high speed in order to overcome their electrical repulsion and attain a sufficiently small separation (less than one-trillionth of a centimeter) so that the short-range strong force dominates. For the production of useful amounts of energy, a large number of nuclei must undergo fusion.

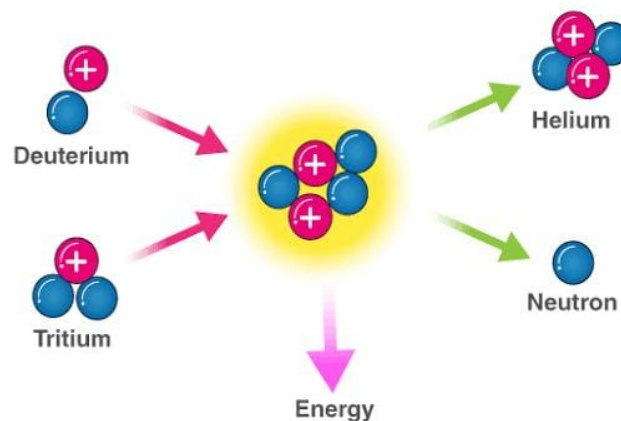
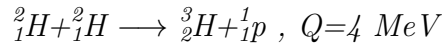
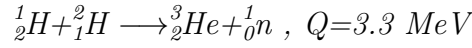
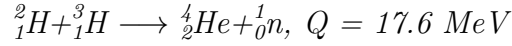


Figure 1.3:

Fusion can involve many different elements in the periodic table. However, researchers working on fusion energy applications are especially interested in the deuterium-tritium (DT) fusion reaction. DT fusion produces a neutron and a helium nucleus. In the process, it also releases much more energy than most fusion reactions.



This are called deuterium- deuterium reactions. If the end product is stable, then the energy released will be more.



This is called the deuterium-tritium reaction. In this case, neutron get the 17.6 MeV energy. This reaction is a source for fast neutrons.

Fusion of 4 protons to form to form ${}^4_2\text{He}$ is responsible for the thermonuclear energy released in stars.

1.2.4 Nuclear fission

In the fission process, certain heavy elements, such as some forms of Uranium, are split when a neutron strikes them. When they split, they release energy in the form of kinetic energy (heat) and radiation. The process not only produces energy but also additional neutrons that can be used to fission other Uranium nuclei and start a chain reaction. Fission is a nuclear reaction in which an atomic nucleus splits, or fission, into fragments, usually two fragments of comparable mass, with the release of large amount of energy in the form of heat and radiation.

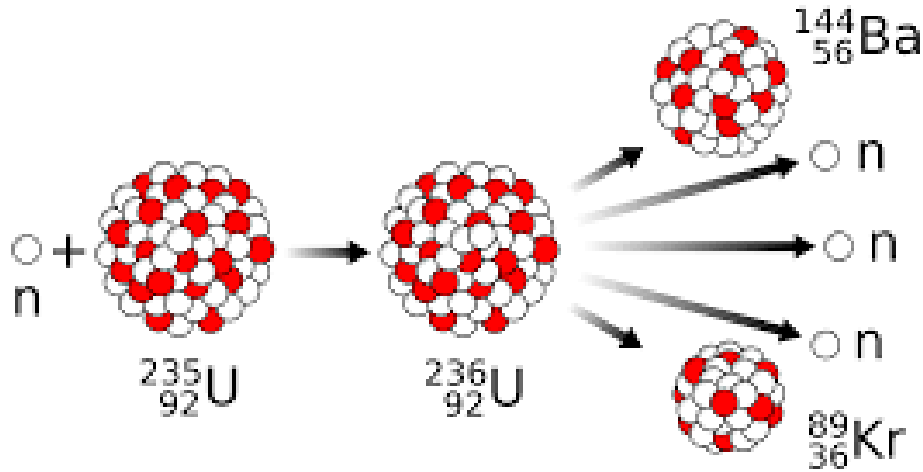


Figure 1.4:

The neutron was discovered by Chadwick in 1932. Enrico Fermi and his co-workers exposed many elements to neutron capture. In 1939, Hans and Strassmann showed that barium is produced when a uranium nucleus is irradiated with a neutron. In 1939, Meitner and Frisch proposed that the uranium nuclei after a neutron capture is highly unstable and may split into two, which is called as fission. Fission process is important only for heavy nuclei (thorium and beyond). A fission process produces two heavy fragments and several neutrons. These neutrons can produce further fission process. Thus chain reaction can occur very rapidly. Using a nuclear reactors, controlled nuclear reactions can be produced to obtain energy for producing electricity.

Chapter 2

NUCLEAR FISSION

2.1 DISCOVERY OF NUCLEAR FISSION

The discovery of nuclear fission was one of the results of attempts to make transuranium elements of atomic number greater than 92 by means of (n, γ) reactions followed by beta decay of the product nucleus. These attempts eventually succeeded. The interpretation of early experiments was difficult, however, because of unexpected results it could be explained only in terms of nuclear fission. A radioactive element formed by a nuclear reaction is usually available only in a very small amount, possibly as small as 10^{-12} g and cannot be separated by means of ordinary chemical methods. But the separation can often be made with the aid of a carrier which is a stable substance with chemical properties similar to those of the radioelement. The element and the carrier usually belong to the same subgroup of the periodic system and can undergo similar chemical reactions. If an appreciable amount, perhaps 10 to 100 mg, of the carrier is added to the solution containing the radio element, and if the carrier is then precipitated from the solution by the formation of an insoluble salt, the radio element is precipitated along with the carrier. The carrier and the radioelement can then be separated by means of other chemical reaction.

In the separation of radium from other member so f the uranium series, barium serves as a carrier. Radium and Barium both belong to group IIA of the periodic system and form insoluble sulphates. When a solution of ab barium salt is added to the solution containing a very small amount of radium and a sulphate is added, barium and radium sulphate are precipitated together. A neater method depends on the fact that barium and radium chlorides are precipitated together from concentrated solution of hydrochloric acid. In analogous ways, lanthanum act as a carrier for actinium ($Z=89$). Sometimes a carrier is a stable form of the radioactive element. E.g. Stable iodine can be used as a carrier for radioactive isotope of iodine.

In early experiments on the formation of transuranium elements, uranium was bombarded with neutrons and several different beta activities, distinguished by their half lives, were detected. Carrier techniques were used to separate the elements responsible for activities. but the number and properties of the new radio elements were such that they could not be fitted into a scheme consistent with non- properties of the heavy element and the predicted properties of the transuranium elements. One difficulty in the analysis of the products of the bombardment of the uranium with neutrons led to a remarkable conclusion. In addition to the elements which seem to be real transuranium elements, there were four which were supposed to beta radio active isotopes of radium because they were precipitated wit barium when the latter was used as the carrier. The decay products of these nuclides seem to be isotopes of actinium because they were precipitated with lanthanum, the carrier for actinium. Although these results appear to be consistent, since actinium follows radium in the periodic system, that raised two serious questions. First, the production of an isotope of Radium by the neutron bombardment of uranium it require an $(n, 2\alpha)$ reaction, but this reaction is very unlikely one especially at low neutron energies. Second, further

chemical experiments showed that "radiant" activities could not be separated from the barium carrier, and the daughter activities could not be separated from the lanthanum carrier. In 1939, Hahn and Strassman performed a beautiful and thorough set of experiments which proved beyond a doubt that the "radium" isotopes are really isotopes of barium and the "actinium" isotopes are isotopes of lanthanum. Furthermore, they showed that one of the barium isotopes resulting from the neutron bombardment of uranium could be identified, because of its half life of 86 minute, with the previously known nuclide barium(139) which has the same half life. Similarly, one of the lanthanum isotopes from the neutron bombardment of uranium was identified with the known nuclide lanthanum (140) which has a half life of 40 hr..

The production of nuclides $^{140}_{57}\text{La}$ and $^{139}_{56}\text{Ba}$ from uranium, which has atomic number 92 and an atomic weight of nearly 240, required a hitherto unknown kind of nuclear reaction in which the uranium nucleus is split into fragments which are themselves nuclei of intermediate atomic weight. If such a process really occurs, it should be also possible to find a nuclei with masses between 90 and 100 and atomic numbers of about 35. **Hahn and Strassman** were able to find an active isotope of strontium ($Z=38$) and one of Yttrium ($Z=39$) which met these requirements, as well as isotope of Krypton ($Z=36$) and Xenon ($Z=54$). It was clear from the chemical evidence that uranium nuclei, when bombarded with neutrons can indeed split into two nuclei of intermediate atomic weight. It was then predicted, from the systematics of stable nuclides and from the semi-empirical binding energy formula, that the product nuclei would have very great energies and when produced large number of ion pairs in passive through a gas. When a thin layer of uranium was put in suitable ionisation chamber connected to an amplifier and irradiated with neutrons, great burst of ionisation were observed corresponding to energies upto 100 MeV. These pulse of ionisation are extremely large compared with those single alpha particles and are easy to recognise. Further chemical work then showed that besides the reaction products mentioned above, other elements of medium mass number were formed, including Bromine, Molybdenum, Rubidium, Antimony, Tellurium, Iodine and Caesium. There was, therefore ample chemical and physical evidence for splitting the uranium nucleus and this process is called **fission**.

It is now known that fission can be produced in various nuclides under different conditions, and some of the results will be mentioned. When small samples of separated isotopes of uranium, prepared in a mass spectrograph, were bombarded, it was found that slow neutrons can cause fission of $^{235}_{92}\text{U}$ but not of $^{238}_{92}\text{U}$, fast neutrons again, with energies greater than 1 MeV cause fission of both $^{235}_{92}\text{U}$ and $^{238}_{92}\text{U}$. Thorium and $^{231}_{90}\text{Pa}$ undergo fission but only when bombarded with fast neutrons. Fission can also be produced in Uranium and Thorium by high energy alpha particles, protons, deuterons and gamma rays. The nuclei $^{239}_{94}\text{Pu}$ and $^{233}_{90}\text{Th}$ formed by (n, γ) reactions on $^{238}_{92}\text{U}$ and $^{232}_{90}\text{Th}$ respectively, followed by beta decay of the products, undergo fission when bombarded with slow or fast neutrons, as do other artificial heavy nuclides. Finally some heavy nuclide had been found to undergo spontaneous fission. In this process, the nucleus divides in the ground state without bombarded by particles from outside.

In addition to two large fission fragments, neutrons and gamma rays are emitted. Division into three fragments of comparable size (ternary fission) have been observed but its a very rare event occurring about five times per billion binary fission. Long range alpha particles are sometimes emitted about once in every 400 fissions. The emission of light nuclei, with masses greater than 4 and probably less than 12, is a relatively common event, occurring once in about 80 fission.

2.1.1 Types of fission reactions

The type of fission in which some kind of particle is involved to break up a target nucleus is known as **induced fission**. In the beginning the fission of Uranium by slow neutrons was detected. Later on it was found that the nuclei of other elements of high atomic number could be made to undergo fission. Also that other particles, besides neutrons, could be effective in this respect. The different Uranium isotopes require projectile or different incident energies. The fission of ^{238}U requires neutrons of 1 MeV energy. The fission of Thorium and Protactinium require fast neutrons 9 MeV deuterons produce fission in both Uranium and Thorium, 32 MeV alpha particles and 7 MeV protons are also effective. Till 1947, the fission of any element with atomic number less than 19 was not detected but in the same year the fission Bi, Pb, Tl, Hg, Au, Pt and Ta was achieved but with projectiles of energy of the order 100 MeV, the latter require alpha particles of 400 MeV.

Thus, it's obvious that any method which will make a nucleus sufficiently unstable can be used to produce the fission of nuclides. *Hahby, Shoupp, Stephens and Wells* (1941) produced fission of Uranium by employing gamma ray photons of 6.3 MeV of energy. Such type of fission is known as **photofission**. *Balwich and Koch* (1945) produced fission of Uranium with the help of 8 MeV X-rays. Later on, the photofission of thorium and protactinium was also achieved.

In addition to induced fission, it was found that nuclides with mass number of 230 are more exhibit spontaneous fission, that is in which no projectile particle is involved. The phenomenon was reported by **Petrzhak and Fleror** in 1941. The half life of ^{238}U for spontaneous fission was estimated 10^{16} years, it means that 25 nuclei on an average, undergo spontaneous fission every hour thus this rate is much smaller than the alpha decay rate ^{238}U . Since 1941, the spontaneous fission rate of some 35 nuclides $N \geq 90$ is measured. The observed rates extend from about 3×10^{-4} fissions $\text{gm}^{-1}\text{sec}^{-1}$ for ^{235}U to about 10^{17} fission $\text{gm}^{-1}\text{sec}^{-1}$ for ^{238}Fm . Thus spontaneous fission has been established as another mode of radioactive decay.

There are several types of fission reactions, distinguished by the type of particles that induce the fission and the specific isotopes involved. Here are the main types:

1. Spontaneous fission: Some heavy isotopes have a tendency to undergo fission spontaneously without external stimulation. This occurs due to the instability of their nuclei. For example, ^{238}U and ^{232}Th undergo spontaneous fission. However, spontaneous fission is relatively rare compared to induced fission.

2. Induced Fission: This type of fission occurs when a heavy nucleus absorbs a neutron or other particle, becoming unstable and then splitting into two or more smaller nuclei. The most common type of induced fission is neutron-induced fission. When a neutron collides with a heavy nucleus such as ^{235}U or ^{239}Pu , it can cause the nucleus to split. This process releases additional neutrons and a significant amount of energy.

3. Photofission: In photofission, the nucleus is split by absorbing a gamma-ray photon. This phenomenon is less common compared to neutron-induced fission and is usually observed in heavy nuclei when they absorb high-energy gamma rays.

4. Electron-induced Fission: In some rare cases, nuclei can undergo fission when bombarded with high-energy electrons. This process is less common and typically requires highly energetic electrons, such as those produced in particle accelerators.

5. Proton-induced Fission: Similar to electron-induced fission, heavy nuclei can also undergo fission when bombarded with high-energy protons. This process is also relatively rare and requires high-energy protons, which can be produced in particle accelerators.

6.Symmetric fission:Symmetric fission refers to a type of nuclear fission in which the nucleus splits into two roughly equal-sized fragments. In this process, the total kinetic energy and other properties of the daughter nuclei are approximately symmetrically distributed between the two fragments. This type of fission occurs under certain conditions, often associated with specific isotopes and particular excitation energies.

7.Asymmetric fission:Asymmetric fission is a type of nuclear fission in which the nucleus splits into two fragments of significantly different sizes. In contrast to symmetric fission, where the resulting fragments are roughly equal in size, asymmetric fission results in one larger fragment and one smaller fragment. This type of fission occurs when the nucleus breaks apart unevenly due to various factors such as the asymmetry in the initial shape or excitation energy distribution of the nucleus. Asymmetric fission plays a crucial role in nuclear physics, particularly in understanding the dynamics of nuclear reactions and in applications such as nuclear energy production and the development of nuclear weapons.

8.Binary fission:It refers to nuclear fission reaction where the parent nucleus splits into exactly two daughter nuclei, each with approximately equal mass.

9.Tertiary fission:It refers to nuclear fission reaction where the parent nucleus splits into more than two daughter nuclei.

10.Controlled fission: In the fission process ,large amount of energy is released along with the new neutrons which are so essential to carry on the self sustained nuclear chain reaction.The value of multiplication factor plays a decisive role in starting or not,of a chain reaction and the value of k is detrmined by size and shape of the system in which a nuclear chain reaction is sort.

(i) $k > 1$,the chain reaction takes place almost instantaneously in an uncontrolled manner such as in an atom bomb.The size of the system is called **super critical** in this case.

(ii) $k = 1$,in this case the rate of neutron production is exactly equal to loss of neutrons by absorption,leakage etc.A chain reaction still proceeds over an extended period but does not explode.In such critical chain reactions are very important in industry and research as this can produce nuclear energy in controlled manner,The size of the system pertaining to this is called **critical**.

(iii) $k < 1$,the chain reaction is not a self sustained one and size of the systems in this case is called **sub-critical**

Thus the condition for producing a controlled self-sustained chain reaction and for an uncontrolled self- sustained chain reaction must e carefully chosen.The problem of producing nuclear energy in a controlled manner has been solved in a device called **nuclear reactor**.The nuclear reactor is thus a device in which a fissionable material called **nuclear fuel**,is arranged along with other materials which act as slowing down agents for a fast neutron is called **moderator** and controlling device in a structure which self-sustaining chain reaction could be maintained under controlled conditions.

Each type of fission reaction has its own characteristics and applications. Neutron-induced fission, in particular, is of great interest due to its significance in nuclear reactors and nuclear weapons. Understanding the different types of fission reactions is crucial for various scientific and technological applications, including nuclear energy production, nuclear physics research, and nuclear weapon development.

2.1.2 Fission cross sections and threshold

The probability of fission, as compared with that of other reactions, is a matter of theoretical interest and practical importance. The ^{235}U nucleus may capture a thermal neutron to form the compound nucleus ^{236}U , or the neutron may be scattered. The compound nucleus may either undergo fission, or it may emit γ rays and decay to the ground state of ^{236}U , which emits a 4.5 MeV α -particle and has a half-life of 2.4×10^7 years. The total cross section of ^{235}U for 2200 m/sec neutrons is 698 b, the total absorption cross section is 683 b, and the fission cross section is 577 b. The ratio of the radiative capture cross section to the fission cross section is 0.18, so that the probability of radiative capture is about 18% that of fission. The natural abundance of ^{235}U is only 0.72%, with the result that the fission and radiative capture cross sections of natural uranium are much smaller than those of the separated isotope ^{235}U . In ^{239}Pu the ratio of the capture cross section to that for fission is 0.39, although the fission cross section is greater than in ^{235}U , the ratio of capture to fission is also greater.

Some of the artificially produced radioactive nuclides undergo fission with thermal neutrons, as do some of the shorter-lived naturally occurring radionuclides. It is evident that many heavy nuclides are fissionable, and that the probability of fission varies over a wide range of values. Only ^{235}U , ^{233}U and ^{239}Pu have high cross sections as well as long half-lives, and either occur naturally (^{235}U) or can be produced in significant amounts in practical lengths of time (^{233}U and ^{239}Pu). Hence, only these three fissionable materials are important in the large-scale applications of nuclear fission. In the thermal region σ_f varies approximately as $\frac{1}{v}$, starting at 0.28 eV. There are many closely spaced resonances, with at least 20 resolved resonances below 20 eV. At high energies, the fission cross section is relatively small, only about one barn in the neighborhood of 1 MeV, the fission cross sections of ^{233}U and ^{235}U in the neighborhood of 1 MeV. Some heavy nuclides, for ... , starting at 0.28 eV, there are many closely spaced resonances, with at least 20 resolved resonances below 20 eV. At high energies, the fission cross section is relatively small, only about one barn in the neighborhood of 1 MeV, the fission cross sections of ^{233}U and ^{235}U in the neighbourhood of 1 MeV. Some heavy nuclides, for example, ^{234}U , ^{236}U , and ^{238}U , do not undergo fission with slow neutrons, but only with fast neutrons. Fission is a threshold reaction in these nuclides, and the fission cross section varies with energy in much the same case of ^{233}U bombarded with deuterons, the cross section rises to a value of one barn at about 20 MeV and then increases slowly to a value between 1.5 and 2 b at deuteron energies between 120 and 200 MeV. The cross section for the fission of ^{233}U by α -particles rises to a value of 1.5 b at about 50 MeV and remains at about this value up to energies of 400 MeV.

2.1.3 Fission product

The number of nuclides of intermediate charge and mass are formed when a uranium nucleus undergoes fission. The study of the nuclei formed in fission was evidently a promising source of information about the mechanism of the fission process and offered the possibility that new, hitherto unknown, nuclides might be discovered. The latter possibility became apparent when the neutron-to-proton ratios of the uranium isotopes were compared with those of some of the fission products. The compound nucleus $^{236}_{92}\text{U}$ has 144 neutrons and 92 protons, and the value of the ratio is $144/92 = 1.57$. The values of the ratio for the stable isotopes of some typical fission products (krypton, iodine, xenon, and caesium) vary from 1.17 to 1.52, and are appreciably lower than that for ^{236}U . When that excited nucleus splits into two smaller nuclei, the neutron to proton ratio for at least one of them must be greater than the value compatible with stability. Such an unstable nucleus might be expected to approach stability by electron emission or, if the excitation energy is high enough, by ejection of one or more neutrons, it has been found experimentally that both of these processes occur.

The investigation of the products of the fission of ^{235}U has shown that the range of mass numbers is from 72, probably an isotope of zinc with atomic number 30, to 158, thought to be an isotope of europium with atomic number 63. About 97% of the ^{235}U nuclei undergoing fission yield products which fall into two groups, a "light" group with mass numbers from 85 to 104, and a "heavy" group with mass numbers from 130 to 149. The most probable type of fission, which occurs in about 7% of the total, gives products with mass numbers 95 and 139. There are 87 possible mass numbers between 72 and 158, which may represent the total number of different nuclides formed as direct fission fragments. If this were the case, the uranium nucleus should be capable of splitting in over 40 different ways. More than 60 primary products have actually been detected, so that there are at least 30 different modes of fission, a different pair of nuclei being formed in each mode.

The fission fragments have too many neutrons for stability and most of them decay by electron emission. Each fragment starts a short radioactive series, involving the successive emission of electrons. These series are called **fission decay chains**, and each chain has three members, on the average, although longer and shorter chains occur frequently. The problem of determining the masses and atomic numbers of the fission products and of identifying the members of the many decay chains is an extremely difficult one. Nevertheless, as the result of careful and persistent work, more than 60 chains have been established, and about 200 different radio-nuclides have been assigned to them.

2.1.4 The mass and energy distribution of the fission product

The mass distribution of the fission products is shown most conveniently in the form of a **fission yield curve**, in which the percentage yields of the different products are plotted against mass number. The yield of any given mass can be found by measuring the abundance of a long-lived nuclide near the end of a chain, or that of the stable end product. Yield curve for the fission of $^{235}_{92}\text{U}$ by thermal neutrons and by 14 MeV neutrons are shown in Fig.a, curves for thermal fission of $^{233}_{92}\text{U}$ and $^{239}_{94}\text{Pu}$ are shown in Fig.b. The fission yield for a particular nuclide is the probability (expressed as a percentage) of forming that nuclide or the chain of which it is a member; it may also be regarded as the percentage of fissions yielding the nuclide or chain. Since two nuclei result from each fission, the total yield adds up to 200%. The yields vary from about $10^{-50}\%$ to about 7%, and the range is so large that a logarithmic scale is used for the ordinate of the yield curve. Each curve shows two peaks, corresponding to the light and heavy groups of products. In the case of $^{233}_{92}\text{U}$ the maxima lie near mass numbers 95 and 135; fission induced by slow neutrons is a highly asymmetric process, and division into two equal fragments occurs in only about 0.01% of the fissions. Small "fine-structure" peaks are evident in some of the yield curves; for $^{235}_{92}\text{U}$, these

peaks are at masses 100 and 134. This fine structure is, for the most part, a shell effect, and can be accounted for by a detailed analysis of the neutron binding energies in nuclei with neutron numbers close to those corresponding to closed shells, i.e., 50 and 82 neutrons. The ^{233}U curves illustrate the effect on the mass distribution of increasing the neutron energy above thermal. The greatest change is the increase in the probability of symmetric fission; for fission by 14 MeV neutrons, the increase is about 100-fold. The other changes are a small drop in the peak yields and a moderate increase in the most asymmetric modes of fission, i.e., a rise in the wings of the yield versus mass curve.

The mass distribution of the fission fragments can also be obtained from the distribution of their kinetic energies, which can be determined by measuring the ionization produced in an appropriate ionization chamber. In one type of chamber, the fissile material is placed on one of the electrodes and the ions which result when a fission fragment enters the region between the electrodes are collected. Since the fission fragments occur in pairs, an experiment of this kind measures the energy of one fragment only. In another type of chamber, a very thin foil is made the common cathode of two back-to-back ionisation chambers. The two fragments resulting from neutron bombardment travel in opposite directions into the two chambers and the ionisations they cause are measured simultaneously. The nucleus undergoing fission can be considered to be initially at rest; and, if the neutrons emitted are neglected, the law of conservation of momentum gives

$M_1V_1 = M_2V_2$ where the subscripts refer to the two fission fragments. The energies of the fragments are then in the ratio

$$\frac{E_1}{E_2} = \frac{1/2M_1V_1^2}{1/2M_2V_2^2} = \frac{M_2}{M_1}$$

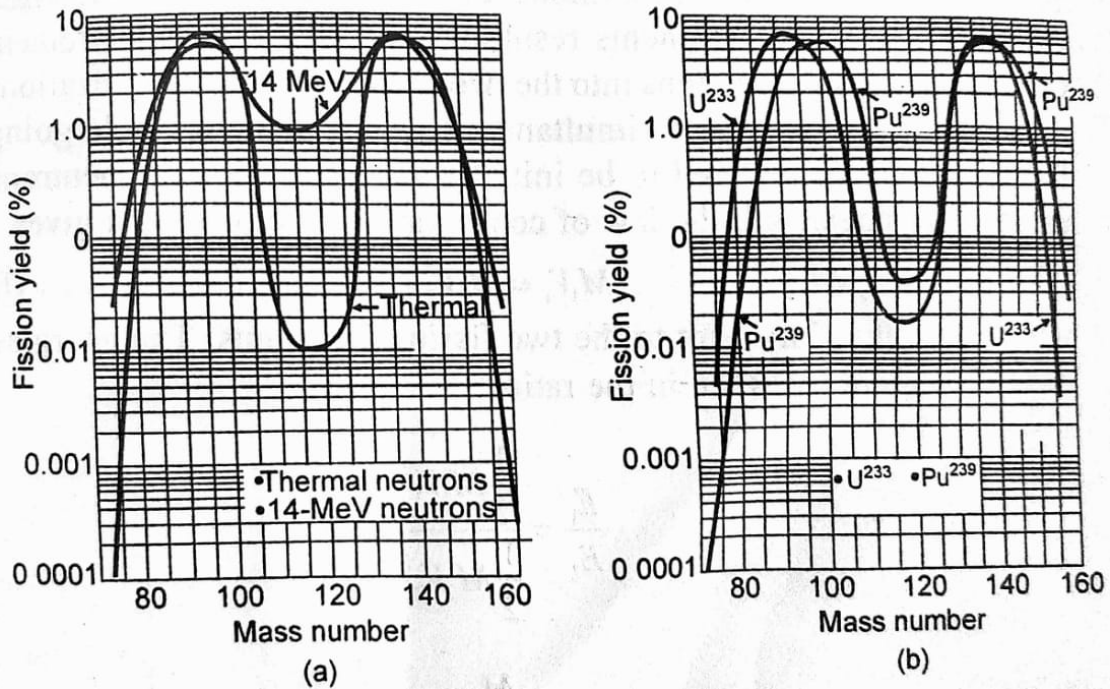
and the masses are inversely proportional to the kinetic energies, when the energy distribution has been measured. The mass distribution is obtained from the above equation.

The energy distribution of the fission products has been measured for the fission of ^{233}U , ^{235}U , ^{239}Pu by thermal neutrons, and for the fission of ^{235}U , ^{238}U , ^{232}Th , and ^{239}Pu by fast neutrons. The energy distribution in the case of the thermal fission of ^{235}U is shown in Figure , this energy spectrum is typical for fission induced by slow neutrons, and the curves for ^{233}U and ^{239}Pu are similar to the curve shown. The peaks near 60 MeV and 95 MeV show the asymmetry of the fission process. The best ionization chamber measurements give 154.7 MeV for the average total kinetic energy of the fragments from ^{235}U . The energy distribution for fission induced by fast neutrons (2.5 and 14 MeV) show the same two peaks, and the maxima differ only slightly from the values for fission induced by slow neutrons. The most important difference in the shapes of the spectra is an increase in the height of the curve in the region between the two peaks when the energy of the bombarding neutrons is in the MeV range. This rise is interpreted as indicating increased probability of symmetrical fission when the neutron energy is high. In the case of ^{235}U , symmetrical fission is about 100 times more probable for 14 MeV neutrons than for thermal neutrons. When the energy of the bombarding neutrons is raised to 45 MeV, there are still two peaks, but the dip between them is small, and the probability of symmetrical fission is still greater than at the lower energies. With 90 MeV neutrons, only one peak is observed, corresponding to division into two equal fragments; at these very high energies symmetrical fission is the most probable mode.

The velocity distribution of the fission fragments has been studied directly by means of a time-of-flight method. A thin foil with a film of ^{233}U , ^{235}U , or ^{239}Pu is irradiated by thermal neutrons, and pairs of fragments are detected by scintillation detectors. One fragment travels only about 1 cm before striking a detector, while the other travels about 350 cm along an evacuated tube. The pulses from the two detectors are shown on the screen of a cathode ray tube and photographed; the velocity of the fragment can be obtained from the distance between the two peaks, and the kinetic energy is then calculated from the velocity. Results obtained with this method show that

the energies are greater than those given by the ionization measurements, and the average total kinetic energy of the fission fragments from U²³⁵ is increased by 12.4 MeV, from 154.7 MeV to 167.1 MeV. This value agrees with the result, 167.1 ± 1.6 MeV, of a calorimetric measurement.

The energy distribution of the fragments from spontaneous fission has also been determined and is similar to that found in neutron-induced fission. The determination of the charge distribution in fission is more difficult than that of the energy distribution because in an ionization chamber or cloud chamber there is no obvious way of determining the nuclear charge at the instant of fission. Radio-chemical methods based on attempts to measure the primary fission yield have been used, these methods are indirect, and the problem has not yet been solved.



2.1.5 Neutron emission in fission process

The comparison of the value of the neutron-to-proton ratio of the compound nucleus (^{236}U) with the values for some of the fission products, indicated also that neutrons might be emitted in fission. The fact that neutrons are emitted was shown in a simple experiment shortly after the discovery of fission. A neutron source was placed at the center of a large vessel, with detectors at various distances from it to determine the neutron density in the vessel. The vessel was filled first with a uranyl sulfate solution, and then with an ammonium nitrate solution for comparison. The average neutron density was found to be greater when uranium was present, showing in a rough way that more neutrons were formed, when fission occurred, than were used up. The average value of the number of neutrons released in fission, usually denoted by ν , has been measured for various fissionable materials. The average number of neutrons released is always greater than two and increases with the energy of the neutrons that induce fission. The number of neutrons released in any one fission process must, of course, be an integer but, since the fissionable nucleus can divide in at least 30 different ways, the average value ν of the number of neutrons does not have to be an integer.

In addition to ν , there is another property of fissionable materials which has practical importance, the average number of neutrons emitted per neutron absorbed by a fissionable nuclide. Radiative capture competes with fission. With ^{235}U , for example, the reaction $^{235}\text{U}(n, \gamma)^{236}\text{U}$ also occurs, and ^{236}U is an α emitter with a half life of 2.4×10^7 years. The ratio of the radiative capture cross section to the fission cross section is usually denoted by α ,

$$\eta = \nu \sigma_f / \sigma_a = \nu \sigma_f / (\sigma_f + \sigma_c) = \nu / (1 + \alpha)$$

and the number of fission neutrons released per neutron absorbed in a fissionable nuclide is given above.

The average number of neutrons emitted per spontaneous fission has been determined for various heavy nuclide. The rate (number of fissions per gram per second) can be measured as well as the total number of neutrons. For ^{238}U , the rate is $(6.90 \pm 0.24) \times 10^{-3}$ fissions/gm/sec, or about 25 fissions/gm/hr, and $\nu = 2.4 \pm 0.2$, close to the value for fission induced by thermal neutrons; for ^{232}Th , $\nu = 2.6 \pm 0.3$, and for ^{252}Cf , $\nu = 3.53 \pm 0.15$.

The neutrons emitted as a result of the fission process can be divided into two classes, **prompt neutron** and **delayed neutrons**. The prompt neutrons, which make up about 99% of the total fission neutrons, are emitted within an extremely short interval of time, possibly as low as 10^{-14} sec, of the fission process. It is thought that the compound nucleus ^{236}U first splits into two fragments, each of which has too many neutrons for stability, and has also the excess energy (6 MeV or more) needed to expel a neutron. The excited, unstable nucleus consequently ejects one or more neutrons within a very short time after its formation; prompt γ -rays are apparently emitted at the same time. These ideas are consistent with the results of experiments on the angular correlation between - the direction of the neutrons and that of the fragment, the neutrons being emitted preferentially in the same direction as the fragments.

The delayed neutrons, which constitute about 0.64% of the total neutrons from the fission of ^{235}U , are emitted with gradually decreasing intensity for several minutes after the actual fission process. Six well defined groups of delayed neutrons have been observed by studying the rate of decay of the neutron intensity. The rate of decay of each group is exponential, just as for other forms of radioactive change, and a specific half-life can be assigned to each group, as well as a mean life and decay constant. From the intensity, the fraction β_i which the group constitutes of the total (prompt and delayed) fission neutrons can be determined. Three additional low-intensity delayed neutron groups from uranium have been reported, with half-lives of 3, 12, and 125 min and yields, per fission, of 5.8×10^{-8} , 5.6×10^{-10} and 2.9×10^{-10} respectively.

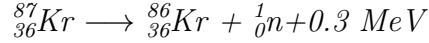
The same groups of delayed neutrons are found in the fission of different heavy nuclides by slow or by fast neutrons, but the yields vary from nuclide to nuclide. Of the three important fissionable materials, ^{235}U has the greatest yield, 0.64%, about three times that of ^{239}Pu . Although the yield of delayed neutrons is less than one percent of the total number of neutrons emitted, the delayed neutrons have a strong influence on the time-dependent behavior of a chain-reacting system based on fission and play an important part in the control of the system.

The mechanism for the emission of delayed neutrons is understood. Since some of the fission products are rich in neutrons and are very unstable with respect to β -emission, a product with Z protons and N neutrons may have a β -decay energy greater than the binding energy of the last neutron in the daughter product with $Z + 1$ protons and $N - 1$ neutrons. In the β -decay, the product nucleus can be left either in the ground state or in one of the many excited states. If the excitation energy of one of the excited states of the $Z + 1$, $N - 1$ nucleus is greater than the binding energy of the last neutron, de-excitation may occur by the emission of a neutron, leaving a nucleus with $Z + 1$ protons and $N - 2$ neutrons. The neutron emission will be delayed and will appear to have the half-life of the beta-decaying nuclide (Z , N), i.e., that of the delayed neutron precursor. Several of the delayed neutron groups have been related to the decay of bromine and iodine fission products. Thus, ^{87}Br is the precursor of the 55-sec delayed neutron emitter. Iodine-137, with a half-life of 24 sec, is associated with the 22-sec group. This group may consist of two or more precursors with similar half-lives, and may also contain ^{88}Br which has a half-life of 16.3 sec.

Prompt and delayed neutrons

The neutrons emitted in the fission of any target (fissionable) can be distinctly divided into two classes: (i) **Prompt neutrons** (ii) **delayed neutrons**

Except for a very small fractions (about 0.75%) of the total fission neutrons, all fission neutrons are emitted virtually instantaneously, the time delay, if any certainly being less than 10^{-14} sec of fission process. These neutrons are called **prompt neutrons**. In terms of compound nucleus formation theory, we may think that compound nucleus first splits into two fragments in which the proton to neutron ratio is too large to allow them to be stable and they quickly expel some of the neutrons almost instantaneously. However for the production of controlled nuclear energy such as in a **Nuclear Reactor**, the other class of neutrons called delayed neutrons are of extreme importance. Delayed neutrons amount to less than 1% to all the fission neutrons and are emitted with a time lag of several seconds to more than one minute of fission even as a result of the radioactive β -decay of the fission fragments. When the excitation energy of the daughter nucleus after β -decay is greater than the neutron separation energy, subsequent de-excitation takes place in the form of neutron emission with a half life almost identical with that of the preceding β -decay. The best known example is that of the $^{87}_{35}\text{Br}$, which in its ground state is stable against neutron decays but is unstable against β -decay. About 30% of $^{87}_{35}\text{Br}$ decays to the ground state and about 70% to a 5.4 MeV excited state of $^{87}_{36}\text{Kr}$. The 5.4 MeV level is up by about 0.3 MeV from the level corresponding to the binding energy of the last viz. (51st) neutron in $^{87}_{36}\text{Kr}$. Therefore, about 2% of the $^{87}_{36}\text{Kr}$ nuclei decay to $^{86}_{36}\text{Kr}$ through neutron emission according to transition:



Rest 98% of $^{87}_{36}\text{Kr}$ decays through γ emission to the ground state of $^{87}_{36}\text{Kr}$. The delayed neutrons are delayed entirely by the 55 sec half life of their β -ray parent $^{87}_{36}\text{Br}$. $^{87}_{36}\text{Kr}$ has 51 neutrons and when a neutron is emitted, closed shell of 50 neutrons is left. Similarly $^{137}_{56}\text{I}$ decays to $^{137}_{54}\text{Xe}$ by β emission and when $^{137}_{54}\text{Xe}$ decays through neutron emission, a closed shell of 82 neutrons is left. At least six distinct group of delayed neutrons, each group with its own characteristic half life and decay constant, have been identified.

2.1.6 Energy distribution of neutrons emitted in fission

The energies of the prompt neutrons emitted in fission vary from values less than 0.05 MeV to more than 17 MeV. The neutrons from the thermal fission of $^{235}_{92}\text{U}$ have been studied in great detail, and this case provides a good example of the methods and results. In the energy range 0.05 to 0.7 MeV, a cloud chamber was used containing hydrogen gas and water vapor. A thin foil of $^{235}_{92}\text{U}$ was bombarded with a beam of thermal neutrons from the thermal column of a nuclear reactor. The fission neutrons were allowed to enter the cloud chamber, where they collided with protons, and the range of the recoil protons was determined by measuring the track lengths. The energy distribution of the recoil protons was obtained from the range measurements, and the energy distribution of the neutrons was obtained from that of the recoil protons. The results are shown in Fig.2.1, where the relative number of neutrons in a given energy interval is plotted against the energy and compared with an empirical formula which will be given below. The neutron energy spectrum between 0.4 MeV and 17 MeV was studied by measuring, with coincidence counters, the range distribution of recoil protons from hydrogenous materials. The neutron intensity shows a broad maximum near 0.75 MeV, and decreases nearly exponentially at energies greater than 2 MeV. The energy distribution between 0.075 MeV and 17 MeV can be described by the empirical formula,

$$N(E) = \exp(-E) \sinh(2E)^{1/2}$$

where $N(E)$ represents the relative number of neutrons per unit energy range as a function of the neutron energy. The experimental results between 0.4 MeV and 17 MeV are shown in Fig.2.2 and agree with the empirical formula. The solid curve of Fig.2.1 represents the same

empirical formula, and the agreement is good also for the low energy neutrons. The average value of the neutron energy is 2.0 ± 0.1 MeV. The fission neutron spectra of ^{235}U and ^{239}Pu , in the range from 0.5 to 8 MeV, have been studied with nuclear emulsion plates. The spectrum for the neutrons from ^{239}Pu shows a maximum in the 0.6 to 0.8 MeV region and the empirical formula of above equation, fits the experimental results; the average energy is again 2.0 MeV.

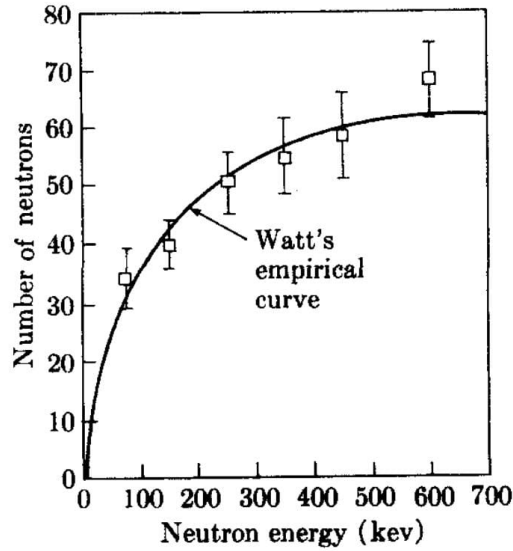


Figure 2.1: Energies of fission neutron:the low energy region

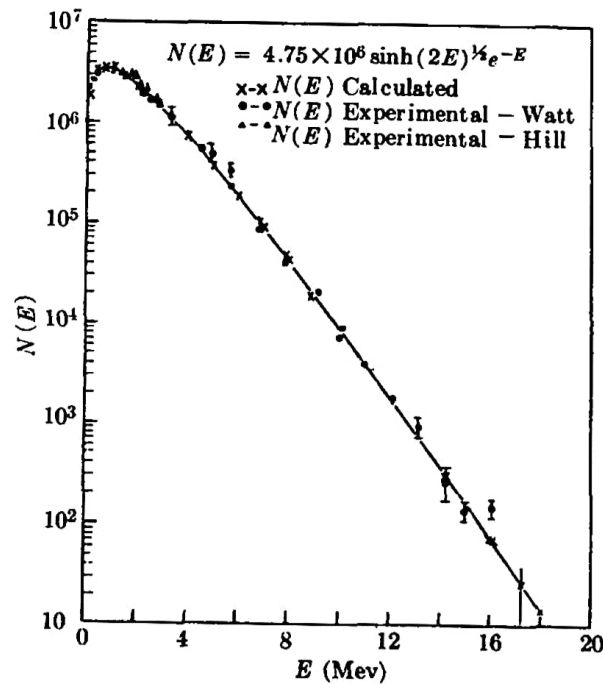
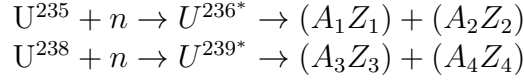


Figure 2.2: Energies of fission neutron:spectrum from 0.4 MeV to 17 MeV

2.1.7 Fissile and fissionable nuclides

In neutron induced fission of heavy nuclei, each fission gives rise to new free neutrons capable of initiating subsequent fission i.e., a possibility arises for producing a self-sustaining chain reaction. In distinction from fission due to γ -ray quanta where the target nucleus is fissioned, when the process is initiated by neutrons a nucleus with the mass number a unit higher is fissioned. For example, when neutrons are absorbed by $^{235}_{\text{U}}$ or $^{238}_{\text{U}}$ nuclei,

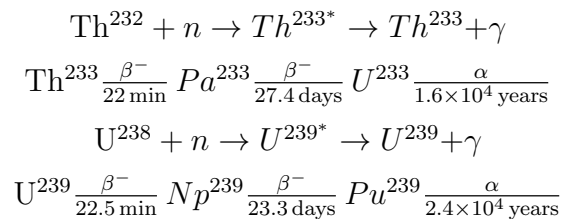


The compound nuclei of $^{236}_{\text{U}}$ and $^{239}_{\text{U}}$ are, in fact, fissioned. To ascertain the possibility of fissioning of original nuclei, one should compare the excitation energies of compound nuclei produced in neutron capture with the values of energy barriers. The minimum excitation energy of compound nucleus is the bond energy of neutron joining the nucleus. If this bond energy is higher than the energy barrier, then fission is possible only on condition that the kinetic energy of neutron is high enough to exceed, when summed with bond energy, the value of the barrier. The bond energies of neutrons in nuclei which are compound in fissioning of most significant heavy nuclides are listed below:

Nuclide	Threshold energy, MeV
Th^{233}	5.07
U^{234}	6.77
U^{236}	6.40
U^{239}	4.76
Pu^{240}	6.38

The bond energy of the paired neutron is always higher than that of unpaired neutron. For this reason, the bond energy of neutron in the nuclei of $^{234}_{\text{U}}$, $^{236}_{\text{U}}$ and $^{240}_{\text{Pu}}$ appears to be higher than the value of energy barrier of the fission, and the nuclei of $^{233}_{\text{Th}}$ and $^{239}_{\text{U}}$ it is lower because the value of the barrier differs but slightly in nuclei similar in the composition. This circumstance responsible for the possibility of fission $^{233}_{\text{U}}$, $^{235}_{\text{U}}$ and $^{239}_{\text{Pu}}$ by neutrons of any energies. Such nuclides are called **fissile**. On the contrary, $^{232}_{\text{Th}}$ and $^{238}_{\text{U}}$ can be fissioned by neutrons only of sufficiently high kinetic energy. Fission thresholds are 1.2 MeV in $^{232}_{\text{Th}}$ and above 1 MeV in $^{238}_{\text{U}}$. These nuclides appear to be unstable of sustaining a chain reaction and are called **threshold**. Among heavy nuclides, any other nuclides with an odd number of neutrons in the composition of nuclei are fissile, and those with an even number of neutrons are threshold. For example, $^{240}_{\text{Pu}}$ like $^{238}_{\text{U}}$ is threshold, while $^{241}_{\text{Pu}}$ is fissile. Nuclides formed in radioactive capture of neutrons $^{233}_{\text{U}}$ and $^{235}_{\text{U}}$, and in $^{234}_{\text{U}}$ and $^{236}_{\text{U}}$, respectively are threshold nucleus.

$^{232}_{\text{Th}}$ and $^{238}_{\text{U}}$ can be reprocessed into fissile nuclides, and as they exist in their normal form, they are considered to be suitable for reprocessing. The idea of reprocessing of fissionable nuclide is in joining one more neutron to the nucleus which is even in number of its neutrons:



^{233}Th and ^{239}U are produced as a result of radiative capture of neutrons in ^{232}Th and ^{238}U , respectively. The former rapidly decay producing short-lived products. After two successive β -decays, ^{233}U and ^{239}Pu are produced, which are odd in number of their neutrons and, therefore, are fissile nuclides. They are α active but has sufficiently long half-lives to be considered stable from the point of view of accumulation and storage. Accumulation of fissile ^{233}U and ^{239}Pu nuclides is usually achieved in nuclear reactors, which always have a great excess of free neutrons.

The five heavy nuclides are most significant for nuclear engineering. Three of them, ^{232}Th , ^{235}U and ^{238}U , are found in nature, and two ^{233}U and ^{239}Pu are produced from fissionable nuclides. Of fissile nuclides, ^{235}U alone is present in its natural form. Natural uranium contains 0.7% of ^{235}U .

2.1.8 The energy released in fission

One of the most striking properties of the fission process is the magnitude of the energy released per fission, which is about 200 MeV as compared with several MeV for other nuclear reactions. An estimate of the amount of energy released per fission can be made by considering the binding energy curve, Fig. 2.3. The value of the average binding energy per particle has a broad maximum of about 8.4 MeV in the range of mass numbers from 80 to 150, and it has been shown that nearly all of the fission products have mass number in this range. The average binding energy per particle is about 7.5 MeV in the neighborhood of uranium. Hence, the average binding energy per particle is about 0.9 MeV greater in the fission products than in the compound nucleus (^{236}U), and the excess, 0.9 MeV, is liberated in the fission process. The total amount of energy released per fission should be roughly equal to the product of the number of particles (236) multiplied by the excess binding energy per particle (0.9 MeV), or approximately 200 MeV. The total energy release per fission can also be calculated from the : nuclear masses of (^{236}U) and a typical pair of fission products. It has been shown that the fission products with the greatest yields have mass numbers near $A = 95$ and $A = 139$. If $^{95}_{42}\text{Mo}$ and $^{139}_{57}\text{La}$ are taken as a pair of stable products at the ends of the chains for their respective masses, their combined mass is $138.955 \text{ amu} + 94.946 \text{ amu} = 233.900 \text{ amu}$, as given by the semi-empirical mass formula. The corresponding mass number is 234, and 2 neutrons are apparently released in this particular fission process. When 2.018 amu are added for these two neutrons, the products of the reaction have a total mass of 235.918 amu. The mass of ^{235}U from the semi-empirical mass formula is 235.124 amu, so that the mass of the compound nucleus ^{236}U is close to $235.124 + 1.009 = 236.133 \text{ amu}$. The mass excess that is converted to energy is equal to $236.133 \text{ amu} - 235.918 \text{ amu} = 0.215 \text{ amu}$. Since one amu is equivalent to 931 MeV, the energy liberated in the process is $931 \times 0.215 = 198 \text{ MeV}$. Although there are at least 30 different ways in which the nucleus can divide, the mass excess is approximately the same for all of these processes, and 200 MeV is a good value for the average amount of energy released per fission, as calculated in this way.

The predicted value of about 200 MeV can be compared with experimental values. The total amount of energy released per fission is the sum of the kinetic energy of the fission fragments, the kinetic energy of the emitted neutrons, the kinetic energy of the prompt γ -rays, and the total energy of the decay processes in the fission decay chains. The average value of the total kinetic energy of the fission fragments from the thermal fission of ^{235}U is 167 MeV; the uncertainty in this number is about 5 MeV. The average value of the kinetic energy carried off by the neutrons is equal to the product of the average number of neutrons emitted per fission and the average kinetic energy of the neutrons; for the thermal fission of ^{235}U , this product is $2.5 \times 2.0 \text{ MeV} = 5 \text{ MeV}$. The kinetic energy of the prompt γ -rays is in the neighborhood of 5 to 8 MeV. The average energy of all radiations (β -rays, γ -rays, and neutrinos) of the fission products is $21 \pm 3 \text{ MeV}$; about half of this energy escapes in the form of neutrinos, and the other half is divided, approximately equally, between the β -rays and the γ -rays.

Fission of ^{235}U can be studied easily using thermal neutrons. ^{236}U can break up into large number of pair of fission fragments. In the present work, thermal neutrons induces fission of ^{235}U into $^{139}_{56}\text{Ba}$

and $^{94}_{36}\text{Kr}$ is studied. For proton induced fission, cross-section is less. Fission can be used in the production of electricity in a controlled manner.

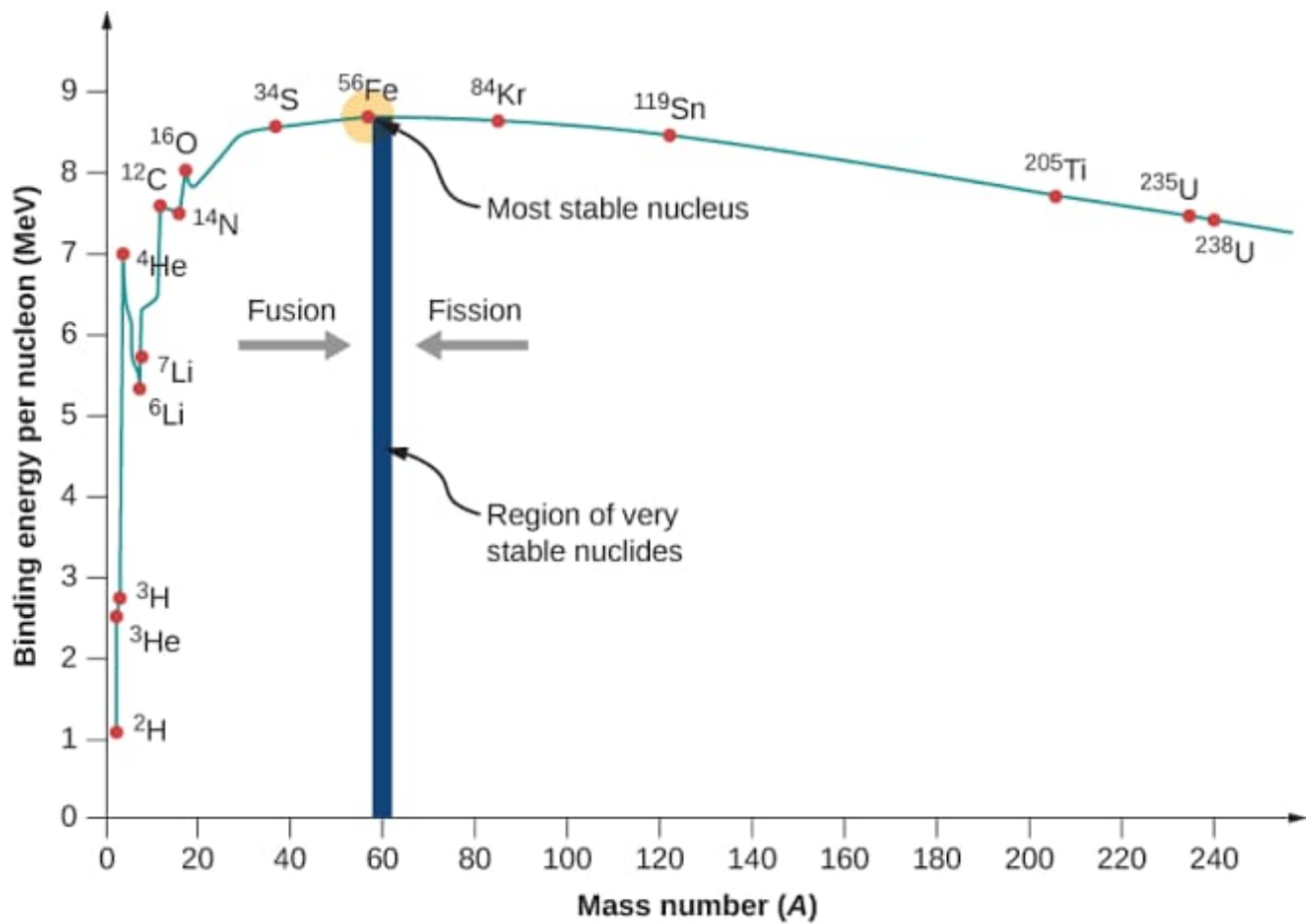


Figure 2.3: Binding energy per nucleon as a function of mass number

Chapter 3

THEORY

3.0.1 The theory of the fission process

It is not surprising, in view of the remarkable properties of the fission process, that a great deal of

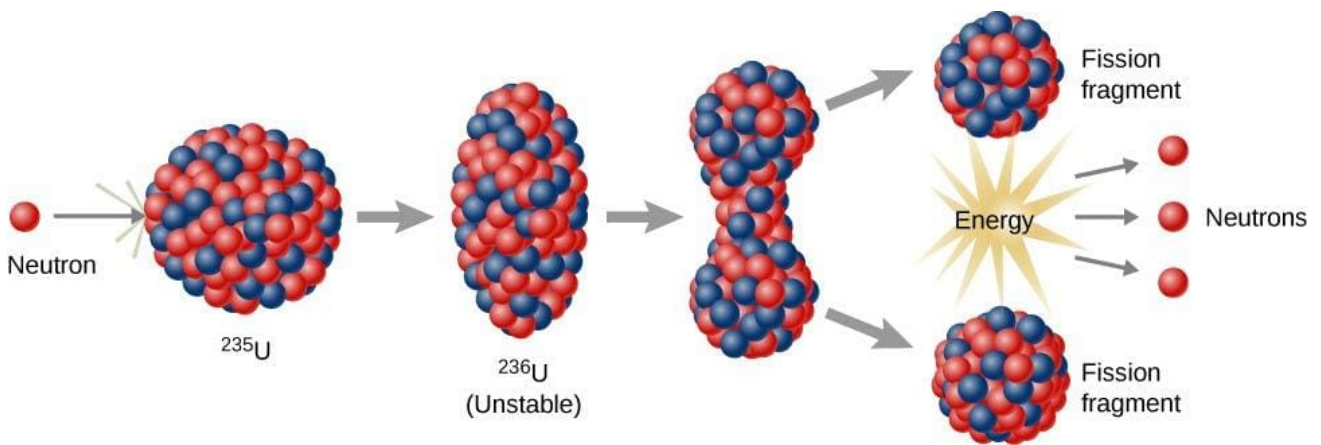


Figure 3.1: Possible steps in the process of nuclear fission according to liquid-drop model.

effort has been put into theoretical studies of the process. The first thorough treatment was that of Bohr and Wheeler who accounted for many of the properties of fission on the basis of the liquid-drop model of the nucleus. This model was used in calculating the energy release in fission from the atomic masses of the fissile and product nuclei as given by the semi-empirical mass formula; that formula was derived by considering that a charged nucleus is analogous to a liquid drop. Bohr and Wheeler showed that the liquid-drop model could be used to describe the process of fission in some detail, and made successful predictions about the existence of spontaneous fission, and about the ability of various heavy nuclei to undergo fission with slow or fast neutrons.

In the spherical, liquid-drop nucleus, the shape of the drop depends on a balance involving the surface tension forces and the Coulomb repulsive forces. If energy is added to the drop, as in the form of the excitation energy resulting from the capture of a slow neutron, oscillations are set up within the drop; these tend to distort the spherical shape so that the drop may become ellipsoidal in shape. The surface tension forces tend to make the drop return to its original shape, while the excitation energy tends to distort the shape still further. If the excitation energy is sufficiently large, the drop may attain the shape of a dumbbell. The Coulomb repulsive forces may then push the two "bells" apart until the dumbbell splits into two similar drops, each of which then becomes spherical in shape. The sequence of steps leading to fission is shown in Fig.3.1. If the excitation energy is not large enough, the ellipsoid may return to the spherical

shape, with the excitation energy being liberated in the form of γ -rays, and the process is one of radiative capture rather than fission.

The potential energy of the drop in the different stages of Fig.3.1 can be calculated under the Bohr-Wheeler theory, as a function of the degree of deformation of the drop. The form of the results is shown in Fig.3.2 where the potential energy E is plotted against a parameter: which is a measure of the degree of deformation. At $r = 0$, which represents

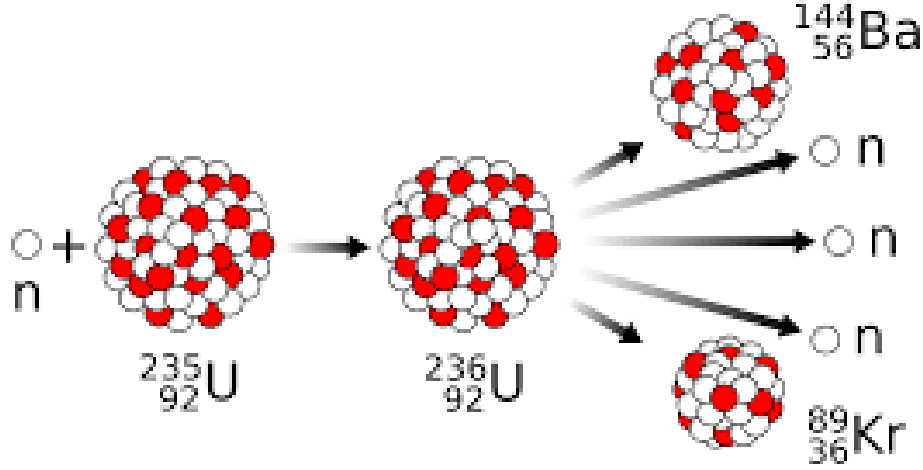


Figure 3.2: Potential energy of two fission fragments as a function of distance between centers and configuration at contact, where the potential energy is 197 MeV.

the initial spherical nuclear drop, there is available an amount of energy E_o , given by ,

$$E_o = c^2[M(A, Z) - M(A_1, Z_1) - M(A_2, Z_2)],$$

where (A, Z) represents the nucleus that may undergo fission, and the subscripts 1 and 2 stand for the possible final products. The value of E_o corresponds to the ground state of the compound nucleus formed when the target nucleus captures a neutron, but does not include the excitation energy resulting from the neutron capture.

The meaning of the graph can now be seen more easily if the process which would be the reverse of fission is considered. When the drop has split into two fragments, r is the distance between their centers; if R_1 and R_2 are the radii of the two product drops, $r = R_1 + R_2$ is the value of the deformation parameter for which the two drops just touch. For values of r smaller than $R_1 + R_2$, r represents the degree of deviation of the original drop from its spherical shape. For values of r greater than $R_1 + R_2$, the energy is just the electrostatic energy resulting from the mutual repulsion of the two positively charged nuclear fragments. The value of E in this region, $\frac{Z_1 Z_2 e^2}{R_1 + R_2}$ and E increases as the distance between the two fragments is decreased; the energy at $r = \infty$: is taken equal to zero. When the two fragments just touch, at $r = R_1 + R_2$, the Coulomb energy is denoted by E_c and is

$$E_c = \frac{Z_1 Z_2 e^2}{R_1 + R_2}$$

When r is smaller than $R_1 + R_2$, the energy depends not only on the electrostatic forces, but also on the surface tension forces; the calculation of E is then complicated, and three different curves (I, II, and III) are shown for the possible variation of E in this region. The shapes of these curves and the values of E_o are related to the mass of the nucleus, that is, to the value of A in equation above. Stable nuclei with values of A somewhat greater than 100 are of type I,

with E_o about 50 MeV smaller than E_c . Nuclei like those of uranium, plutonium, or thorium are of type II, for which $E_c - E_o$ is about 6 MeV. For still heavier nuclei, E_o may be greater than E_c as in case III. Nuclei of type III should undergo fission spontaneously and would not be expected to last long.

On a classical basis, nuclei of type II should be stable with respect to fission, but according to quantum mechanics, there is a certain probability that they will undergo spontaneous fission. The fragments may "leak through" the barrier represented by E_c , in the same way that α -particles do in α -decay. In ^{238}U , about 25 spontaneous fissions occur per gram per hour, and the half-life for this process is about 10^{17} years.

The activation energy $E_c - E_o$ needed to induce fission in nuclei of type II has been calculated from the Bohr-Wheeler theory, and can be compared with the excitation energy resulting from the capture of a neutron, or other particle, or a γ -ray. The excitation energy of a compound nucleus can be computed with the aid of the semi-empirical binding energy (or mass) formula, or from the masses of the nuclei involved if they are : known. When the compound nucleus is formed by the capture of a slow neutron, the excitation energy differs by a negligible amount from the binding energy of the last neutron.

In ^{233}U , ^{235}U ^{239}Pu the excitation energy is considerably greater than the activation energy, and these nuclei would be expected to undergo fission with thermal neutrons; they do, and the thermal fission cross sections of ^{235}U and ^{239}Pu are large, 577 b and 742 b, respectively. In ^{238}U and ^{232}Th , the excitation energy is smaller than the activation energy and these nuclei do not undergo fission with thermal neutrons, as predicted. To induce fission, the neutrons must have considerable kinetic energy, and fission should be a threshold reaction in these nuclei, as it is. In ^{231}Pa and ^{237}Np , the excitation energy is greater than the activation energy, and these nuclides should undergo fission with thermal neutrons. Early experiments indicated that they do not; recent measurements show that there is some thermal fission, but the cross sections are very small (about 10^7b) compared with those of ^{235}U and ^{239}Pu . The fact that the calculated values of the excitation energy are greater than those of the activation energy for ^{231}Pa and ^{237}Np does not agree with the very low probabilities of fission by thermal neutrons. It is found generally that the excitation energy is greater for target nuclei with odd numbers of neutrons than for nuclei with even numbers of neutrons, and this effect is related to the presence of the "odd-even" term in the 'binding energy formula. Fission with thermal neutrons occurs much more often in nuclei with an odd number of neutrons than in nuclei with an even number of neutrons. Nearly all of the target nuclei with large values of the cross section for fission by thermal neutrons have odd numbers of neutrons, while nearly all of the nuclei with very small cross sections have even numbers of neutrons. There is, therefore, some correlation between the values of the excitation energy and the occurrence of thermal fission, in qualitative agreement with the predictions of the Bohr-Wheeler theory. There are some obvious exceptions; $^{240}_{96}\text{Cm}$ has a very large cross section nearly (20000b) and $^{230}_{92}\text{U}$ and $^{232}_{92}\text{U}$ has a very large cross section (25 b and 80 b, respectively) ; each of these nuclei has an even number of neutrons. The nuclides $^{231}_{91}\text{Pa}$ $^{237}_{93}\text{Np}$ have low excitation energies, corresponding to their even numbers of neutrons, but their activation energies are still lower, so that they should undergo fission with slow neutrons inspite of their odd neutron numbers. The theory based on the liquid-drop model gives incorrect values of the activation energy in these cases.

The theory gives a qualitative, and sometimes quantitative, picture of the process and has made possible the description of some of the properties of fission, but it also has some serious defects in addition to those already mentioned. According to the theory, the most probable mode of division of the liquid-drop nucleus is into two equal fragments, so that it fails to account for the observed asymmetry of the process; it also yields values for photo-fission thresholds and for spontaneous fission rates which are wrong quantitatively, and whose variations from nuclide to nuclide do not agree with experiment.

Attempts have been made to modify the liquid-drop theory of fission in order to overcome these difficulties. The application of the shell model has been suggested and has had some success in accounting for the asymmetry. Another promising approach seems to be that of Hill and Wheeler, which is based on the new collective model of the nucleus.

3.1 NUCLEAR MODELS

3.1.1 Liquid drop models

A great deal of importance has been attached to the mass and binding energy of the nucleus, and a formula which would allow the calculation of nuclear masses or binding energies would be very useful. Such a formula has been developed and is called **semi-empirical mass formula** or the **semi-empirical binding energy formula**. It is related to the liquid drop model of the nucleus. Some of the properties of nuclear forces (saturation, short range), which have been deduced from approximately linear dependence of the binding energy and the volume on the number of particles in the nucleus, are analogous to the properties of the forces which hold a liquid drop together. Hence, a nucleus may be considered to be analogous to a drop of incompressible fluid of very high density 10^{14} g/cm³. This idea has been used, together with other classical ideas such as electrostatic repulsion and surface tension, to set up a semi-empirical formula for the mass or binding energy of a nucleus in its ground state. The formula has been developed by considering the different factors which affect the nuclear binding, and weighting these factors with constants derived from theory where possible and from experimental data where theory cannot help.

The main contribution to the binding energy of the nucleus comes from a term proportional to mass number A , since the volume of the nucleus is also proportional to A , this term may be regarded as a **volume energy**.

$$E_v = a_v A$$

The coulomb energy between the protons tends to lower the binding energy and its effect appears as a term with minus sign. The total coulomb energy of a nucleus of charge Z , and its effect on the binding energy is represented by the term,

$$E_c = E_c^0 \left(1 - \frac{1}{5} \alpha^2 - \frac{4}{105} \alpha^3 \right)$$

$$E_c^0 = a_c Z(Z-1)A^{-1/3}$$

$$\alpha = \sqrt{\frac{5}{4\pi}} \times \beta$$

The binding energy is also reduced because the nucleus has a surface. Particles at the surface interact, on the average again, only with half as many other particles as do particles in the interior of the nucleus. In the volume energy equation, it was assumed that every nucleon has the same access to other nucleons and it is necessary, therefore again, to subtract a term proportional to the surface area of the nucleus. This **surface energy** is represented as

$$E_s = E_s^0 \left(1 + \frac{2}{5} \alpha^2 - \frac{4}{105} \alpha^3 \right)$$

$$E_s^0 = a_s(1 - K_s \times I^2)A^{2/3}$$

and it is analogous to the surface tension of a liquid.

The binding energy formula needs another term to represent so called **symmetry effect**. For a given value of A there is a particular value of Z which corresponds to most stable nuclide. For light nuclides, where the coulomb effect is small, this value is $A/2$, as is seen from the fact that the number of protons and neutrons are equal in the most abundant light nuclides. In the absence of coulomb effect a departure from the condition $Z=A/2$ would tend to lead to instability and a smaller value of the binding energy. A term proportional to some power of the **neutron excess** ($A-2Z$) would represent the magnitude of this effect. The second power is chosen because the term then vanished for $Z=A/2$, as does its derivative with respect to Z . The later condition corresponds to the maximum value of the binding energy in the absence of coulomb energy. A detailed study of symmetry effect shows that it is inversely proportional to A , with the result that the **symmetry energy** can be written as,

$$E_{\text{sym}} = a_{\text{sym}} \times \frac{(A-2Z)^2}{A}$$

During the discussion of stable nuclides, that those with even number of protons and neutrons are the most abundant and the most stable. Nuclei with odd number of both protons and neutrons are least stable, while nuclei for which either the proton or the neutron number is even are intermediate in stability. It has been shown that this "**odd-even effect**" can be represented by a term E_{pair} whose value depend in following way on the numbers of protons Z and neutrons N in the nucleus.

$$E_{\text{pair}} = \begin{cases} a_p A^{-3/4} & \text{when } N \text{ and } Z \text{ are even} \\ -a_p A^{-3/4} & \text{when } N \text{ and } Z \text{ are odd} \\ 0 & \text{otherwise} \end{cases}$$

The formula for the binding energy is obtained by combining all of the terms into an equation,

$$V_{LD} = E_v - E_s - E_c + E_{\text{sym}} + E_{\text{pair}}$$

$$V_{LD} = a_v A - E_s^0 \left(1 + \frac{2}{5}\alpha^2 - \frac{4}{105}\alpha^3\right) - E_c^0 \left(1 - \frac{1}{5}\alpha^2 - \frac{4}{105}\alpha^3\right) + a_{\text{sym}} \times \frac{(A-2Z)^2}{A} + E_{\text{pair}}$$

3.1.2 The shell or, independent particle model

The nuclear shell model is one of the most important and useful models of nuclear structure. Its conception and name come from the results of empirical correlations of certain nuclear data. It has become apparent in recent years that many nuclear properties vary periodically in a sense similar to that of the periodic system of the elements. Most of these properties show marked discontinuities near certain even values of the proton or neutron number. The experimental facts indicate that especially stable nuclei result when either the number of protons Z or the number of neutrons $N = A - Z$ is equal to one of the numbers 2, 8, 20, 50, 82, 126. These numbers are commonly referred to as **magic numbers**. The magic numbers of neutrons and protons have been interpreted as forming closed shells of neutrons or protons in analogy with the filling of electron shells in atoms, and the neutron and proton shells appear to be independent of each other. It has been shown that the existence of the closed neutron and proton shells corresponding to the magic numbers can be described theoretically in terms of a particular nuclear model.

One of the most important pieces of evidence for the magic numbers comes from the study of the stable nuclides. The nuclei for which Z and N are 2 or 8, ${}^4_2\text{He}$ and ${}^{16}_8\text{O}$ respectively, are more

stable than their neighbors, as can be seen from the binding energies. Evidence for some of the other magic numbers has been obtained from very careful studies of the details of the binding energy curve in the region of intermediate and heavy nuclei. When the average binding energy per nucleon (obtained from mass spectrographic measurements and nuclear disintegration data) is plotted against the mass number, the curve is not really smooth but has several kinks in it. These kinks or breaks correspond to sudden increases in the value of the binding energy per nucleon. They have been found to occur at ${}_{82}^{208}\text{Pb}$ ($Z = 82$, $N = 126$), ${}_{58}^{140}\text{Ce}$ ($Z = 58$, $N = 82$), ${}_{50}^{120}\text{Sn}$ ($Z = 50$, $N = 70$), and ${}_{38}^{88}\text{Sr}$ ($Z = 38$, $N = 50$). In other studies, masses have been determined in the neighborhoods of $Z = 20$ and 28 , and $N = 20$ and 28 , and maxima have been found at these values for the binding energy per nucleon. As in other properties, the evidence for the special stability of nuclei with N or Z equal to 28 is less marked than that for the magic numbers. Along with the numbers 14 and 40 , 28 is sometimes called a **semimagic number**.

Stability is usually related not only to high binding energy, but also to high natural abundance; in all cases where the binding energies are known, a definite correlation is found between the binding energies and the abundances. The relative stability of different elements is also indicated by the number of stable isotopes per element. A graph of this number is shown in Fig. as a function of Z . The lightest element with more than two stable isotopes is oxygen ($Z = 8$); the first element with four stable isotopes is sulfur, whose heaviest isotope is ${}_{16}^{36}\text{S}$ with 20 neutrons. Five isotopes occur first in calcium ($Z = 20$); even more appear in molybdenum, of which the lightest is ${}_{42}^{92}\text{Mo}$ with 50 neutrons. Ten isotopes occur only once—in tin with 50 protons. Figure represents the number of stable nuclides having a given number of neutrons; such nuclei are sometimes called isotones. The peaks at 20 , 28 , 50 , and 82 neutrons are clear.

The energies of the α -particles emitted by heavy radioactive nuclei provide strong evidence for the magic numbers $Z = 82$ and $N = 126$. The fact that lead ($Z = 82$) is the stable end product of the three natural radioactive series points to the special nature of nuclei with $Z = 82$. The nuclides ${}_{85}^{213}\text{At}$ and ${}_{84}^{212}\text{Po}$, which have 128 neutrons, emit exceptionally energetic α -particles, 9.2 MeV and 8.78 MeV, respectively; ${}_{84}^{212}\text{Po}$ has the extremely short half-life of 3×10^{-7} sec, while that of ${}_{85}^{213}\text{At}$ has not yet been measured. In both of these cases, the daughter nucleus has 126 neutrons; there seems to be a strong tendency for nuclei with 128 neutrons to emit an α -particle and change into much more stable nuclei with 126 neutrons. When the parent nucleus has 126 neutrons, the α -energies should be particularly low, since the decay starts from a nucleus with low energy (high binding energy). These expectations are borne out by the experimental data; ${}_{84}^{210}\text{Po}$ and ${}_{85}^{211}\text{At}$, with 126 neutrons, have relatively long half-lives and low α -energies; ${}_{83}^{209}\text{Bi}$ ($N = 126$) is stable, whereas the neutron-richer isotopes ${}_{83}^{210}\text{Bi}$ and ${}_{83}^{211}\text{Bi}$ are α -radioactive.

Similar relations have been found in β -decay. The energies of the emitted β -particles are especially large when the number of neutrons or protons in the product nucleus corresponds to a magic number. The results of the investigations on α -decay and β -decay show that there are energy discontinuities of about 2 MeV in the neighborhood of the magic numbers 28 , 50 , 82 , and 126 . Since the average binding energy per particle in a nucleus is about 8 MeV, the effect in energy is about 25% .

The absorption cross sections for neutrons of about 1 MeV in energy are unusually low for nuclei containing 50 , 82 , or 126 neutrons. The cross sections for these nuclei are lower by a factor of about fifty than those of neighboring nuclei; the low cross section is an indication of large level spacing in the compound nucleus formed by the neutron capture. Similar evidence has been obtained from thermal neutron capture cross sections. The observed values are unusually small for nuclei with magic neutron numbers, and nuclei with 50 protons (tin) and 82 protons (lead) have smaller capture cross sections than their neighbors.

APPLICATIONS OF SHELL MODEL

The nuclear shell model does not only explain the specific properties of the magic nuclei, but in some cases can be used to calculate spins and magic moments of nuclei with unfilled nucleon shells. This model helps one to understand the law of abundance of nuclear isomers among all the nuclides because it allows one to determine the mass number region in which one should expect a great difference in the mechanical moment of the first excited ground states of a nucleus. At the same time, the nuclear shell model like any other nuclear model is not universal or useful in calculating the characteristics of nuclei whose form markedly deviates from the spherical one.

DISADVANTAGES OF SHELL MODEL

The following problems remain inexplicable in terms of shell model of a spherical nucleus:

1. In even-even nuclei, both protons and neutrons tend to pair up to form spin-zero ground states, which are particularly stable due to the pairing energy associated with spin pairing. Excited states in even-even nuclei often involve breaking these proton-neutron pairs and promoting nucleons to higher energy levels. This process requires overcoming the pairing energy, making these excited states less stable and more difficult to describe accurately.

2. The ground states of odd nuclei in the range $150 \leq A \leq 190$ and at $A \leq 220$ contain unpaired nucleons, which adds complexity to the nuclear structure, leading to deviations in properties from those predicted by the shell model. The shell model is based on the assumption of nucleon pairing, where nucleons are paired in spin and spatial states to achieve greater stability. In odd nuclei, the presence of unpaired nucleons disrupts this pairing, leading to deviations in the energy levels and properties of the nucleus. The presence of unpaired nucleons results in a non-zero total angular momentum (J), which affects the nuclear properties. The magnetic properties of odd nuclei are not fully captured by the shell model, leading to discrepancies between predicted and observed values. In odd nuclei with $A < 220$, variations in the neutron-proton ratio may lead to isospin effects, where the presence of unpaired nucleons affects the overall stability and properties of the nucleus.

3. The nuclear quadrupole moment characterizes the deviation of the nuclear charge distribution from spherical symmetry. It arises due to the distribution of protons and neutrons within the nucleus, which may not be perfectly spherically symmetric.

4. Probability of radiative transitions: Radiative transitions occur when a nucleus emits or absorbs electromagnetic radiation, such as gamma rays. These transitions involve changes in the energy states of nucleons within the nucleus, often accompanied by the emission or absorption of photons.

3.1.3 The collective nuclear model

The liquid drop model can account for the behavior of the nucleus as a whole, as in nuclear reactions and nuclear fission. In the latter, certain nuclei actually divide into two smaller nuclei and the division can be described in terms of the deformation of a drop; an explanation in terms of the motion of a single nucleon, or any number of nucleons, seems impossible. The liquid-drop model is said to describe the collective behavior of the nucleus, and the excitation of the nucleus is treated as surface oscillations, elastic vibrations, and other such **collective** modes of motion. On the other hand, many nuclear phenomena seem to show that nucleons behave as individual, and nearly independent, particles. Hence there are two entirely different ways of regarding nuclei, with a basic contradiction between them. The reasonable conclusion is that the two different models or pictures are incomplete parts of a larger or more general model. The problem of the electric quadrupole moments has led to such a model, which includes both the liquid-drop and independent-particle aspects; this new model is called the **collective model**. It

is assumed, in this model, that the particles within the nucleus exert a centrifugal pressure on the surface of the nucleus as a result of which the nucleus may be deformed into a permanently non-spherical shape; the surface may undergo oscillations (liquid-drop aspect). The particles within the nucleus then move in a non-spherical potential like that assumed in order to account for the quadrupole moments. Thus, the nuclear distortion reacts on the particles and modifies somewhat the independent-particle aspect. The nucleus is regarded as a shell structure capable of performing oscillations in shape and size. The result is that the collective model can be made to describe such drop-like properties as nuclear fission while at the same time preserving the shell model characteristics and, in fact, improving on the earlier shell model.

The simplest type of collective motion which has been identified experimentally is connected with rotations of deformed nuclei. If the rotational collective motion is sufficiently slow, it will not affect the internal structure of the nucleus. According to classical physics, the rotational energy is proportional to the square of the angular velocity ω , so that

$$E_{rot} = \frac{1}{2}I\omega^2$$

The parameter I denotes an effective moment of inertia of the nucleus which can be calculated if a specific model is assumed for the internal structure. Rotational energy levels are obtained when the angular momentum is quantised and the result is, for even-even nuclei,

$$E_{rot} = \frac{\hbar^2}{2I}J(J+1)$$

where J is the total angular momentum quantum number of the nucleus. In the case of a spheroidal nucleus, the deformation is symmetric with respect to reflection in the nuclear center. As a result, J is restricted to even values, 0, 2, 4, 6, . . . , and the parity should be even. According to the theory, the deformation (deviation from spherical shape) should be greatest and the rotational levels most easily observed for nuclei with numbers of nucleons far from closed shell.

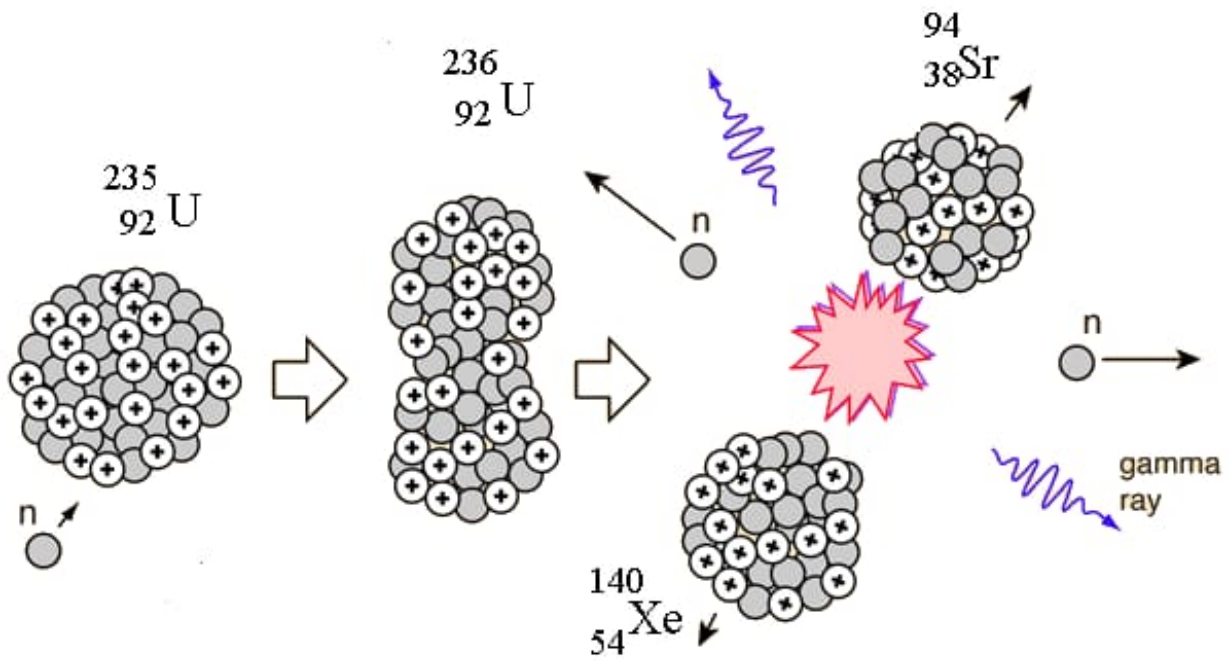


Figure 3.3: Scheme of fission of Uranium into Xenon and Strontium

Chapter 4

TABULATIONS

4.1 CRITICAL ENERGY AS A FUNCTION Z^2/A

$$y = \left(\frac{98}{135}\right) \cdot (1 - x^3 + 3x^2 - 3x) + \left(\frac{11368}{34425}\right) \cdot (x^4 - 4x^3 + 6x^2 - 4x + 1)$$

x	y
0	1.056
0.05	0.891
0.1	0.746
0.15	0.618
0.2	0.507
0.25	0.411
0.3	0.328
0.35	0.258
0.4	0.199
0.45	0.151
0.5	0.111
0.55	0.079
0.6	0.055
0.65	0.036
0.7	0.022
0.75	0.013
0.8	0.006
0.85	0.0026
0.9	0.00075
0.95	0.000092
1.0	0
1.05	-0.00000868
1.1	-0.000693
1.15	-0.0023
1.2	-0.00528
1.25	-0.010054

Table 4.1:

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4.1.1 CRITICAL ENERGY AS A FUNCTION OF Z^2/A

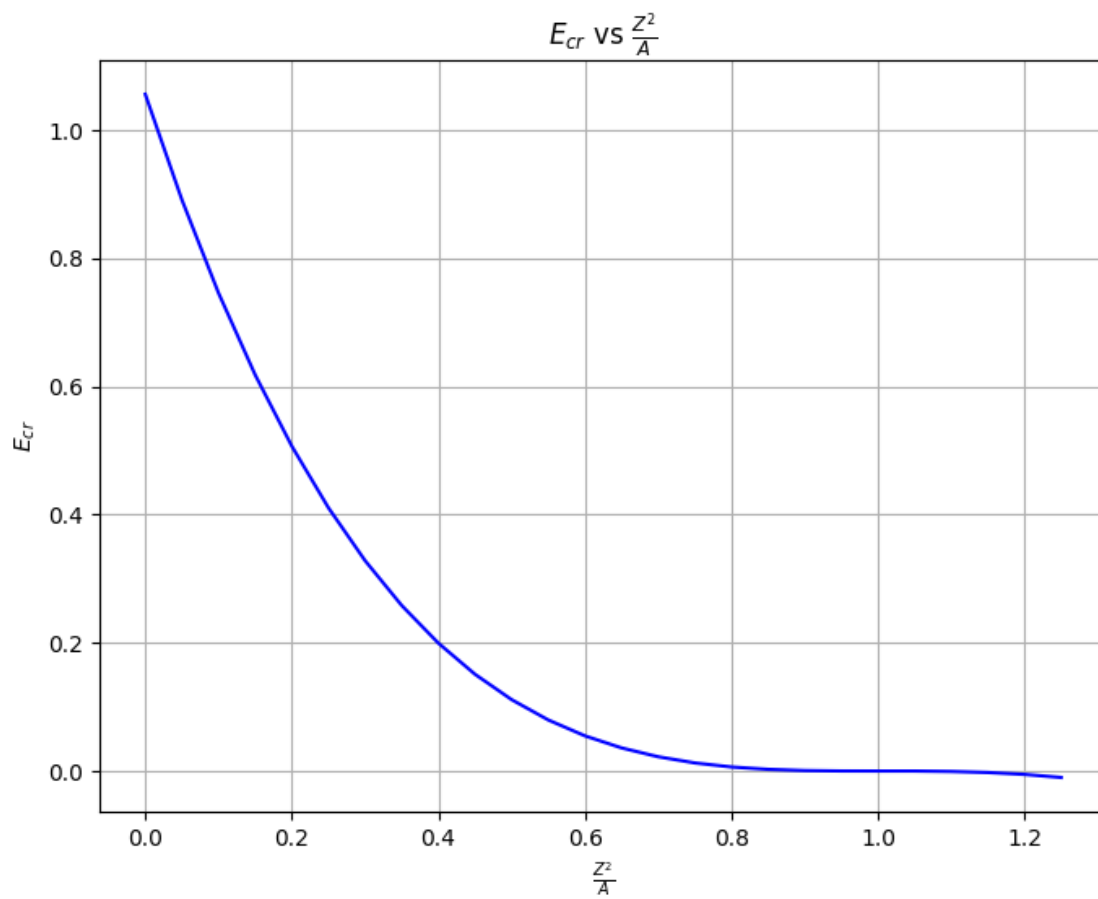
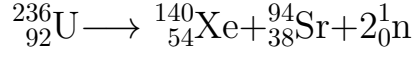


Figure 4.1:

4.2 LIQUID DROP ENERGY OF INDIVIDUAL FRAGMENTS ($^{140}_{54}\text{Xe}$) AS A FUNCTION OF DEFORMATION PARAMETER



$$E_{\text{L.D}} = E_v - E_s - E_c + E_{\text{sym}} + E_{\text{pair}}$$

$$V_{\text{L.D}} = a_v A - E_s^0 \left(1 + \frac{2}{5}\alpha^2 - \frac{4}{105}\alpha^3\right) - E_c^0 \left(1 - \frac{1}{5}\alpha^2 - \frac{4}{105}\alpha^3\right) + a_{\text{sym}} \times \frac{(A-2Z)^2}{A} + E_{\text{pair}}$$

$$\alpha = \sqrt{\frac{5}{4\pi}} \times \beta$$

β_1	$E_{\text{L.D}}$
-0.9	1117.815
-0.8	1118.467
-0.7	1131.447
-0.6	1137.38
-0.5	1142.096
-0.4	1145.635
-0.3	1148.23
-0.2	1150.84
-0.1	1151.97
0	1152.37
0.1	1151.84
0.2	1150.75
0.3	1148.88
0.4	1146.35
0.5	1142.24
0.6	1139.68
0.7	1136.52
0.8	1134.13
0.9	1132.23

Table 4.2:

4.2.1 LIQUID DROP ENERGY OF INDIVIDUAL FRAGMENTS ($^{140}_{54}\text{Xe}$) AS A FUNCTION OF DEFORMATION PARAMETER

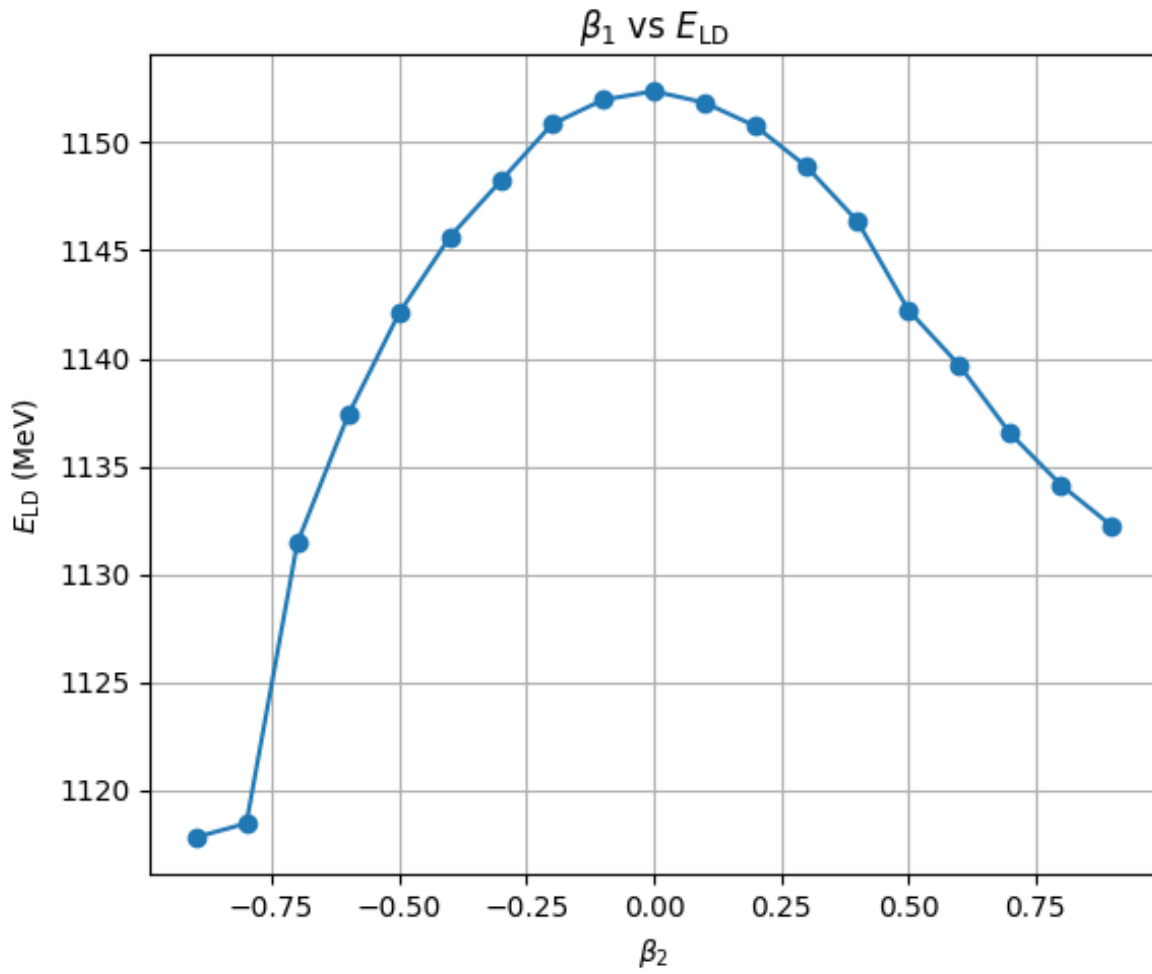
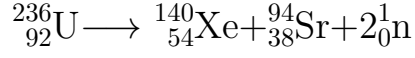


Figure 4.2:

4.3 LIQUID DROP ENERGY OF INDIVIDUAL FRAGMENTS($^{94}_{38}\text{Sr}$) AS A FUNCTION OF DEFORMATION PARAMETER.



$$E_{LD} = E_v - E_s - E_c + E_{\text{sym}} + E_{\text{pair}}$$

$$V_{LD} = a_v A - E_s^0 \left(1 + \frac{2}{5}\alpha^2 - \frac{4}{105}\alpha^3\right) - E_c^0 \left(1 - \frac{1}{5}\alpha^2 - \frac{4}{105}\alpha^3\right) + a_{\text{sym}} \times \frac{(A-2Z)^2}{A} + E_{\text{pair}}$$

$$\alpha = \sqrt{\frac{5}{4\pi}} \times \beta$$

β_2	$E_{(L.D)}$
-0.9	780.52
-0.8	785.77
-0.7	792.346
-0.6	798.806
-0.5	802.925
-0.4	806.346
-0.3	809.406
-0.2	811.256
-0.1	812.226
0	812.596
0.1	812.236
0.2	811.196
0.3	809.66
0.4	806.665
0.5	803.01
0.6	798.866
0.7	792.93
0.8	785.63
0.9	779.132

Table 4.3:

4.3.1 LIQUID DROP ENERGY OF INDIVIDUAL FRAGMENTS($^{94}_{38}\text{Sr}$) AS A FUNCTION OF DEFORMATION PARAMETER

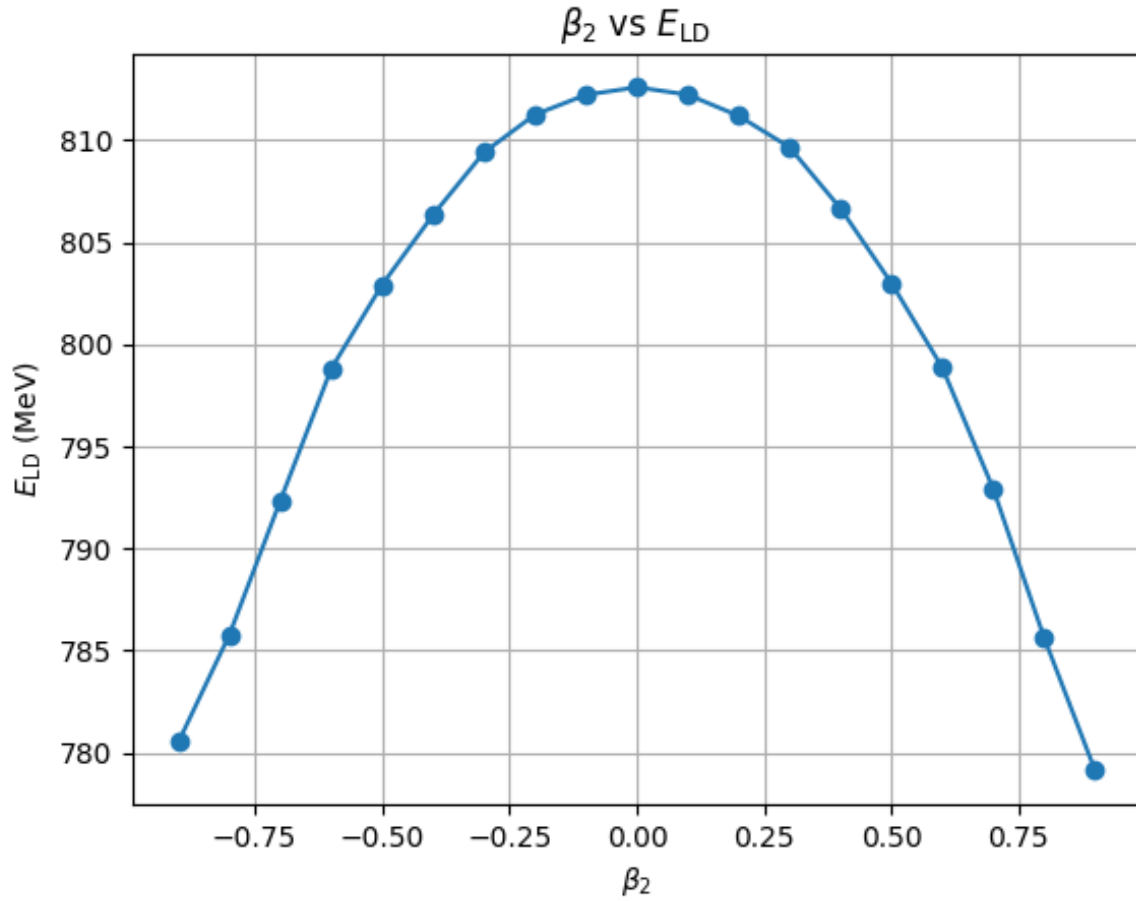
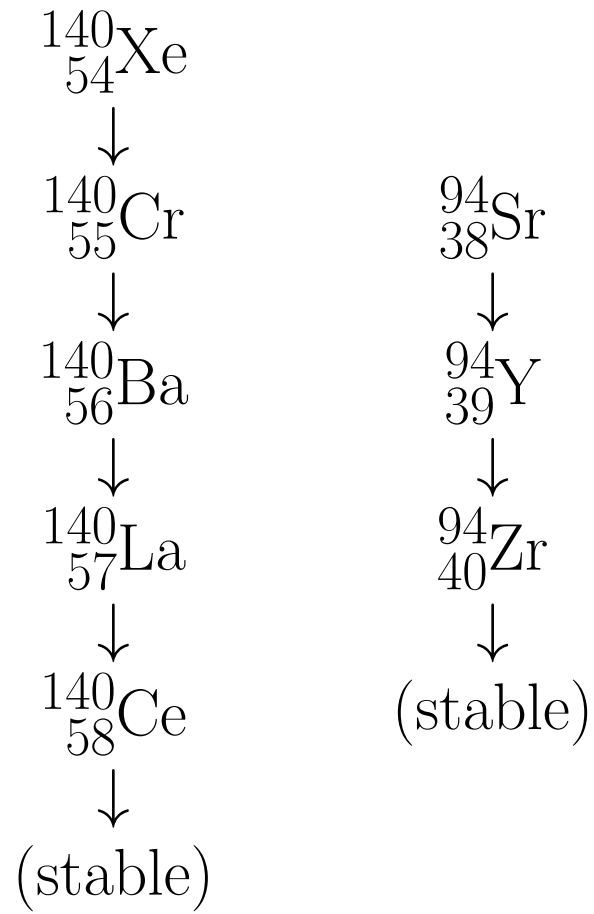


Figure 4.3:

4.4 DECAY CHAIN OF $^{140}_{54}\text{Xe}$ and $^{94}_{38}\text{Sr}$



4.5 Z^2/A for $^{140}_{54}\text{Xe}$ DECAY CHAIN

N	Z^2/A
54	20.82
55	21.607
56	22.4
57	23.20
58	24.02

Table 4.4:

4.6 Z^2/A for $^{94}_{38}\text{Sr}$ DECAY CHAIN

N	Z^2/A
38	15.36
39	16.18
40	17.02

Table 4.5:

4.6.1 Z^2/A for $^{140}_{54}\text{Xe}$ DECAY CHAIN

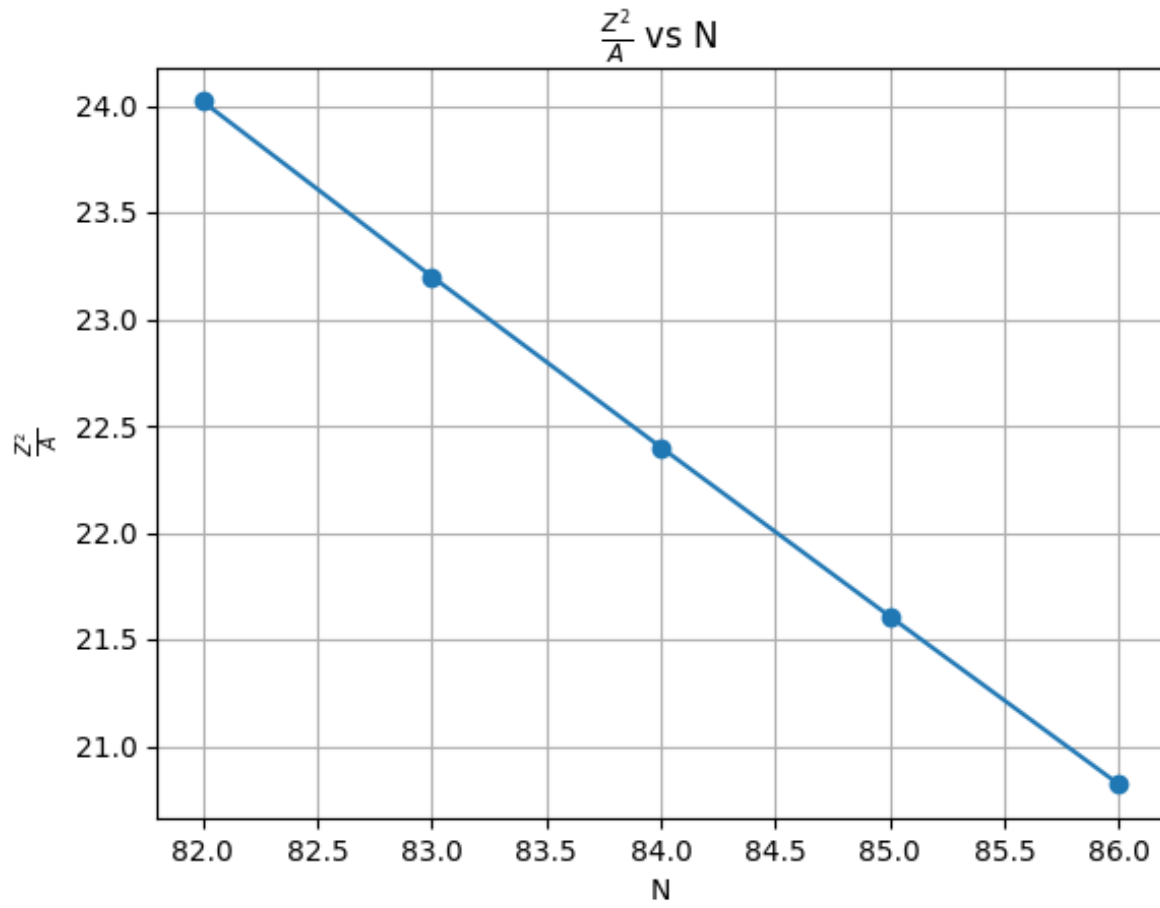


Figure 4.4:

4.6.2 Z^2/A for $^{94}_{38}\text{Sr}$ DECAY CHAIN

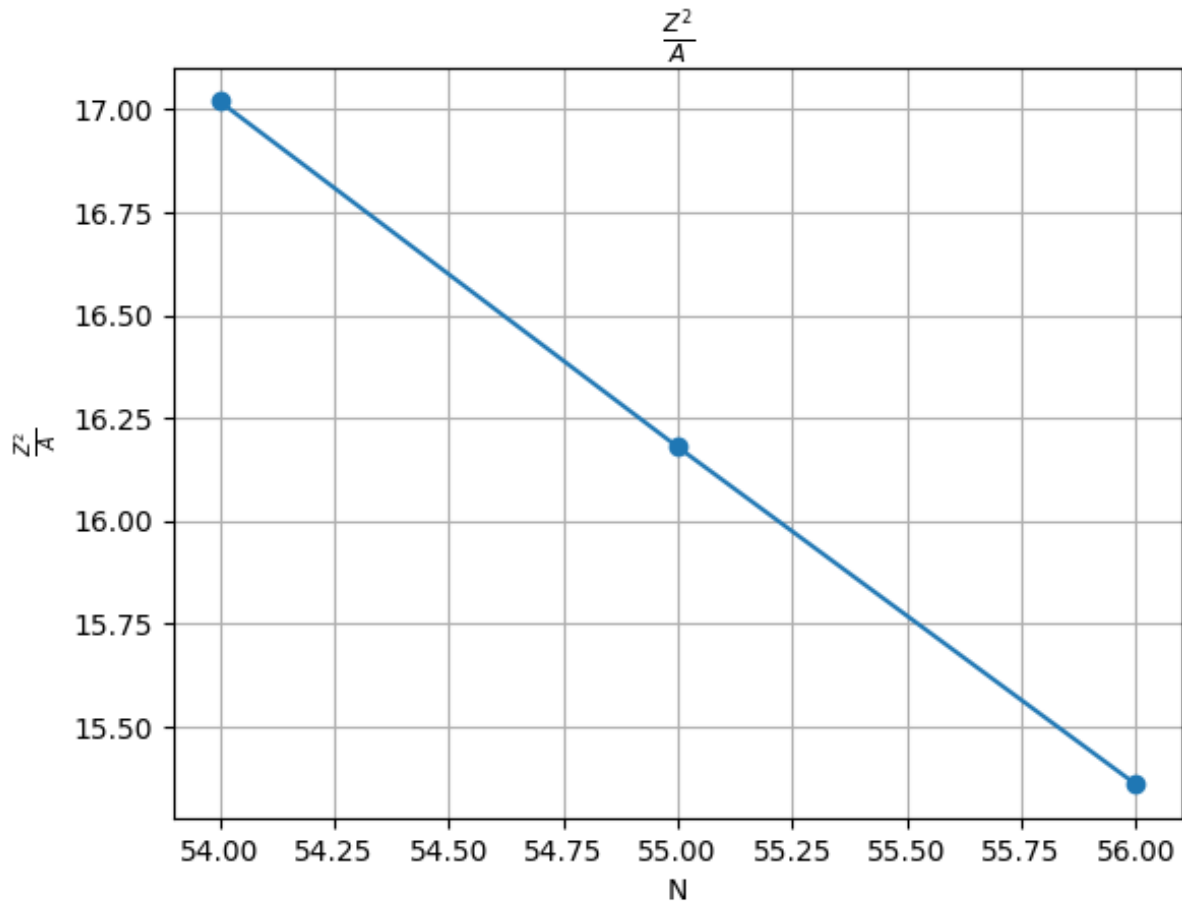


Figure 4.5:

4.7 N-Z for $^{140}_{54}\text{Xe}$ DECAY CHAIN

N	Z
86	54
85	55
84	56
83	57
82	58

Table 4.6:

4.8 N-Z for $^{94}_{38}\text{Sr}$ DECAY CHAIN

N	Z^2/A
56	38
55	39
54	40

Table 4.7:

4.8.1 N-Z for $^{140}_{54}\text{Xe}$ DECAY CHAIN

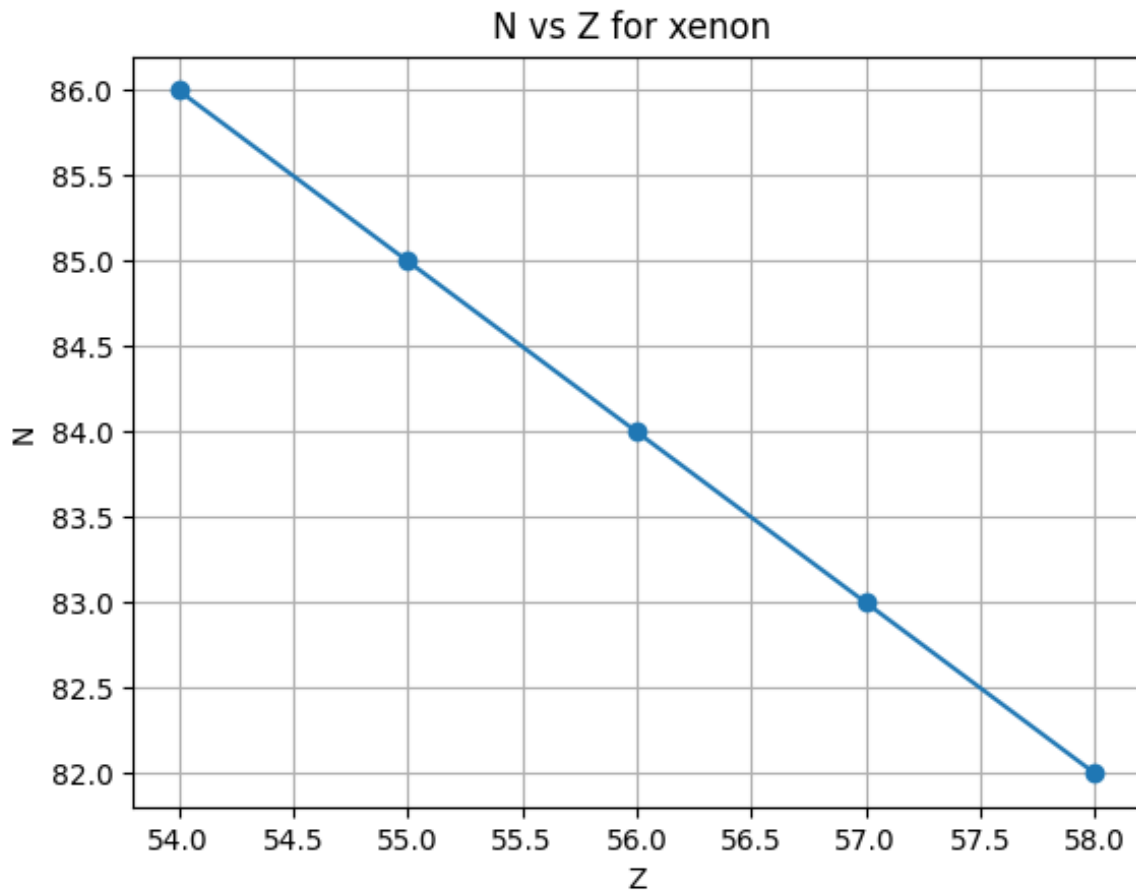


Figure 4.6:

4.8.2 N-Z for $^{94}_{38}\text{Sr}$ DECAY CHAIN

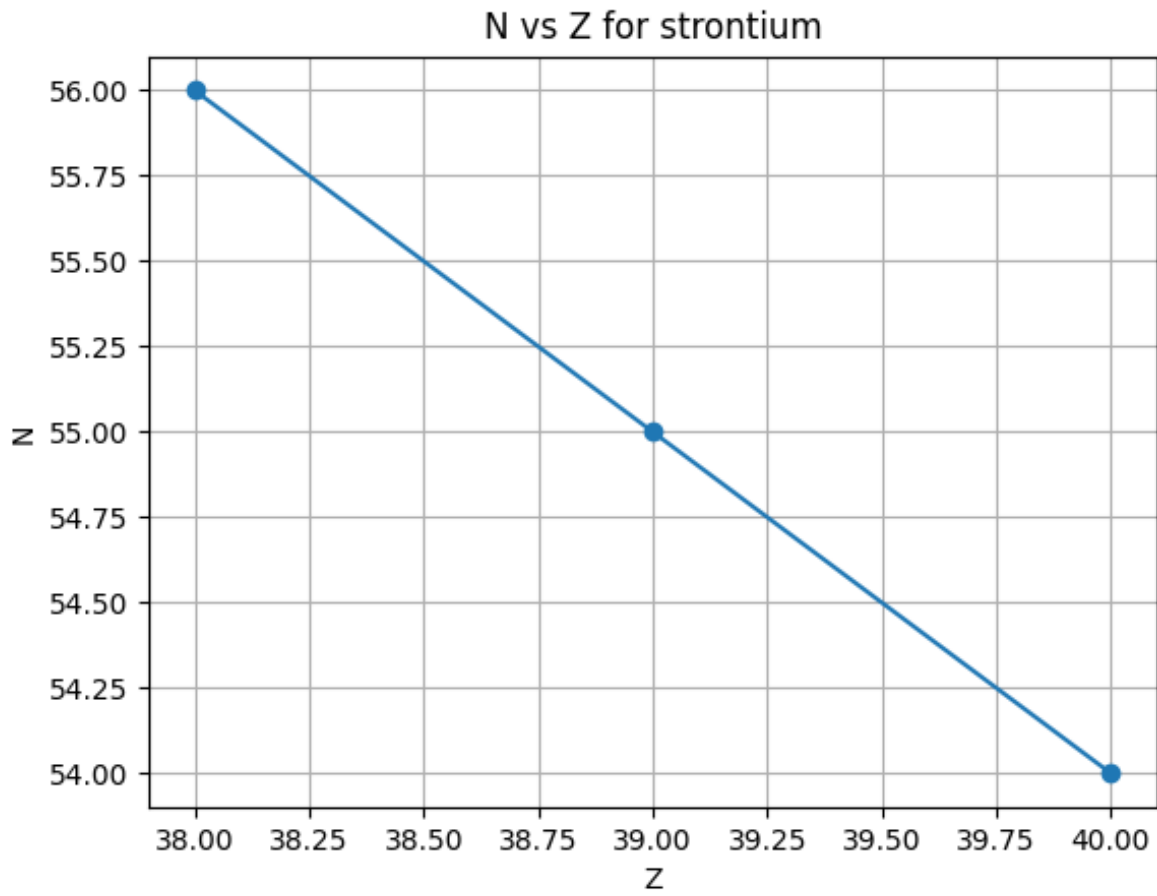
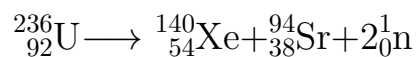


Figure 4.7:

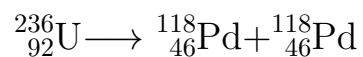
4.8.3 ENERGY RELEASED IN FISSION



Nucleus/Nuclide	No. of nucleons	B.E/A (Mev)	Total B.E (Mev)	Combined B.E (MeV)	Energy released (MeV)
$^{236}_{92}\text{U}$	236	7.66	1810.032	1810.032	175.83
$^{140}_{54}\text{Xe}$	140	8.34	1168.52	1985.86	
$^{94}_{38}\text{Sr}$	94	8.6	817.34		

Table 4.8:

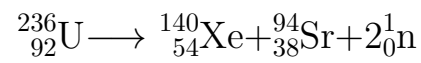
ENERGY RELEASED IN A SYMMETRIC FISSION



Nucleus/Nuclide	No. of nucleons	B.E/A (Mev)	Total B.E (Mev)	Combined B.E (MeV)	Energy released (MeV)
$^{236}_{92}\text{U}$	236	7.66	1810.032	1810.032	194.936
$^{118}_{46}\text{Pd}$	118	8.49	1002.484	2004.968	
$^{118}_{46}\text{Pd}$	118	8.49	1002.484		

Table 4.9:

4.8.4 $\frac{Z}{A}$ AND $\frac{Z^2}{A}$ FOR FISSILE NUCLEUS AND FISSION PRODUCTS



Nucleus/Nuclides	Z/A	Z^2/A
${}_{92}^{236}\text{U}$	0.3898	35.864
${}_{54}^{140}\text{Xe}$	0.3857	20.82
${}_{38}^{94}\text{Sr}$	0.404	15.36

Table 4.10:

4.9 CHANGE IN RADIUS OF NUCLEI AS A FUNCTION OF DEFORMATION PARAMETER

$$R_1(\theta) = R_{01} [1 + \beta_i Y_{20}(\theta)]$$

$$R_{01} = 1.2536A_1^{1/3} - 0.80012A_1^{-1/3} - 0.0021444A_1^{-1} = 6.336$$

$$R_2(\theta) = R_{02} [1 + \beta_i Y_{20}(\theta)]$$

$$R_{02} = 1.2536A_2^{1/3} - 0.80012A_2^{-1/3} - 0.0021444A_2^{-1} = 5.514$$

$$Y_{20}(\theta = 0) = \frac{1}{4}\sqrt{5}/\pi(3\cos\theta^2 - 1) = 0.63$$

$$Y_{20}(\theta = 90) = \frac{1}{4}\sqrt{5}/\pi(3\cos\theta^2 - 1) = -0.31$$

β	$R_1(\theta = 0^\circ)$	$R_1(\theta = \frac{\pi}{2}^\circ)$	$R_2(\theta = 0^\circ)$	$R_2(\theta = \frac{\pi}{2}^\circ)$
-0.9	2.740	8.08	2.38	7.04
-0.8	3.139	7.89	2.73	6.87
-0.7	3.53	7.70	3.080	6.70
-0.6	3.93	7.507	3.42	6.53
-0.5	4.33	7.31	3.77	6.36
-0.4	4.73	7.11	4.12	6.19
-0.3	5.133	6.91	4.46	6.02
-0.2	5.53	6.72	4.81	5.85
-0.1	5.93	6.52	5.16	5.68
0	6.33	6.33	5.51	5.51
0.1	6.72	6.13	5.85	5.33
0.2	7.12	5.93	6.20	5.16
0.3	7.52	5.74	6.55	4.99
0.4	7.92	5.54	6.89	4.82
0.5	8.32	5.34	7.24	4.65
0.6	8.72	5.15	7.59	4.48
0.7	9.121	4.95	7.93	4.31
0.8	9.52	4.76	8.28	4.14
0.9	9.91	4.56	8.63	3.97

Table 4.11:

4.9.1 CHANGE IN RADIUS OF NUCLEI AS A FUNCTION OF DEFORMATION PARAMETER

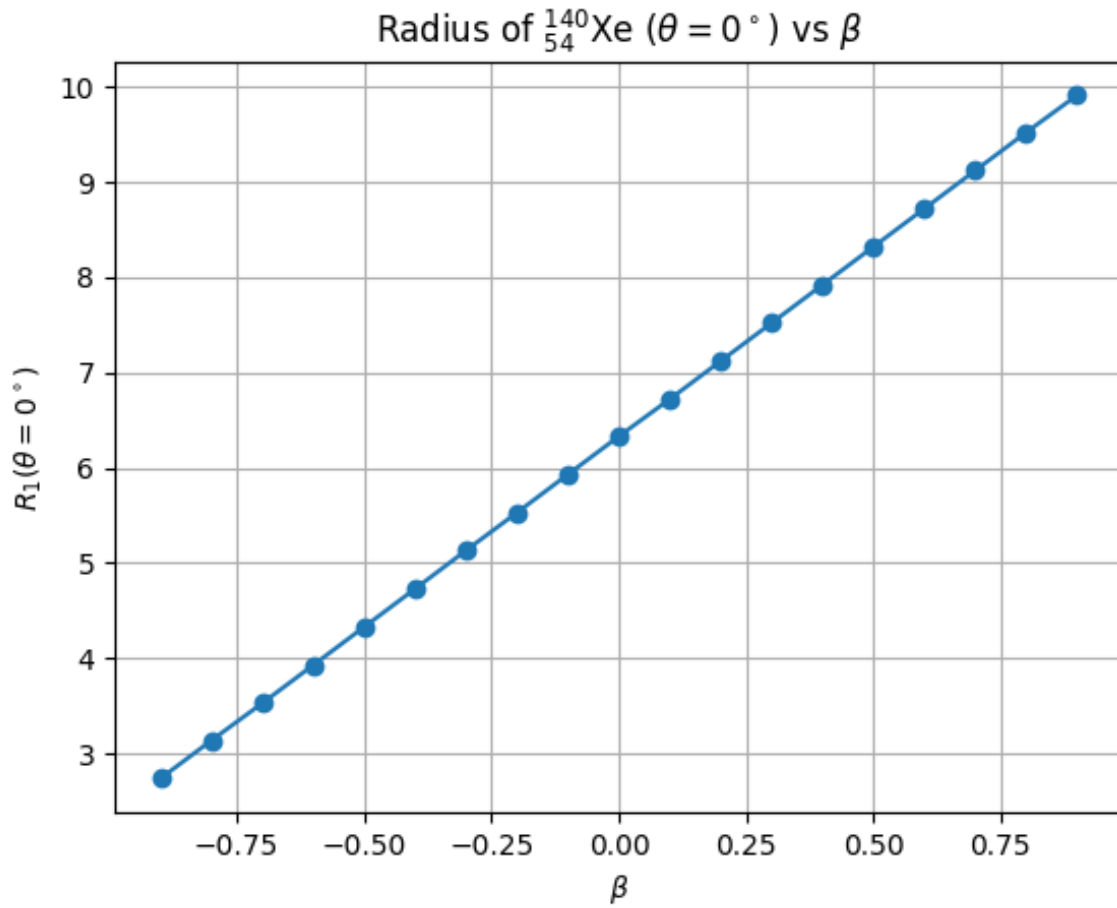
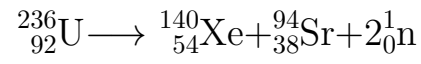


Figure 4.8:

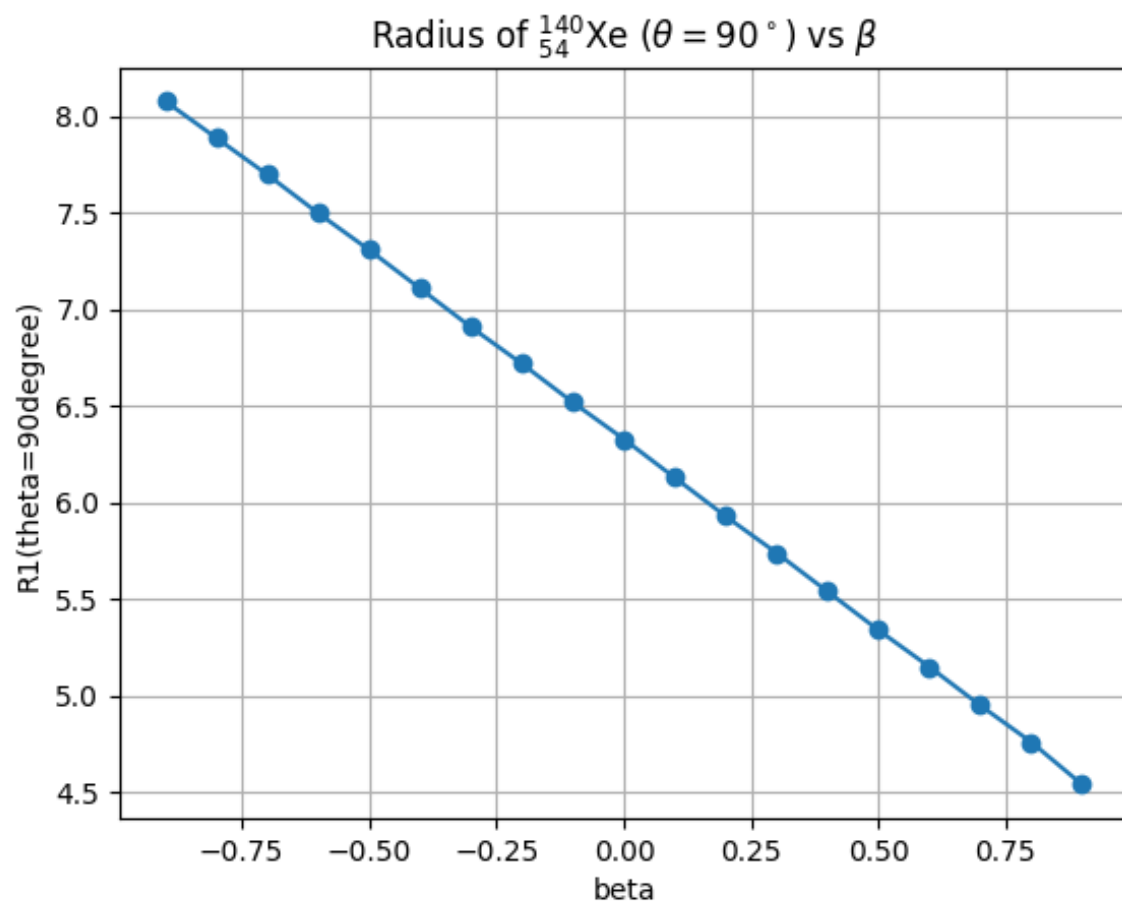


Figure 4.9:

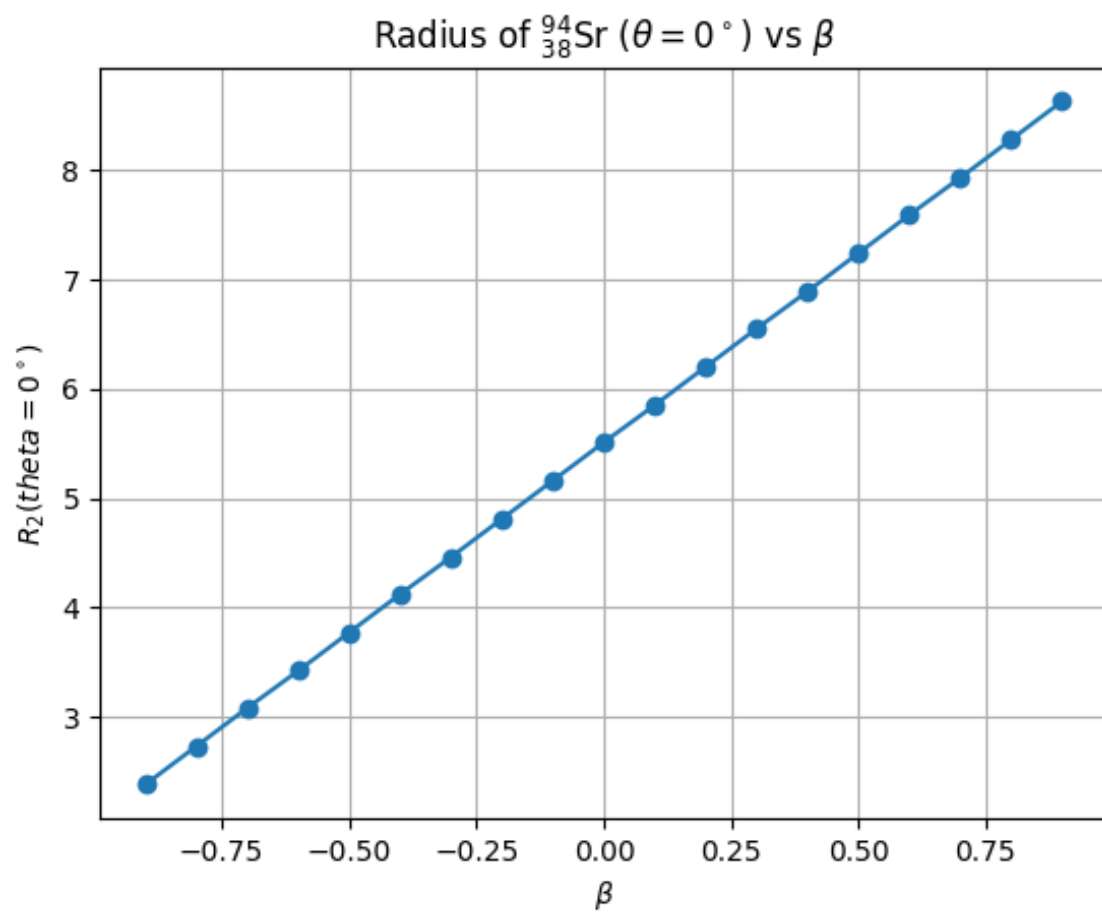


Figure 4.10:

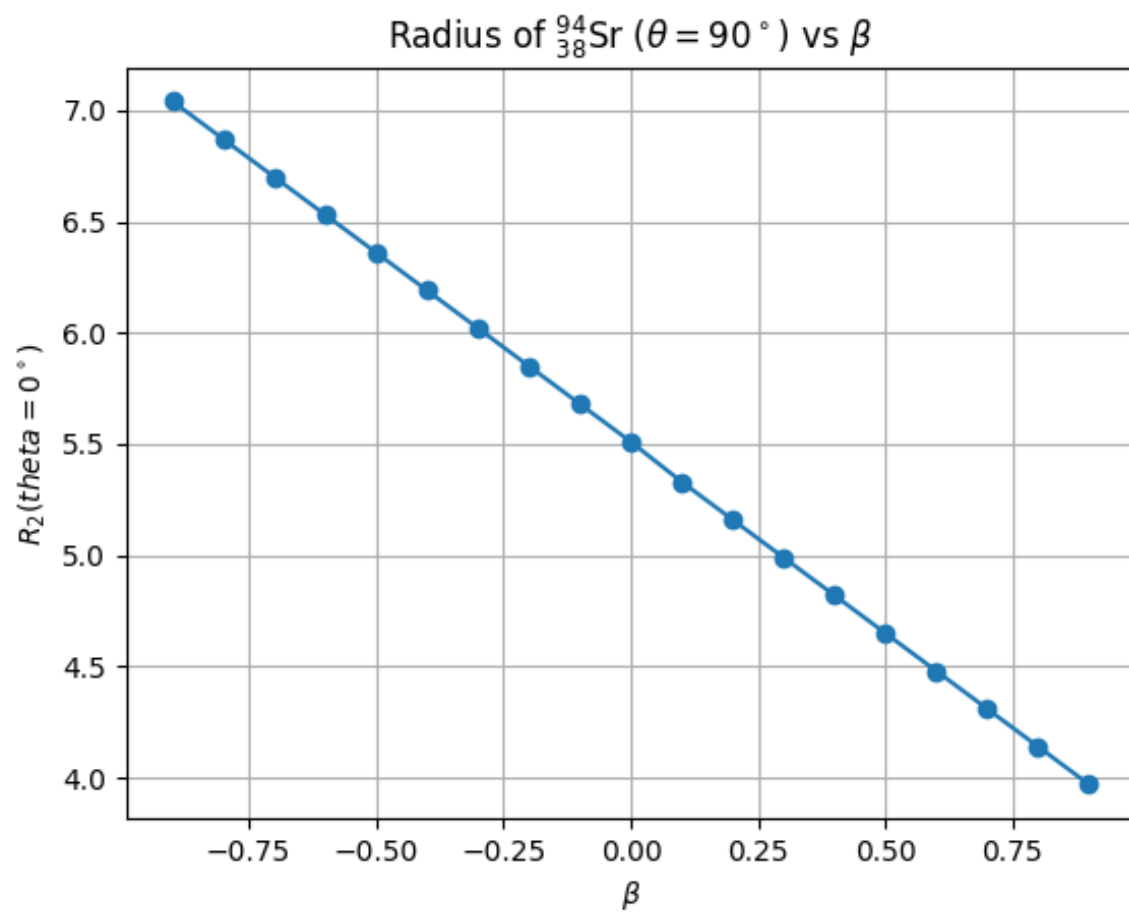


Figure 4.11:

4.10 AVERAGE DISTANCE AS A FUNCTION OF DEFORMATION PARAMETER

$$S_1 = \frac{R_1^2(\pi/2)R_2^2(\pi/2)}{R_1^2(\pi/2)R_1(0)+R_2^2(\pi/2)R_1(0)}$$

$$S_{01} = \frac{R_{01}R_{02}}{R_{01}+R_{02}}$$

β_i	S_1	S_{01}/S_1
-0.9	10.28	0.28
-0.8	8.99	0.327
-0.7	7.23	0.406
-0.6	6.1838	0.4757
-0.5	5.31	0.553
-0.4	4.608	0.638
-0.3	4.016	0.7331
-0.2	3.520	0.835
-0.1	3.093	0.950
0	2.728	1.077
0.1	2.407	1.221
0.2	2.12	1.386
0.3	1.77	1.661
0.4	1.66	1.771
0.5	1.478	1.989
0.6	1.31	2.244
0.7	1.15	2.556
0.8	1.02	2.882
0.9	0.904	3.25221

Table 4.12:

4.10.1 AVERAGE DISTANCE AS A FUNCTION OF DEFORMATION PARAMETER

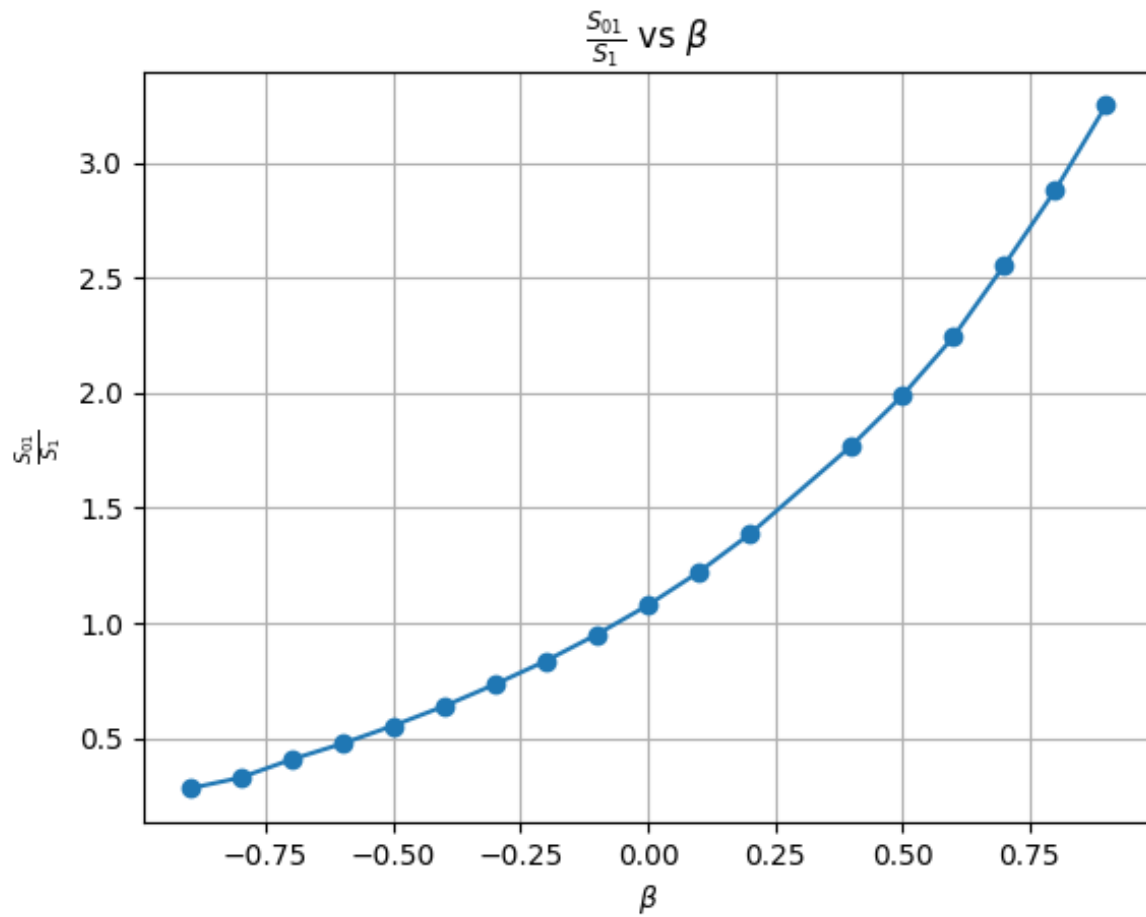


Figure 4.12:

4.11 PROJECTILE ENERGY AND EXCITATION ENERGY OF NEUTRON

$$U_{FN} = \left(1 - \frac{m_n}{m_{C_N}}\right) E_n + B_n(A_{C_N}, Z_{C_N})$$

$$B_n = 7.6 \times 236 = 1790.415 \text{ MeV}$$

$$m_n = 1.008665 \text{ amu}$$

$$m_{C_N} = 236.045566201 \text{ amu}$$

$$\overline{N} = N_{C_N} - \gamma$$

$E_n(\text{Mev})$	$U_{FN}(n, f)$
0.025	1790.4398
0.5	1790.9128
1.0	1791.4107
1.5	1791.9085
2.0	1792.4064
2.5	1792.9043
3.0	1793.4021
3.5	1793.9000
4.0	1794.3979
4.5	1794.8957
5.0	1795.3936
5.5	1795.8915
6	1796.3893

Table 4.13:

4.11.1 PROJECTILE ENERGY AND EXCITATION ENERGY OF NEUTRON

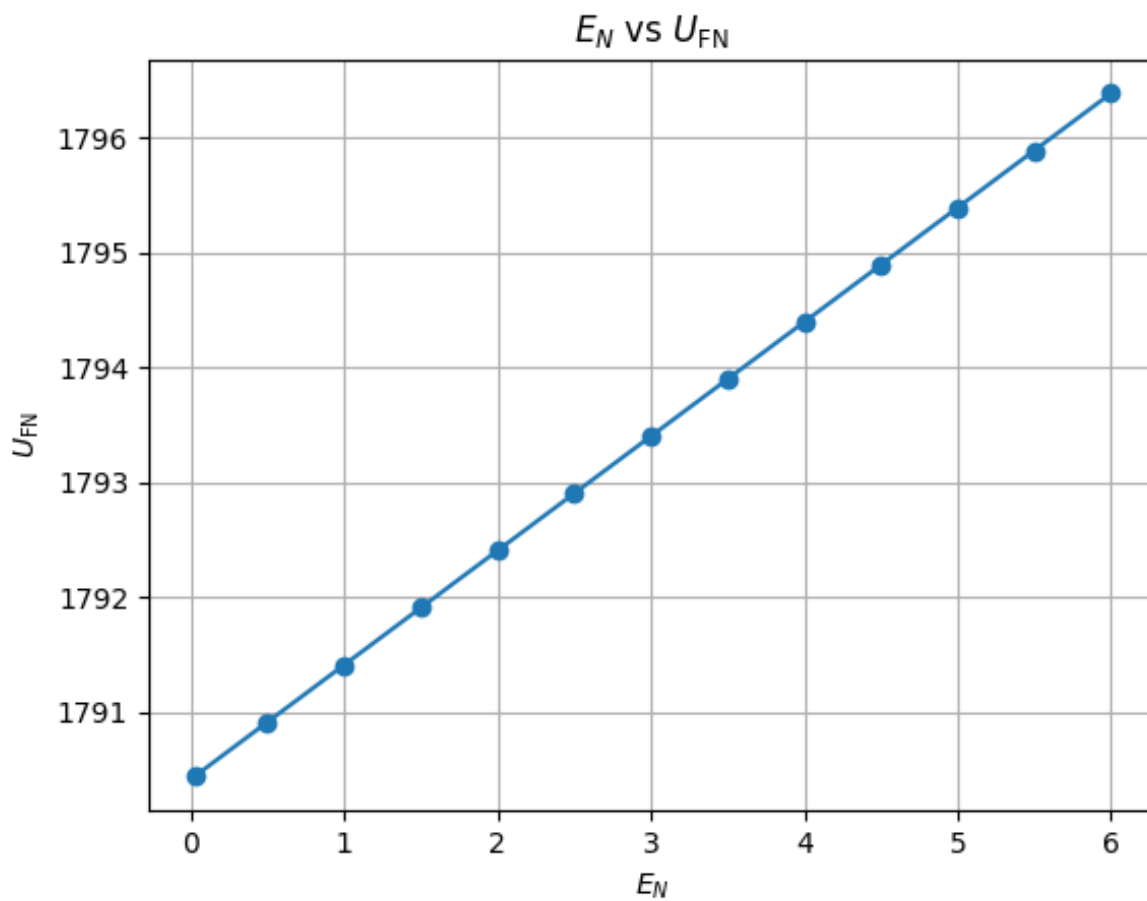


Figure 4.13:

4.12 NUCLEAR POTENTIAL AND YIELD

$$\text{Nuclear Potential, } V(N, A) = V_{\text{mac}}(N) + V_{\text{sn}} \exp(-\gamma \epsilon(N))$$

$$V_{\text{mac}} = C_{\text{mac}} \left(N - \frac{\bar{N}_{C_N}}{2} \right)^2 + V_0$$

$$V_{\text{sh}}(N) = C_{\text{in}} \left[\exp \left(-\frac{(N - N_{\text{in}})^2}{\sigma_{\text{in}}^2} \right) + \exp \left(-\frac{(N - \bar{N}_{\text{in}})^2}{\sigma_{\text{in}}^2} \right) \right] + C_{\text{out}} \left[\exp \left(-\frac{(N - N_{\text{out}})^2}{\sigma_{\text{out}}^2} \right) + \exp \left(-\frac{(N - \bar{N}_{\text{out}})^2}{\sigma_{\text{out}}^2} \right) \right]$$

$$C_{\text{mac}} = \frac{8}{N_{C_N}^2} \times 10^{7.1693 - 0.26602 \left(\frac{Z_{C_N}^2}{A_{C_N}} \right)} + 0.00283 \left(\frac{Z_{C_N}}{A_{C_N}} \right)^2$$

$$\epsilon(N) = E^* - [V_{\text{mac}}(N) + V_{\text{sh}}(N)]$$

$$\bar{N}_{C_N} = N_{C_N} - \nu = 141.5, \nu = 2.5$$

$$V_0 = 5.0$$

$$C_{\text{in}} = -9.31$$

$$C_{\text{out}} = -14.2 \text{ MeV}$$

$$\sigma_{\text{in}} = -0.662 A_{\text{CN}} + 164 = 7.768$$

$$\sigma_{\text{out}} = 5.13$$

$$N_{\text{in}} = 82$$

$$N_{\text{out}} = 90.5$$

$$E^* = 6.517 \text{ MeV}$$

$$\text{Yield, } Y(N, A, E) = \exp \left(2 \sqrt{\bar{a}} (E - [V_{\text{mac}}(N) + V_{\text{sn}}(N) \exp(-\gamma \epsilon(N))]) \right)$$

$$\bar{a} = -11.6 A_{\text{CN}} + 2799 = 61.4$$

A	N	V_{mac}	$V_{Sh}(N)$	ϵ_N	V(N,A)	Y(N,A)
80	48.813	8.431	-13.238	11.204	1.63	$7.682 \cdot 10^{14}$
82	50.034	8.008	-15.811	14.16	1.25	$2.857 \cdot 10^{15}$
84	51.254	7.716	-17.163	15.864	1.10	$6.012 \cdot 10^{15}$
86	52.474	7.365	-17.174	16.206	0.83	$1.123 \cdot 10^{16}$
88	53.695	7.066	-16.101	15.434	0.64	$2.183 \cdot 10^{16}$
90	54.359	6.768	-14.486	14.114	0.52	$3.269 \cdot 10^{16}$
92	56.135	6.531	-12.919	12.814	0.474	$3.810 \cdot 10^{16}$
94	57.356	6.267	-11.673	11.813	0.470	$3.859 \cdot 10^{16}$
96	58.576	6.053	-10.775	11.139	0.468	$3.885 \cdot 10^{16}$
98	59.799	5.860	-10.037	10.604	0.474	$3.810 \cdot 10^{16}$
100	61.017	5.669	-9.270	10.100	0.52	$3.269 \cdot 10^{16}$
102	62.237	5.517	-8.342	9.25	0.65	$2.109 \cdot 10^{16}$
104	63.457	5.371	-7.247	8.293	0.89	$1.00 \cdot 10^{16}$
106	64.678	5.261	-6.034	7.197	1.26	$2.755 \cdot 10^{15}$
108	65.898	5.159	-4.843	6.00	1.72	$5.528 \cdot 10^{14}$
110	67.118	5.065	-3.785	5.118	2.22	$8.670 \cdot 10^{13}$
112	68.339	5.032	-2.959	4.344	2.66	$1.456 \cdot 10^{13}$
114	69.563	5.020	-2.438	3.855	3.069	$4.135 \cdot 10^{12}$
116	70.779	5.5.012	-2.461	3.97	3.07	$4.227 \cdot 10^{12}$
118	72.004	5.00	-2.85	3.801	3.16	$2.70 \cdot 10^{12}$
120	73.220	5.031	-3.007	4.381	2.645	$1.587 \cdot 10^{13}$
122	74.441	5.068	-3.825	5.154	2.19	$9.739 \cdot 10^{13}$
124	75.661	5.178	-4.806	6.15	1.70	$6.041 \cdot 10^{14}$
128	78.102	5.381	-7.214	8.343	0.88	$1.023 \cdot 10^{16}$
130	79.322	5.516	-8.304	9.295	0.64	$2.183 \cdot 10^{16}$
132	80.542	5.617	-9.216	10.105	0.52	$3.269 \cdot 10^{16}$
134	81.767	5.850	-10.074	10.631	0.473	$3.822 \cdot 10^{16}$
136	82.983	6.053	-10.709	11.243	0.470	$4.076 \cdot 10^{16}$
138	84.208	6.279	-11.636	11.854	0.470	$4.081 \cdot 10^{16}$
140	85.423	6.543	-12.191	12.864	0.48	$3.732 \cdot 10^{16}$
142	86.644	6.712	-14.4726	14.178	0.52	$3.269 \cdot 10^{16}$
144	87.864	7.000	-16.120	15.477	0.65	$2.109 \cdot 10^{16}$
146	89.085	7.311	-17.109	16.215	0.84	$1.152 \cdot 10^{16}$
148	90.305	7.732	-17.102	15.807	1.10	$6.00 \cdot 10^{15}$
150	91.525	8.085	-15.623	14.135	1.27	$2.659 \cdot 10^{15}$

Table 4.14:

4.12.1 NUCLEAR POTENTIAL AS A FUNCTION OF NEUTRON NUMBER

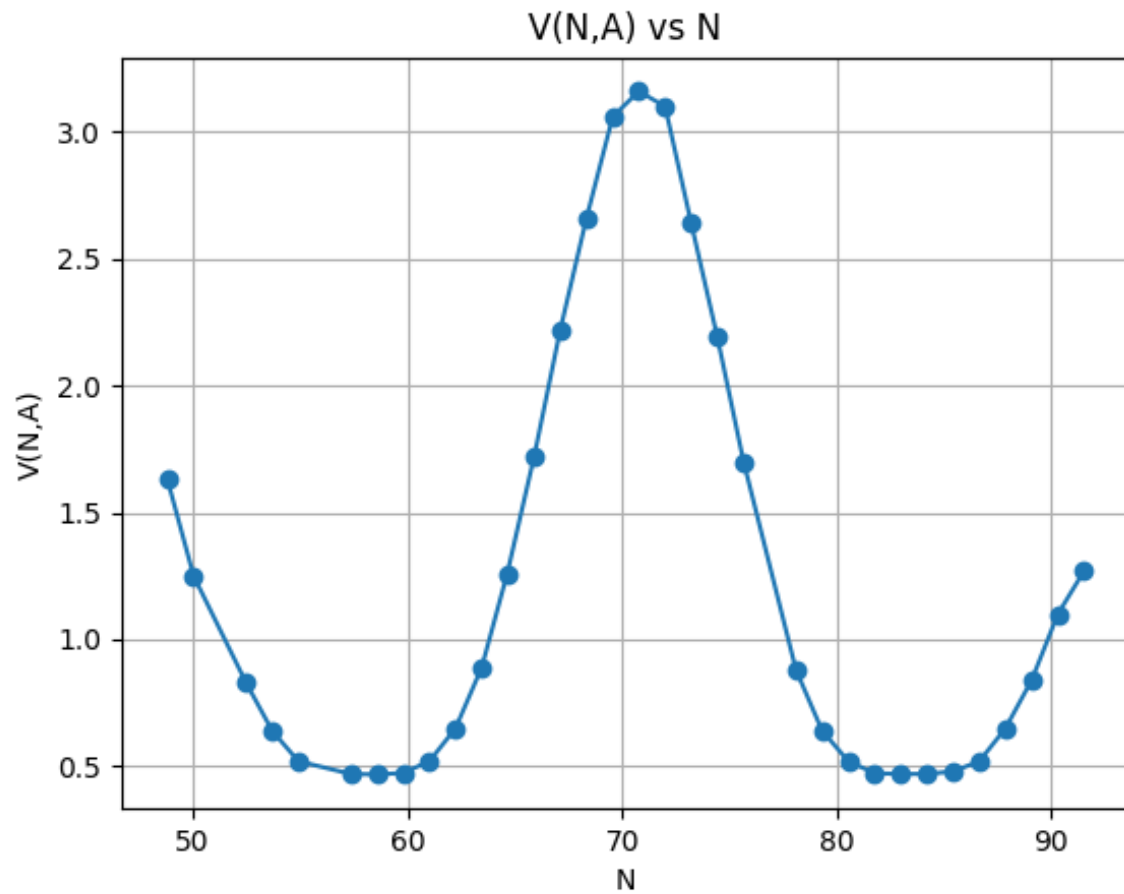


Figure 4.14:

4.12.2 YIELD AS A FUNCTION OF MASS NUMBER

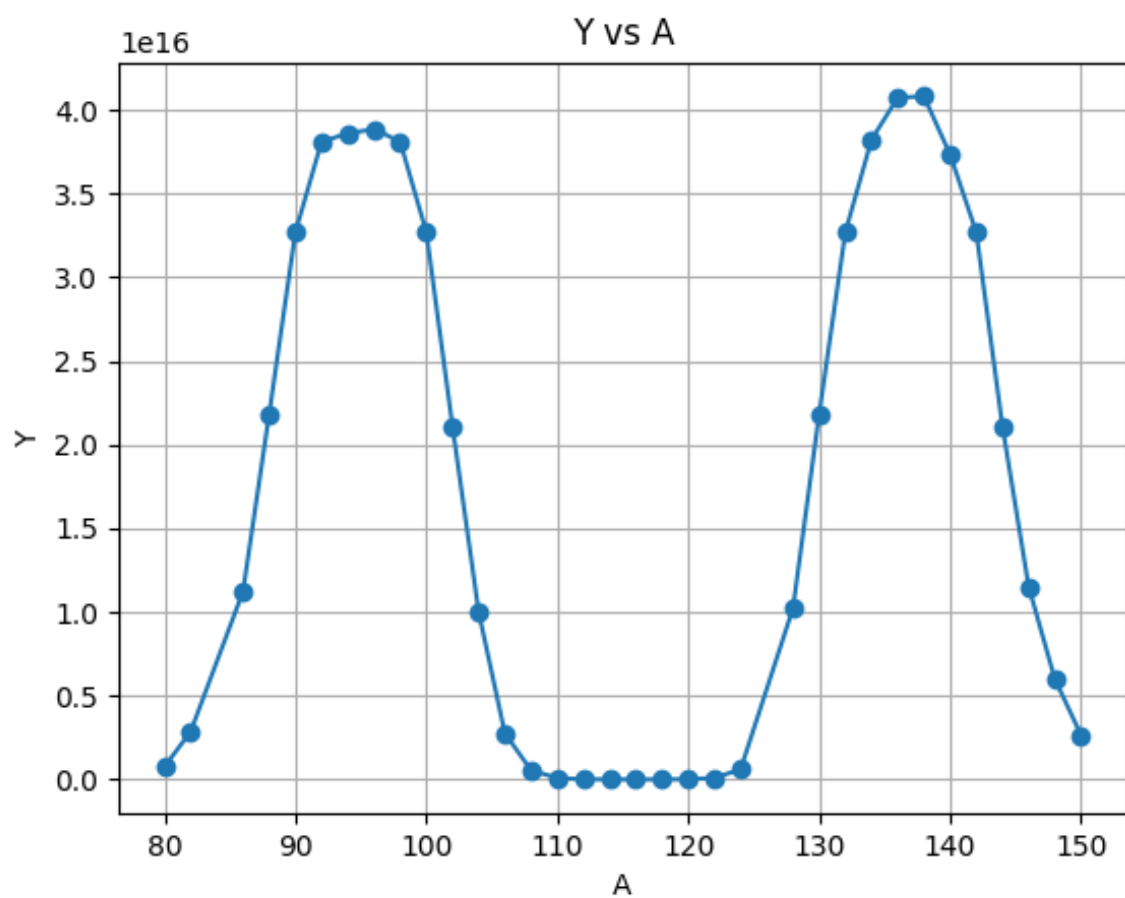


Figure 4.15:

4.12.3 YIELD AS A FUNCTION OF NEUTRON NUMBER NUMBER

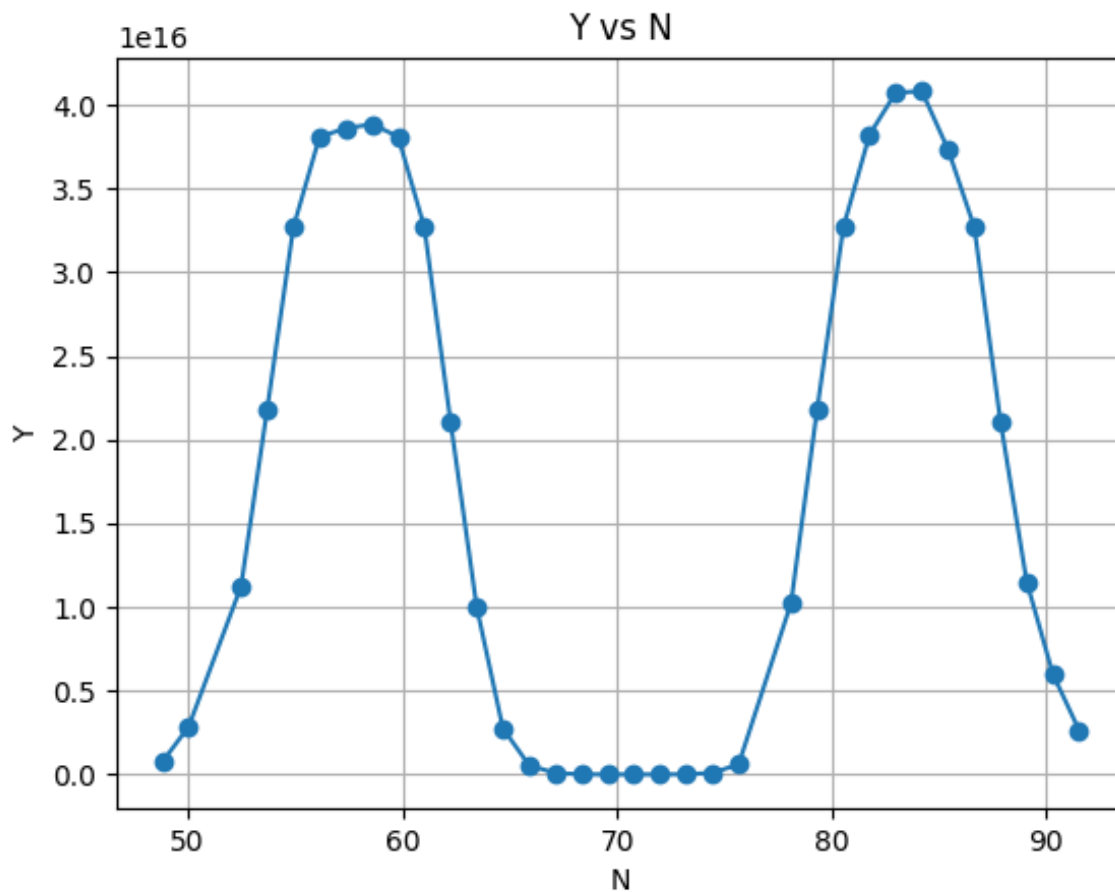


Figure 4.16:

4.12.4 YIELD AS A FUNCTION OF ATOMIC NUMBER

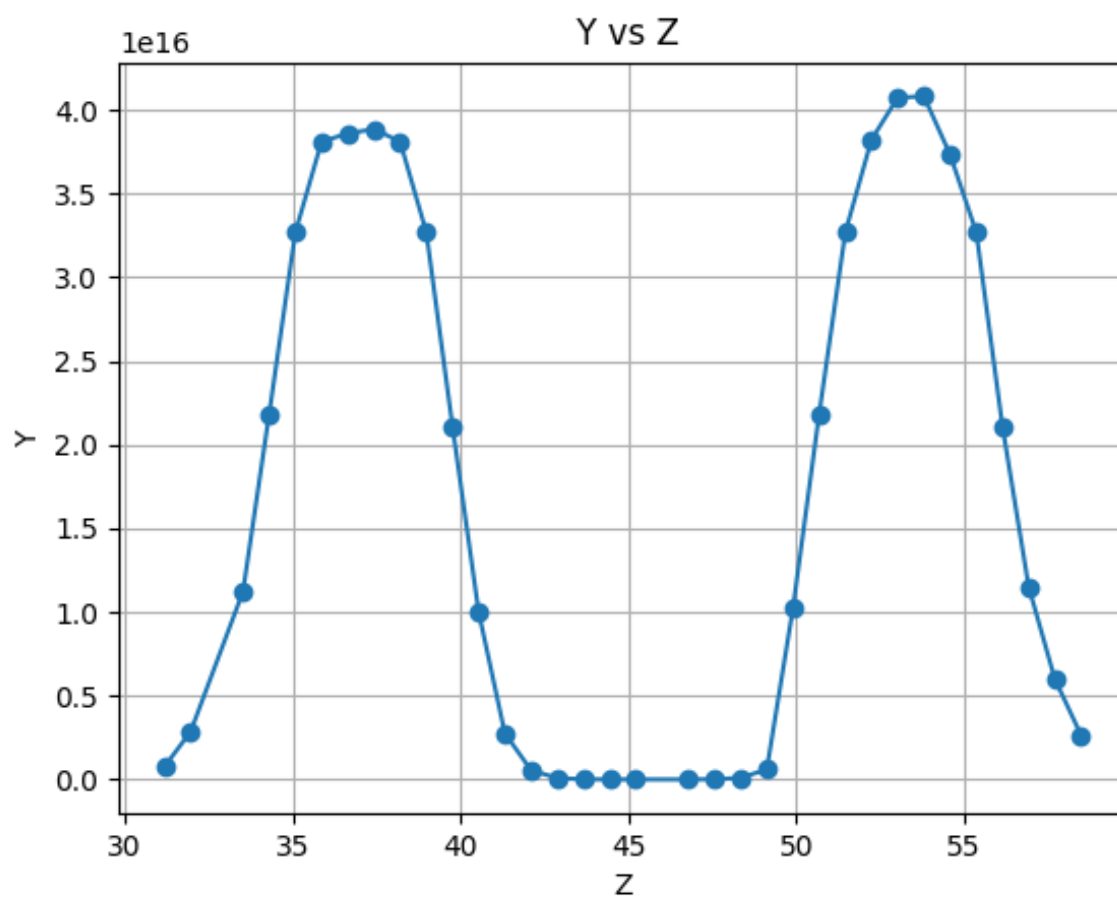
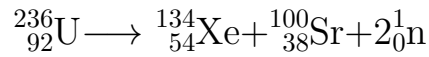


Figure 4.17:

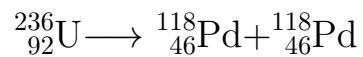
4.13 ENERGY RELEASED IN FISSION



Nucleus/Nuclide	No. of nucleons	B.E/A (Mev)	Total B.E (Mev)	Combined B.E (MeV)	Energy released (MeV)
$^{236}_{92}\text{U}$	236	7.66	1810.032	1810.032	172.33
$^{134}_{54}\text{Xe}$	134	8.46	1134.21	1982.37	
$^{100}_{38}\text{Sr}$	100	8.48	848.16		

Table 4.15:

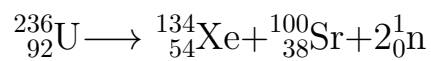
4.14 ENERGY RELEASED IN SYMMETRIC FISSION



Nucleus/Nuclide	No. of nucleons	B.E/A (Mev)	Total B.E (Mev)	Combined B.E (MeV)	Energy released (MeV)
$^{236}_{92}\text{U}$	236	7.66	1810.032	1810.032	194.936
$^{118}_{46}\text{Pd}$	118	8.49	1002.484	2004.968	
$^{118}_{46}\text{Pd}$	118	8.49	1002.484		

Table 4.16:

4.14.1 $\frac{Z}{A}$ AND $\frac{Z^2}{A}$ FOR FISSILE NUCLEUS AND FISSION PRODUCTS



Nucleus/Nuclides	Z/A	Z^2/A
${}_{92}^{236}\text{U}$	0.3898	35.864
${}_{54}^{134}\text{Xe}$	0.4029	21.76
${}_{38}^{100}\text{Sr}$	0.38	14.44

Table 4.17:

4.15 CHANGE IN THE RADIUS OF NUCLEI AS A FUNCTION OF DEFORMATION PARAMETER

$$R_1(\theta) = R_{01} [1 + \beta_1 Y_{20}(\theta)]$$

$$R_{01} = 1.2536 A_1^{1/3} - 0.80012 A_1^{-1/3} - 0.0021444 A_1^{-1} = 6.2554$$

$$R_2(\theta) = R_{02} [1 + \beta_2 Y_{20}(\theta)]$$

$$R_{02} = 1.2536 A_2^{1/3} - 0.80012 A_2^{-1/3} - 0.0021444 A_2^{-1} = 5.5643$$

$$Y_{20}(\theta = 0) = \frac{1}{4} \sqrt{5/\pi} (3 \cos^2 \theta - 1) = 0.63$$

$$Y_{20}(\theta = 90) = \frac{1}{4} \sqrt{5/\pi} (3 \cos^2 \theta - 1) = -0.31$$

β_1	$R_1(\theta = 0^\circ)$	$R_1(\theta = \frac{\pi}{2}^\circ)$	$R_2(\theta = 0^\circ)$	$R_2(\theta = \frac{\pi}{2}^\circ)$
-0.9	2.7085	7.974	2.443	7.2199
-0.8	3.1026	7.781	2.799	7.044
-0.7	3.496	7.587	3.154	6.869
-0.6	3.890	7.394	3.510	6.694
-0.5	4.284	7.201	3.865	6.519
-0.4	4.679	7.008	4.221	6.344
-0.3	5.073	6.814	4.576	6.169
-0.2	5.467	6.621	4.932	5.994
-0.1	5.861	6.428	5.287	5.819
0	6.255	6.25	5.643	5.64
0.1	6.64	6.041	5.99	5.470
0.2	7.043	5.848	6.35	5.295
0.3	7.437	5.655	6.710	5.120
0.4	7.81	5.461	7.065	4.945
0.5	8.225	5.268	7.421	4.770
0.6	8.619	5.075	7.776	4.595
0.7	9.014	4.882	8.132	4.420
0.8	9.408	4.688	8.487	4.245
0.9	9.802	4.495	8.843	4.070

Table 4.18:

4.15.1 CHANGE IN THE RADIUS OF NUCLEI AS A FUNCTION OF DEFORMATION PARAMETER

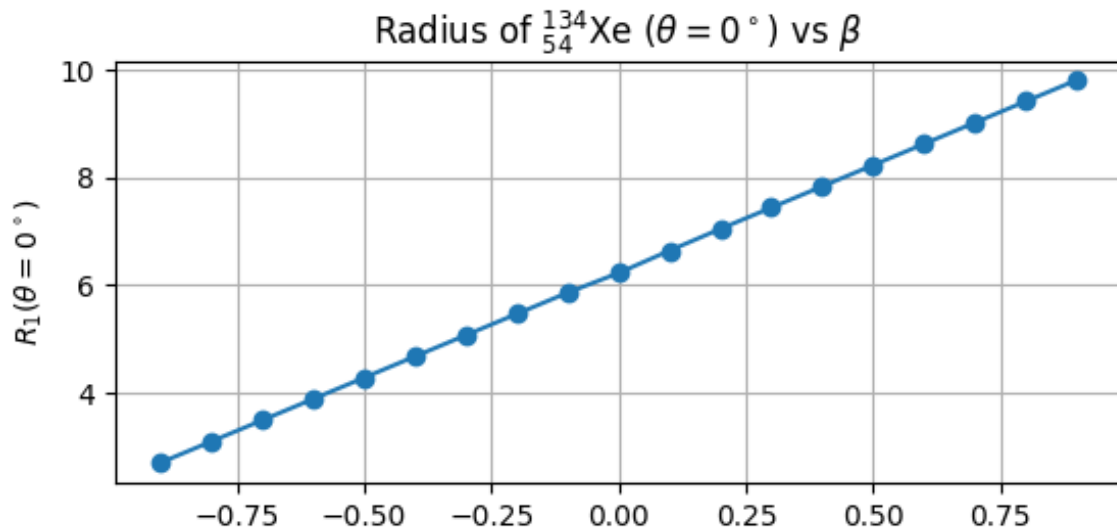
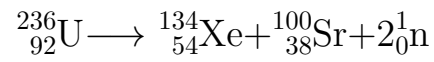


Figure 4.18:

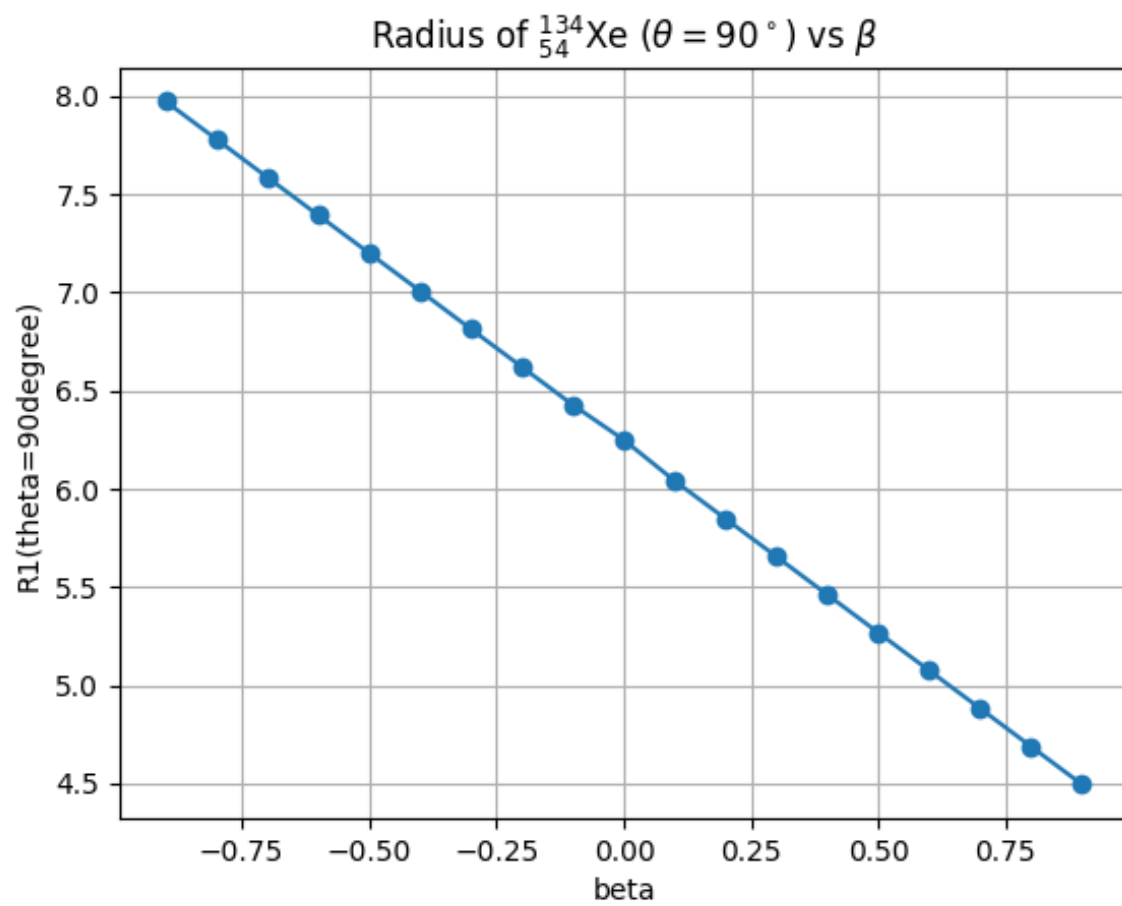


Figure 4.19:

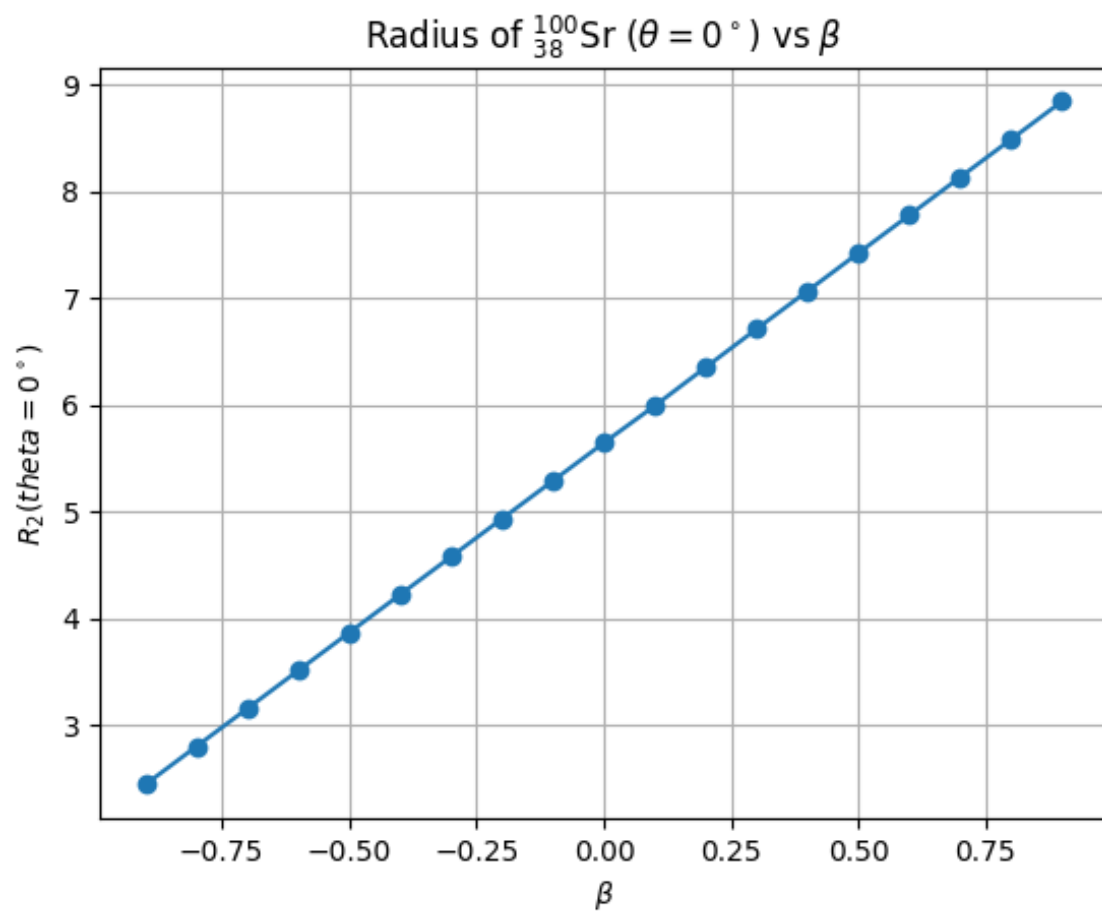


Figure 4.20:

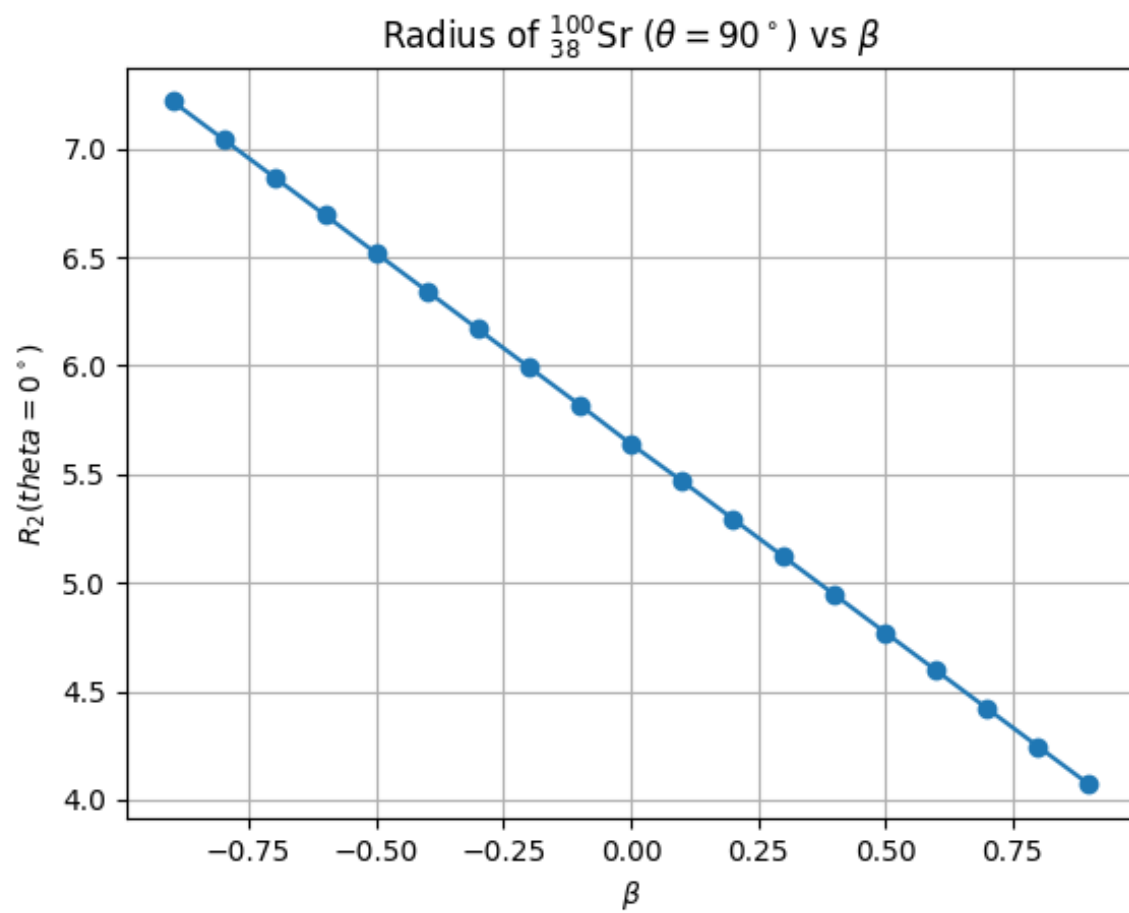
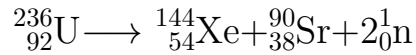


Figure 4.21:

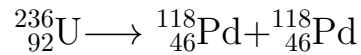
4.16 ENERGY RELEASED IN FISSION



Nucleus/Nuclide	No. of nucleons	B.E/A (Mev)	Total B.E (Mev)	Combined B.E (MeV)	Energy released (MeV)
$^{236}_{92}\text{U}$	236	7.66	1810.032	1810.032	168.52
$^{144}_{54}\text{Xe}$	144	8.24	1187.19	1978.56	
$^{90}_{38}\text{Sr}$	90	8.79	791.37		

Table 4.19:

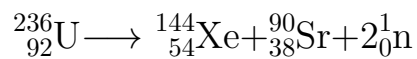
4.17 ENERGY RELEASED IN A SYMMETRIC FISSION



Nucleus/Nuclide	No. of nucleons	B.E/A (Mev)	Total B.E (Mev)	Combined B.E (MeV)	Energy released (MeV)
$^{236}_{92}\text{U}$	236	7.66	1810.032	1810.032	194.936
$^{118}_{46}\text{Pd}$	118	8.49	1002.484	2004.968	
$^{118}_{46}\text{Pd}$	118	8.49	1002.484		

Table 4.20:

4.18 $\frac{Z}{A}$ AND $\frac{Z^2}{A}$ FOR FISSILE NUCLEUS AND FISSION PRODUCTS



Nucleus/Nuclides	Z/A	Z^2/A
${}_{92}^{236}\text{U}$	0.3898	35.864
${}_{54}^{144}\text{Xe}$	0.375	20.25
${}_{38}^{90}\text{Sr}$	0.422	16.04

Table 4.21:

4.18.1 CHANGE IN THE RADIUS OF NUCLEI AS A FUNCTION OF DEFORMATION PARAMETER

$$R_1(\theta) = R_{01} [1 + \beta_1 Y_{20}(\theta)]$$

$$R_{01} = 1.2536 A_1^{1/3} - 0.80012 A_1^{-1/3} - 0.0021444 A_1^{-1} = 6.414$$

$$R_2(\theta) = R_{02} [1 + \beta_2 Y_{20}(\theta)]$$

$$R_{02} = 1.2536 A_2^{1/3} - 0.80012 A_2^{-1/3} - 0.0021444 A_2^{-1} = 5.436$$

$$Y_{20}(\theta = 0) = \frac{1}{4} \sqrt{5} / \pi (3 \cos^2 \theta - 1) = 0.63$$

$$Y_{20}(\theta = 90) = \frac{1}{4} \sqrt{5} / \pi (3 \cos^2 \theta - 1) = -0.31$$

β_1	$R_1(\theta = 0^\circ)$	$R_1(\theta = \frac{\pi}{2}^\circ)$	$R_2(\theta = 0^\circ)$	$R_2(\theta = \frac{\pi}{2}^\circ)$
-0.9	2.777	8.203	2.353	6.944
-0.8	3.181	8.004	2.696	6.776
-0.7	3.585	7.805	3.038	6.608
-0.6	3.989	7.607	3.381	6.439
-0.5	4.393	7.408	3.723	6.271
-0.4	4.797	7.209	4.0661	6.103
-0.3	5.201	7.010	4.408	5.934
-0.2	5.605	6.811	4.751	5.766
-0.1	6.009	6.612	5.09	5.598
0	6.414	6.41	5.43	5.43
0.1	6.818	6.215	5.77	5.261
0.2	7.222	6.016	6.120	5.093
0.3	7.626	5.817	6.463	4.925
0.4	8.030	5.618	6.805	4.756
0.5	8.434	5.419	7.148	4.58
0.6	8.838	5.219	7.490	4.420
0.7	9.242	5.022	7.833	4.251
0.8	9.646	4.823	8.175	4.083
0.9	10.050	4.624	8.518	3.915

Table 4.22:

4.18.2 CHANGE IN THE RADIUS OF NUCLEI AS A FUNCTION OF DEFORMATION PARAMETER

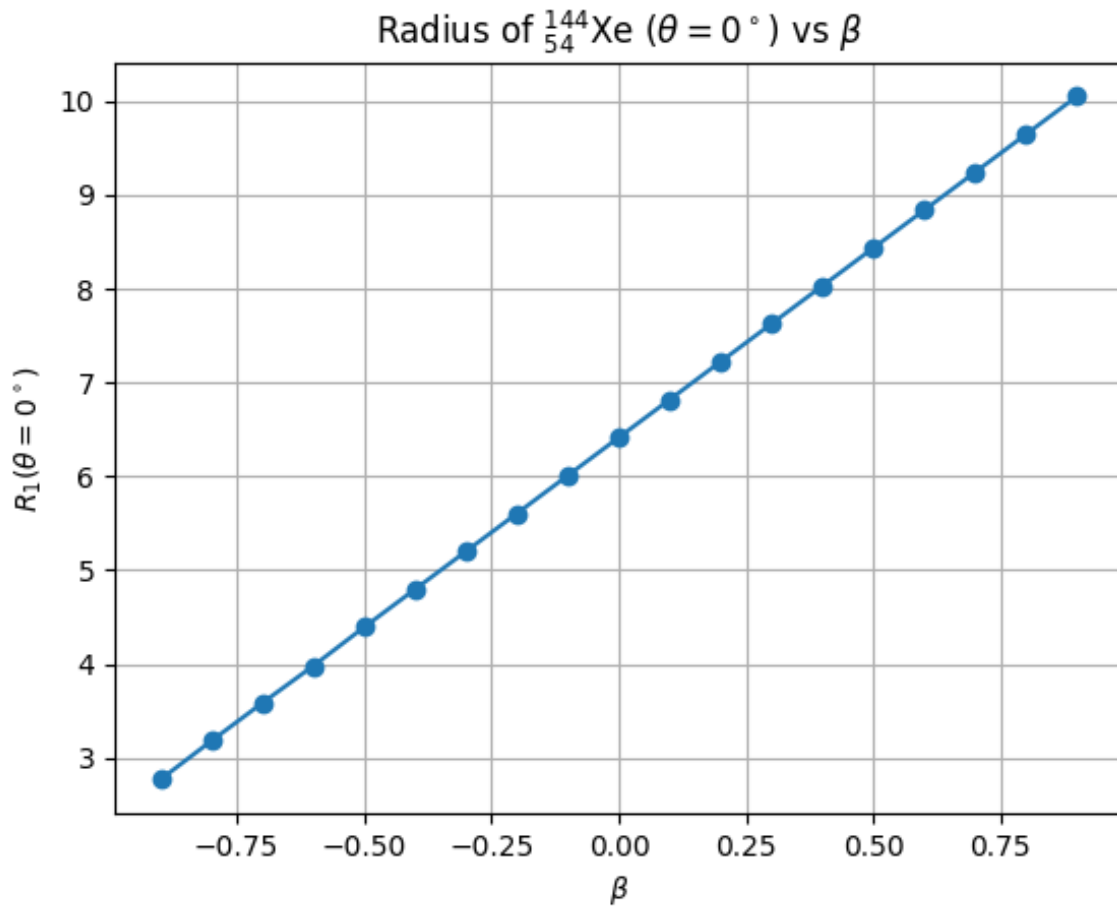
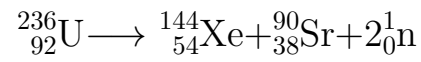


Figure 4.22:

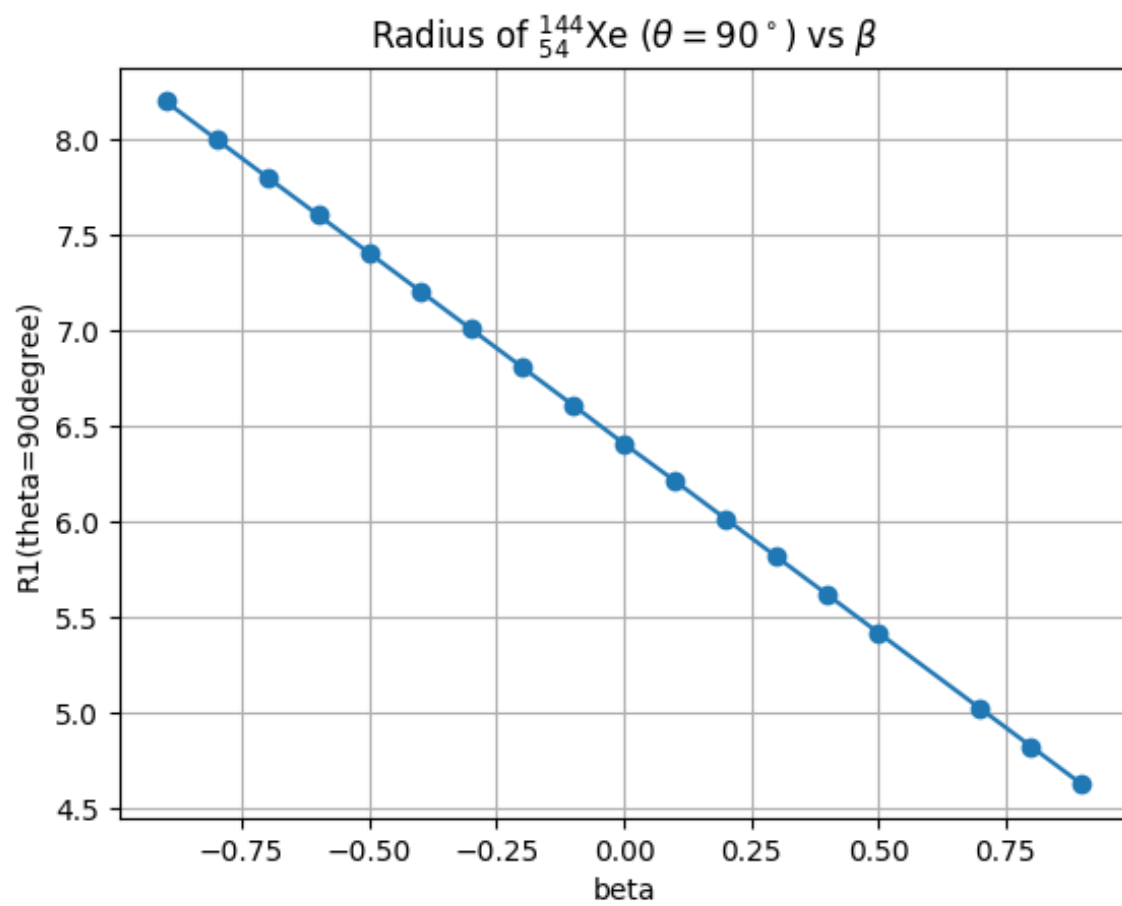


Figure 4.23:

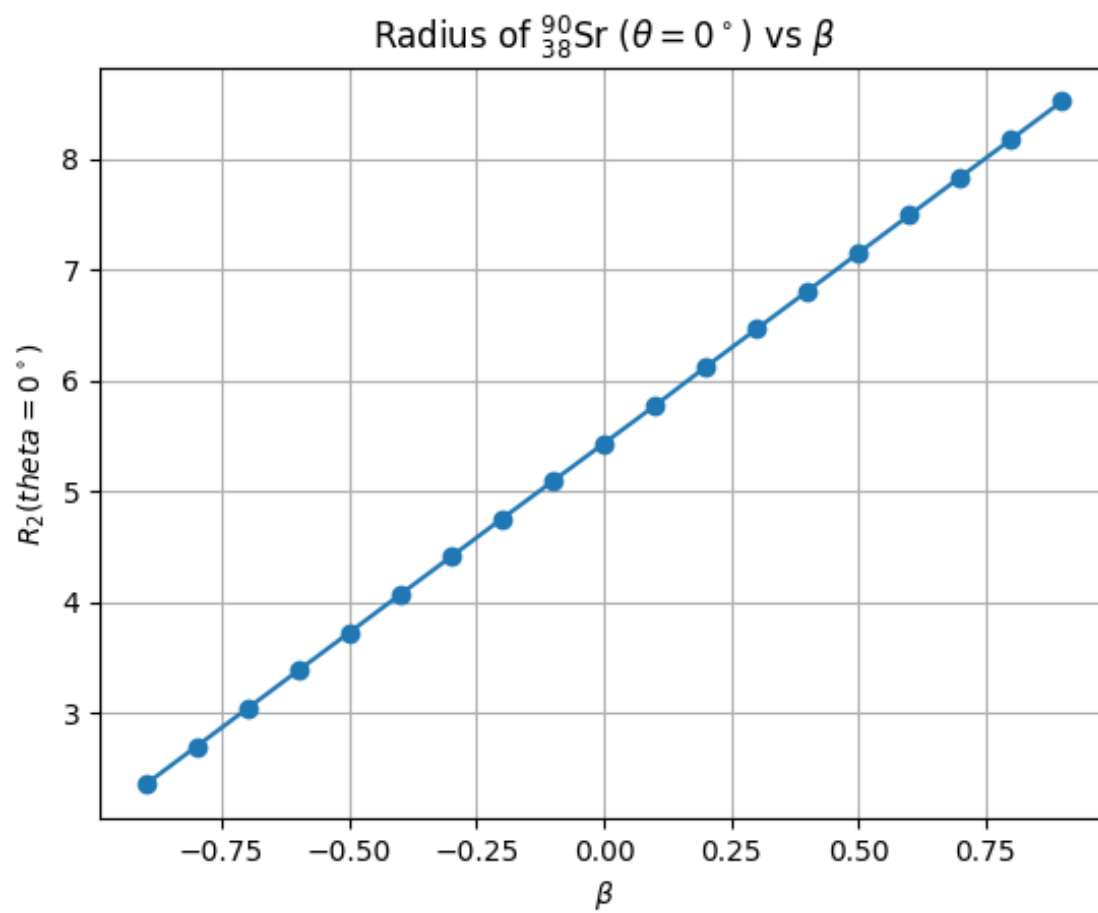


Figure 4.24:

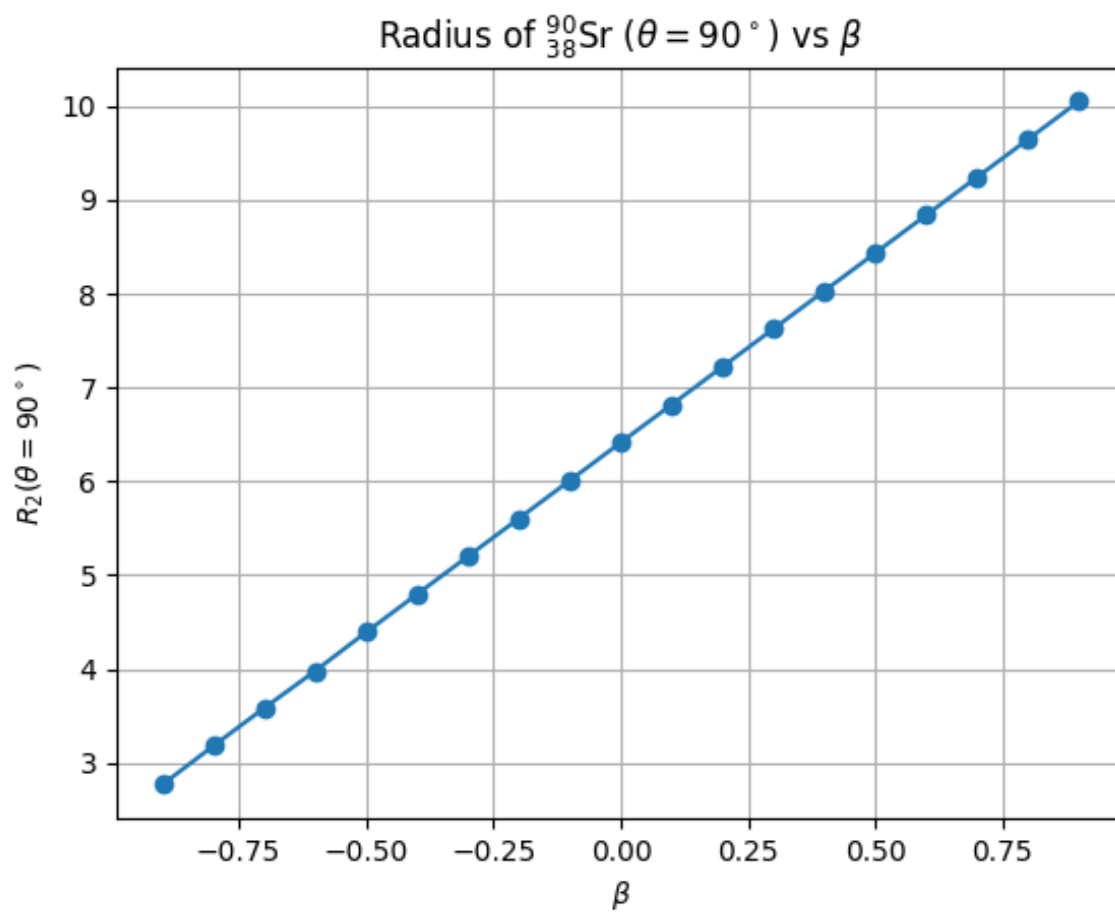


Figure 4.25:

4.19 CHANGE IN SURFACE ENERGY AND POTENTIAL ENERGY

$$\Delta S/S = \left(\frac{A_1^{2/3} + A_2^{2/3}}{A^{2/3}} \right) - 1$$

$$\Delta U/U = \frac{\frac{Z_1^2}{A_1^{1/3}} + \frac{Z_2^2}{A_2^{1/3}}}{\frac{Z^2}{A^{1/3}}} - 1$$

System	$\Delta S/S$	$\Delta U/U$
${}_{92}^{236}\text{U} \rightarrow {}_{54}^{134}\text{Xe} + {}_{38}^{100}\text{Sr} + 2{}^1_0\text{n}$	0.24983	-0.3567
${}_{92}^{236}\text{U} \rightarrow {}_{54}^{140}\text{Xe} + {}_{38}^{94}\text{Sr} + 2{}^1_0\text{n}$	0.24735	-0.3581
${}_{92}^{236}\text{U} \rightarrow {}_{54}^{144}\text{Xe} + {}_{38}^{90}\text{Sr} + 2{}^1_0\text{n}$	0.2452	-0.3585
${}_{92}^{236}\text{U} \rightarrow {}_{46}^{118}\text{Pd} + {}_{46}^{118}\text{Pd}$	0.25992	-0.370039

Table 4.23:

4.20 NEUTRON ENERGY DISTRIBUTION AS A FUNCTION OF TEMPERATURE

$$U \propto \exp\left(-\frac{\epsilon}{T}\right)$$

$$U \propto \sqrt{\epsilon} \times \exp\left(-\frac{\epsilon}{T}\right)$$

$$U \propto \epsilon^2 \times \exp\left(-\frac{\epsilon}{T}\right)$$

$$\epsilon = 0.025 \text{ MeV}$$

T	$\exp\left(-\frac{\epsilon}{T}\right)$	$\sqrt{\epsilon} \times \exp\left(-\frac{\epsilon}{T}\right)$	$\epsilon^2 \cdot \exp\left(-\frac{\epsilon}{T}\right)$
0.025	0.3678	0.05816	2.02 x 10 ⁻⁴
0.25	0.9048	0.14306	2.08 x 10 ⁻⁴
0.5	0.9512	0.1504	1.97 x 10 ⁻⁴
0.75	0.9672	0.1529	1.91 x 10 ⁻⁴
1	0.9753	0.1542	1.86 x 10 ⁻⁴
1.25	0.98019	0.1549	1.82 x 10 ⁻⁴
1.5	0.98347	0.1555	1.799 x 10 ⁻⁴
1.75	0.9858	0.15587	1.77 x 10 ⁻⁴
2	0.9875	0.15614	1.75 x 10 ⁻⁴
2.25	0.9889	0.15636	1.732 x 10 ⁻⁴
2.5	0.990	0.15654	1.71 x 10 ⁻⁴
2.75	0.9905	0.15668	1.70 x 10 ⁻⁴
3	0.99170	0.1568	1.68 x 10 ⁻⁴
3.25	0.9923	0.1569	1.67 x 10 ⁻⁴
3.5	0.9828	0.15698	1.66 x 10 ⁻⁴
3.75	0.9933	0.15706	1.65 x 10 ⁻⁴
4	0.9937	0.1571	1.64 x 10 ⁻⁴

Table 4.24:

4.20.1 NEUTRON ENERGY DISTRIBUTION AS A FUNCTION OF TEMPERATURE

$$U \propto \exp\left(-\frac{\epsilon}{T}\right)$$

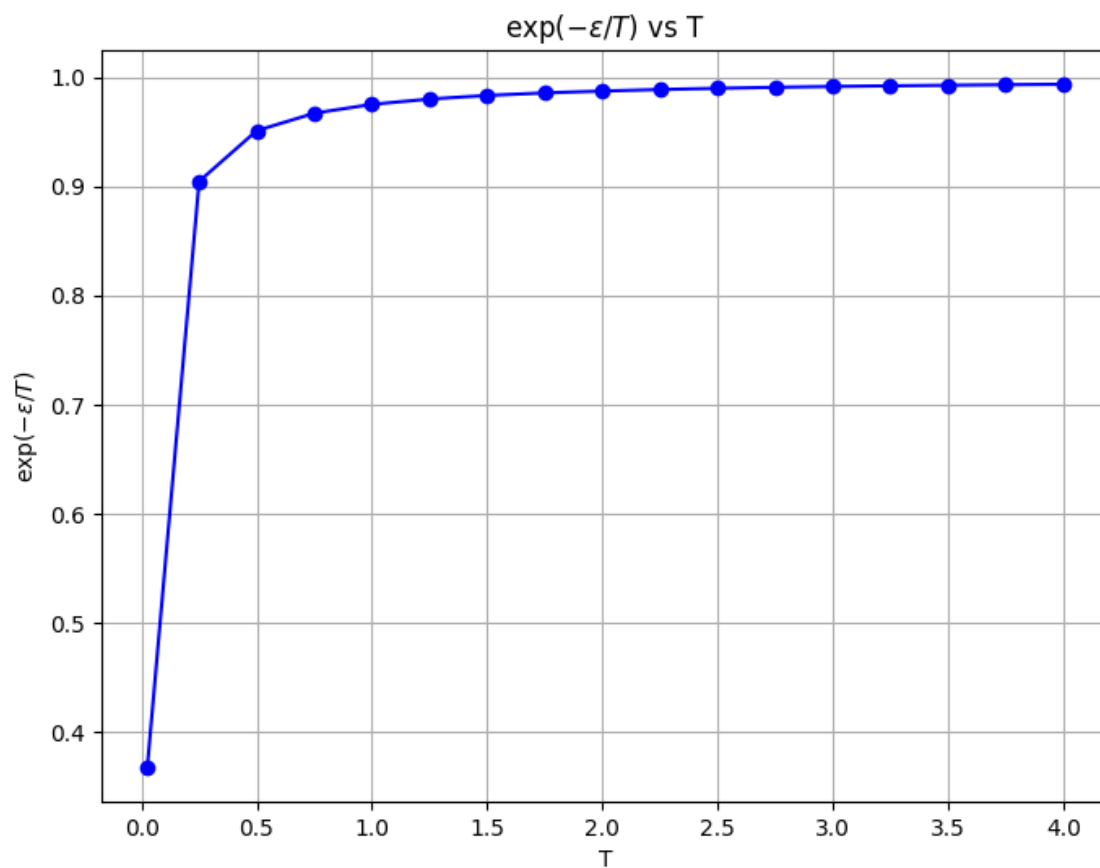


Figure 4.26:

$$U \propto \sqrt{\epsilon} \times \exp\left(-\frac{\epsilon}{T}\right)$$

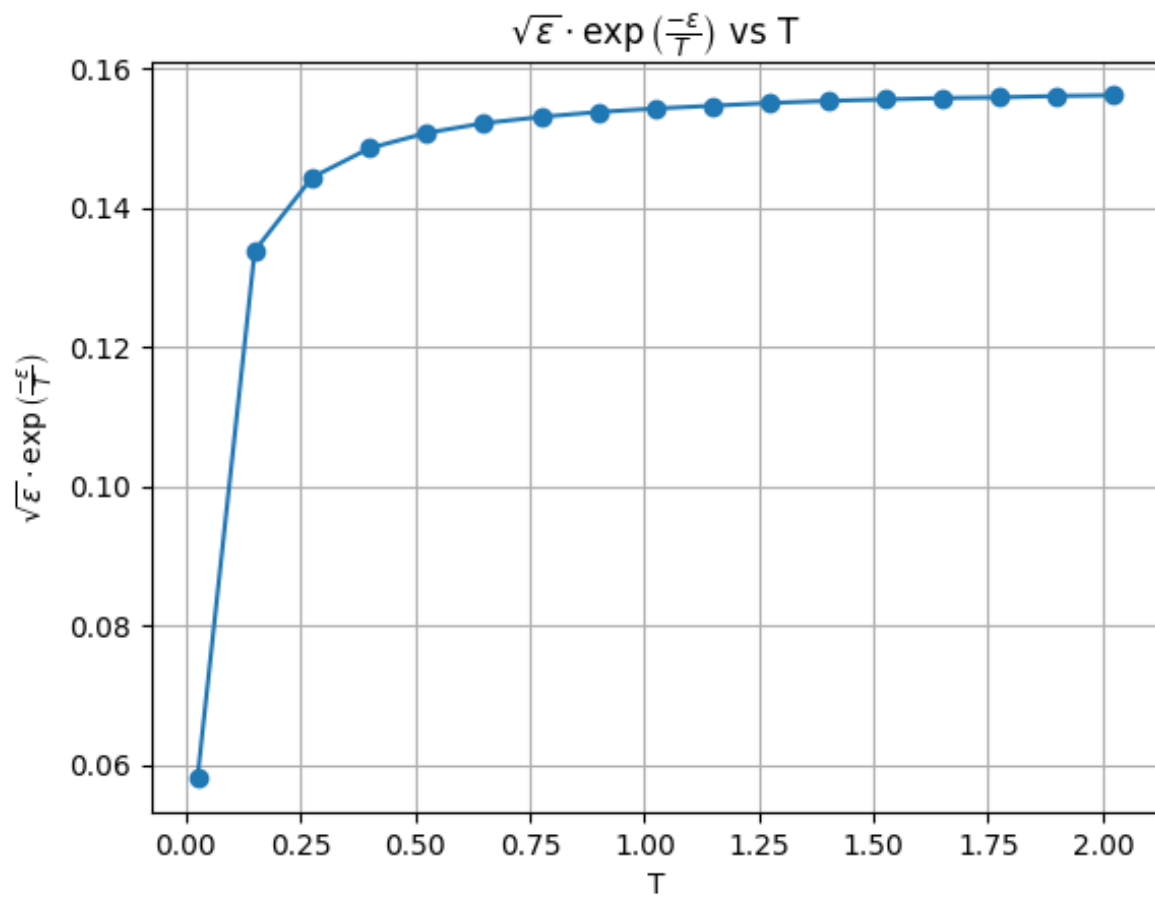


Figure 4.27:

$$U \propto \epsilon^2 \times \exp\left(-\frac{\epsilon}{T}\right)$$

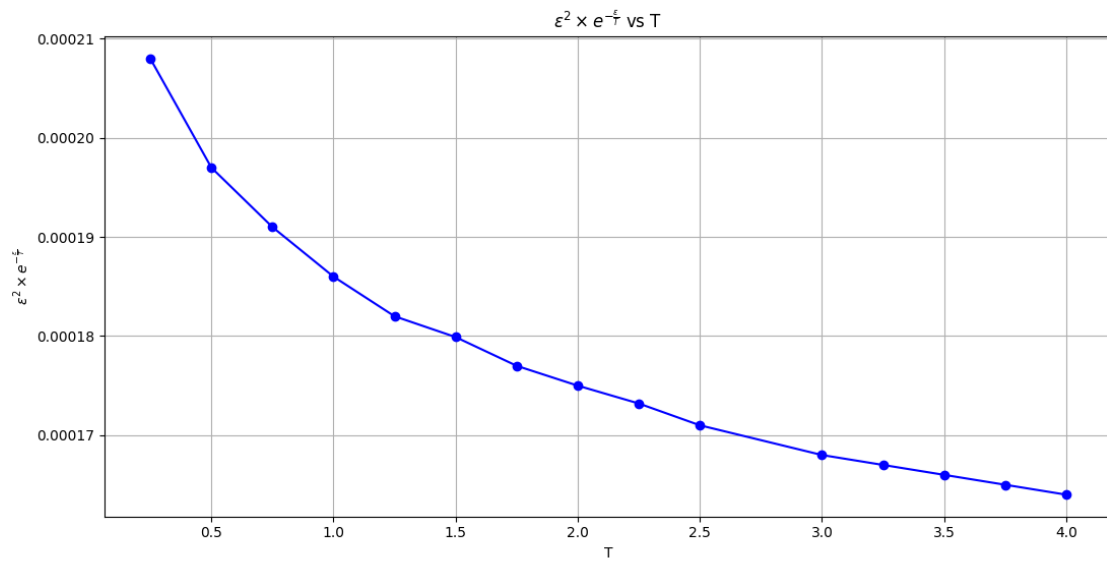


Figure 4.28:

4.21 NEUTRON ENERGY DISTRIBUTION AS A FUNCTION OF ϵ

$$U \propto \exp\left(-\frac{\epsilon}{T}\right)$$

$$U \propto \sqrt{\epsilon} \times \exp\left(-\frac{\epsilon}{T}\right)$$

$$U \propto \epsilon^2 \times \exp\left(-\frac{\epsilon}{T}\right)$$

$T = 0.5 \text{ MeV}$

ϵ	$\exp\left(-\frac{\epsilon}{T}\right)$	$\sqrt{\epsilon} \times \exp\left(-\frac{\epsilon}{T}\right)$	$\epsilon^2 \cdot \exp\left(-\frac{\epsilon}{T}\right)$
0.025	0.9512	0.1504	0.000594
0.0625	0.8824	0.2370	0.00344
0.1025	0.8146	0.2608	0.00855
0.1425	0.75201	0.2838	0.0152
0.1825	0.6941	0.2965	0.02312
0.2225	0.6408	0.3022	0.03172
0.2625	0.5915	0.3030	0.03838
0.3025	0.54607	0.30034	0.04996
0.3425	0.50409	0.2950	0.05913
0.3825	0.4653	0.2877	0.06808
0.4225	0.4295	0.2792	0.07667
0.4625	0.3965	0.2696	0.0848
0.5	0.3678	0.2601	0.09196
0.75	0.2231	0.1932	0.12551
1.0	0.1353	0.1353	0.13533
1.25	0.08208	0.0917	0.1282
1.5	0.0497	0.0609	0.11202
1.75	0.03019	0.0399	0.0924
2.0	0.0183	0.0259	0.07326

Table 4.25:

4.21.1 NEUTRON ENERGY DISTRIBUTION AS A FUNCTION OF ϵ

$$U \propto \exp\left(-\frac{\epsilon}{T}\right)$$

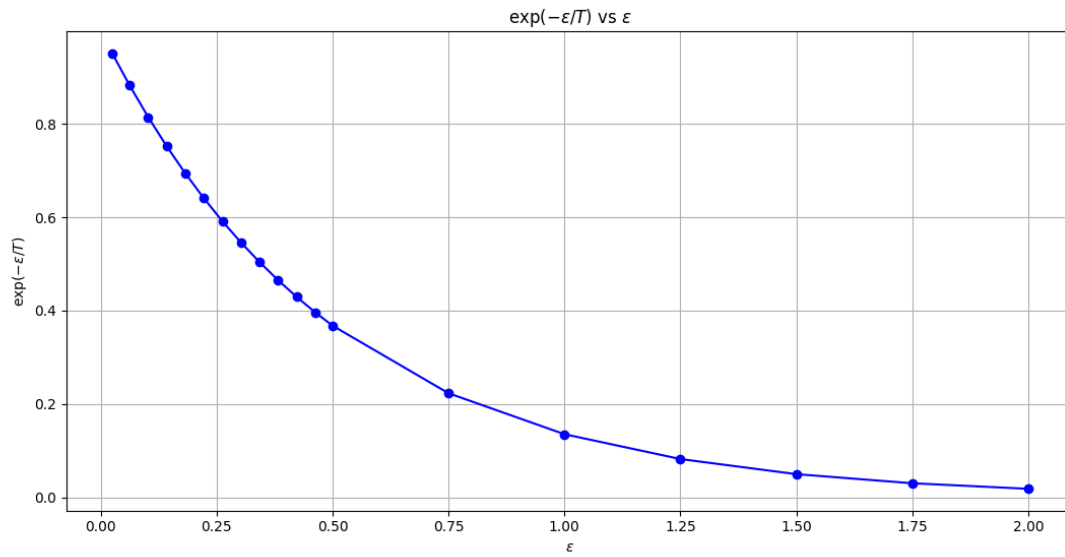


Figure 4.29:

$$U \propto \sqrt{\epsilon} \times \exp\left(-\frac{\epsilon}{T}\right)$$

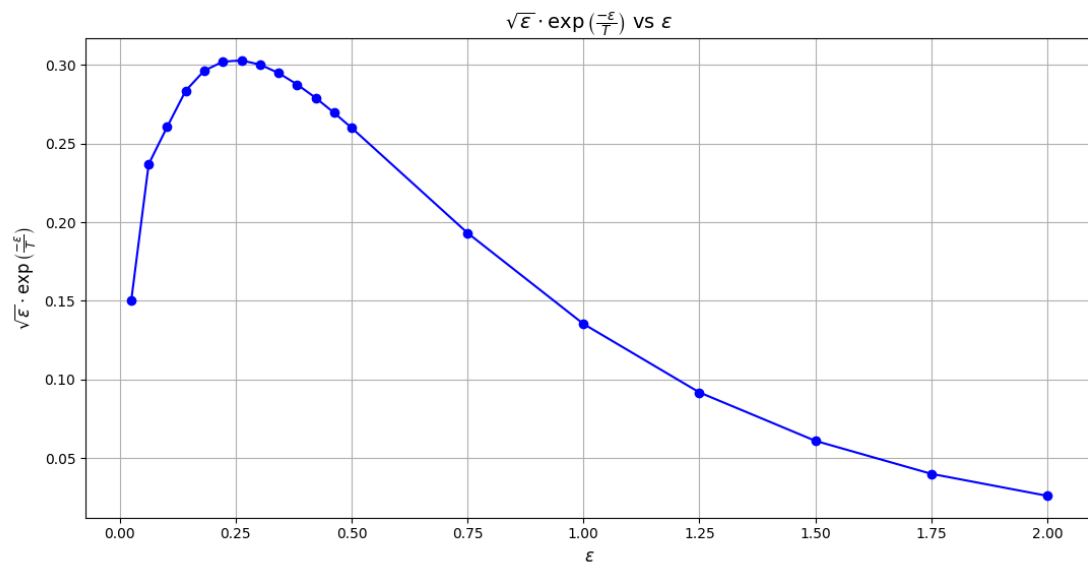


Figure 4.30:

$$U \propto \epsilon^2 \times \exp\left(-\frac{\epsilon}{T}\right)$$

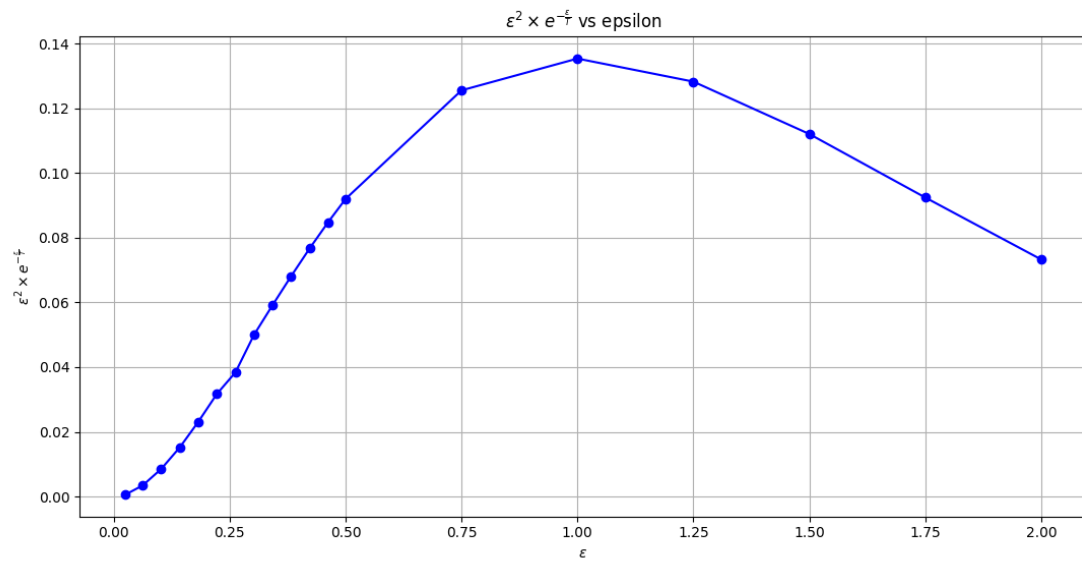


Figure 4.31:

Chapter 5

RESULTS AND DISCUSSIONS

- Critical energy as a function of $\frac{Z^2}{A}$ is measured and plotted.
- The liquid drop energy of individual fission fragments $^{140}_{54}\text{Xe}$, $^{94}_{38}\text{Sr}$ and their isotopes $^{134}_{54}\text{Xe}$, $^{100}_{38}\text{Sr}$, $^{144}_{54}\text{Xe}$, $^{90}_{38}\text{Sr}$ as a function of the β deformation parameter is calculated, tabulated, and plotted.
- The $\frac{Z^2}{A}$ vs N and N vs Z for the decay chain of XENON and STRONTIUM are plotted.
- The amount of energy released in the fission process and the ratio of surface energy and volume energy due to fission is also calculated.
- The change in the radius of the nucleus as a function of the deformation parameter is tabulated and plotted.
- $\frac{Z}{A}$ and $\frac{Z^2}{A}$ for fissile nuclei and fission products are obtained.
- The average distance as a function of the deformation parameter is tabulated and plotted.
- Projectile energy and excitation energy of the neutron are calculated and plotted.
- Nuclear potential and the yield distribution of the mass number of fragments, neutron number of fragments, and proton number of fragments are measured and plotted.
- The neutron distribution curves are plotted at different temperatures of the fissile nucleus.

Chapter 6

CONCLUSION

- Yield of fission fragments distribution is calculated using a quantum mechanical model and the yield distribution is plotted.
- Experimentally measured yield distribution and our calculated distribution are similar.
- The asymmetric mass distribution of fragments can only be explained by the quantum mechanical model. Liquid drop model is a failure in this case.
- Using the quantum mechanical model, we reproduced the experimental yield neutron distribution and it is plotted.

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